Tools for the Optimal Operation and Planning of Power Systems

Lecture 1: Introduction to Operations Research

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Solutions to the proposed exercises

https://github.com/AlvaroGarciaCerezo/CourseOR/tree/main/Lecture%201



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2. Classification of optimization models
3. Linear programming
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5. Nonlinear programming
6. Optimization software
7. Dual variables
8. Reformulation techniques

1. Introduction

Optimization

Introduction

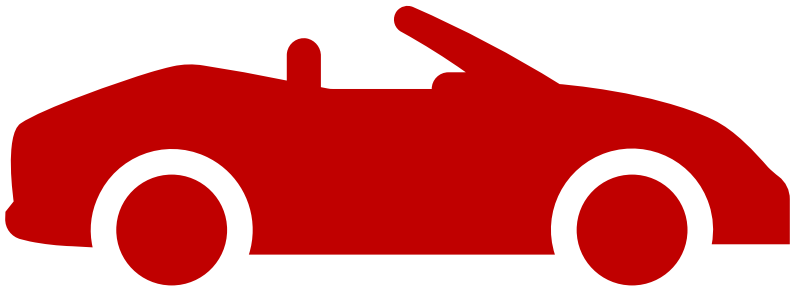
To make the best decision

Optimization

What is the best option?

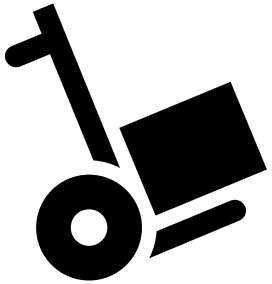


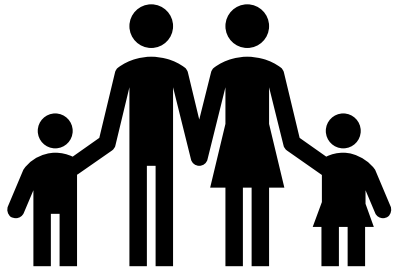


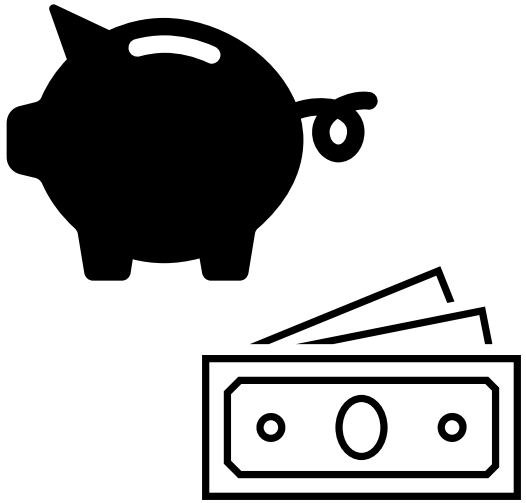


Optimization

It depends









Optimization

To make the best decision
for a given objective function
and subject to the
constraints of the problem

Optimization

Objective function:

- Minimize the cost / maximize the profit.
- Maximize the renewable production.
- Minimize the losses of energy.

Optimization

Constraints:

- Power balance.
- Budget.
- Technical limits.
- Environmental and political limits.

General formulation

$$\min_{x} \quad z = f(x)$$

s.t.:

$$h(x) = 0$$

$$g(x) \leq 0$$

General formulation

$\min_x \quad z = f(x)$ ⟵ Objective function

s.t.:

$h(x) = 0$

$g(x) \leq 0$

General formulation

$$\min_{x} \quad z = f(x)$$

Variables

s.t.:

$$h(x) = 0$$

$$g(x) \leq 0$$

General formulation

$$\min_{x} \quad z = f(x)$$

s.t.: ← Subject to

$$h(x) = 0$$

$$g(x) \leq 0$$

General formulation

$$\min_{x} \quad z = f(x)$$

s.t.:

$\boldsymbol{h(x) = 0}$ ⟵ Equality constraints

$$g(x) \leq 0$$

General formulation

$$\min_{x} \quad z = f(x)$$

s.t.:

$$h(x) = 0$$

$$\boldsymbol{g(x) \leq 0} \longleftarrow \text{Inequality constraints}$$

General formulation

$$\min_x \quad z = f(x)$$

s.t.:

$$h(x) = 0$$

$$g(x) \leq 0$$



$$\max_x \quad z = -f(x)$$

s.t.:

$$h(x) = 0$$

$$g(x) \leq 0$$

Applications to power systems

- Participation in electricity markets.
- Security of power systems.
- Participation in demand response programs.
- Investment in transmission lines, generating units, and storage facilities.

What will you learn?

- To classify optimization models.
- To solve optimization problems graphically.
- To solve optimization problems using computational tools.
- The meaning of dual variables.
- To reformulate nonlinear optimization problems.

2. Classification of optimization models

Classification of optimization problems

What is mathematical optimization?

https://www.youtube.com/watch?v=AM6BY4btj-M

Classification of optimization problems

Model Type	Abbreviation	Terms	Variables
Linear programming	LP	Linear	Continuous
Mixed-integer linear programming	MILP	Linear	Integer and continuous
Nonlinear programming	NLP	Nonlinear and linear	Continuous

Classification of optimization problems

There are other types, such as:

Model Type	Abbreviation	Terms	Variables
Mixed-integer nonlinear programming	MINLP	Nonlinear and linear	Integer and continuous

Classification of optimization problems

LP → Easy to solve.

MILP → It depends on the number of integer variables and the problem addressed.

NLP → (Generally) Hard to solve.

Classification of optimization problems

M. Carrión, R. Domínguez, and G. Oggioni, "Optimal participation of wind power producers in a hybrid intraday market: a multi-stage stochastic approach", *Energy Economics*, vol. 144, art. no. 108303, Apr. 2025, DOI: 10.1016/j.eneco.2025.108303.

LP problem:

- Continuous variables: 3.8 million.
- Constraints: 3.7 million.
- Computational time: lower than 1 min.

Classification of optimization problems

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

MILP problem:

- Continuous variables: 14,337.
- Integer variables: 1,008.
- Constraints: 12,372.
- Computational time: 3 min.

Classification of optimization problems

Á. García-Cerezo, A. Pleite-Paniagua, L. Baringo, and R. Domínguez, "Uncertainty-driven long-term participation of a virtual power plant in the day-ahead market and a grid-balancing demand response auction", *24th Conference of the International Federation of Operational Research Societies (IFORS 2026)*, Vienna, Austria, Jul. 2026.

MILP problem:

- Continuous variables: 1,329,631.
- Integer variables: 129,990.
- Constraints: 2,436,066.
- Computational time: 1 h.

Classification of optimization problems

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024, DOI: 10.1109/TPWRS.2023.3236397.

MILP problem:

- Continuous variables: 1,544,135.
- Integer variables: 272,017.
- Constraints: 3,570,235.
- Computational time: not solved after 168 h (1 week).

Classification of optimization problems

J. Muñoz Pedrosa, "Planning and operational decision-making for an industrial plant with self-consumption", Bachelor's Final Project, Electrical Engineering Degree, Universidad de Castilla-La Mancha, supervised by Á. García-Cerezo and M. Cañas-Carretón, Jul. 2026.

MINLP problem: not solved after 24 h (1 day).

Classification of optimization problems

Not all NLP and MINLP problems are hard to solve:

Model Type	Abbreviation	Terms	Variables
Mixed-integer quadratically constrained quadratic programming	MIQCQP	Quadratic and linear	Integer and continuous

Classification of optimization problems

J. Muñoz Pedrosa, "Planning and operational decision-making for an industrial plant with self-consumption", Bachelor's Final Project, Electrical Engineering Degree, Universidad de Castilla-La Mancha, supervised by Á. García-Cerezo and M. Cañas-Carretón, Jul. 2026.

MINLP problem: not solved after **24 h** (1 day).

MIQCQP problem: **3 min**.

The optimization problem was **THE SAME**.

We only changed **how** the optimization problem **was modeled**.

Convexity

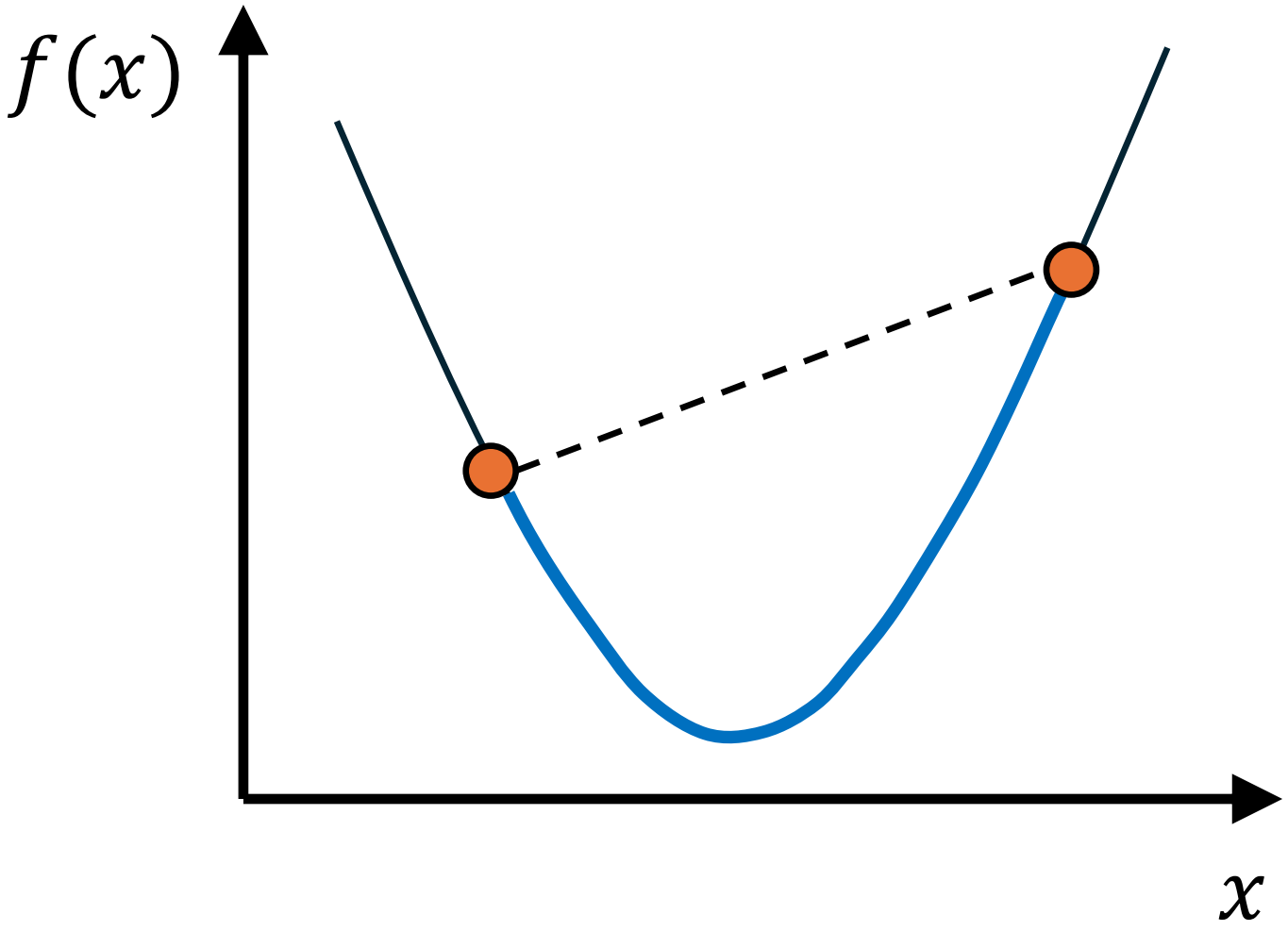
Convex problem

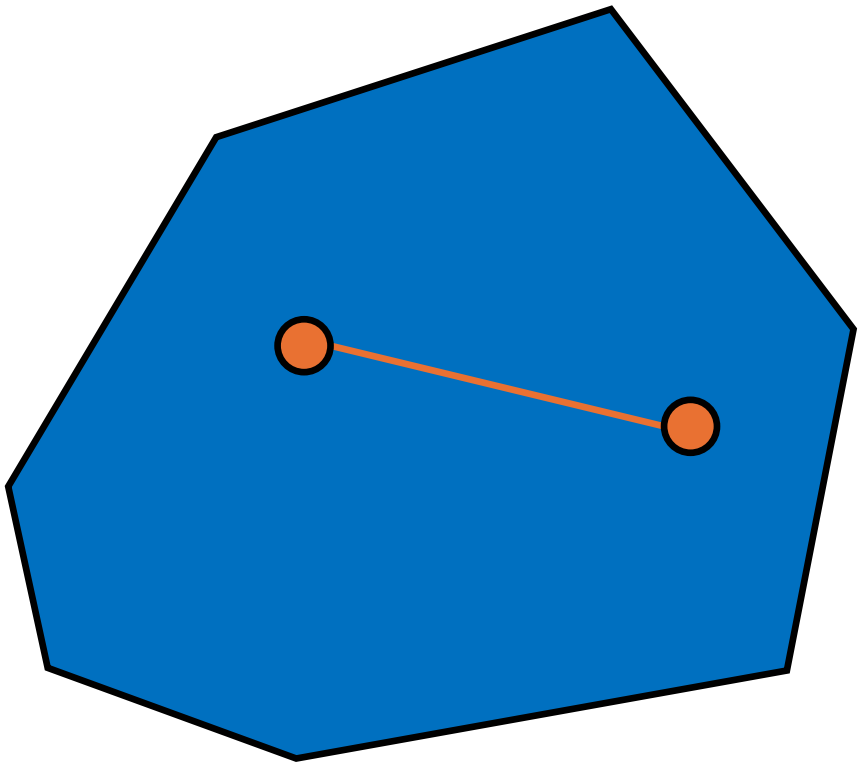


Nonconvex problem

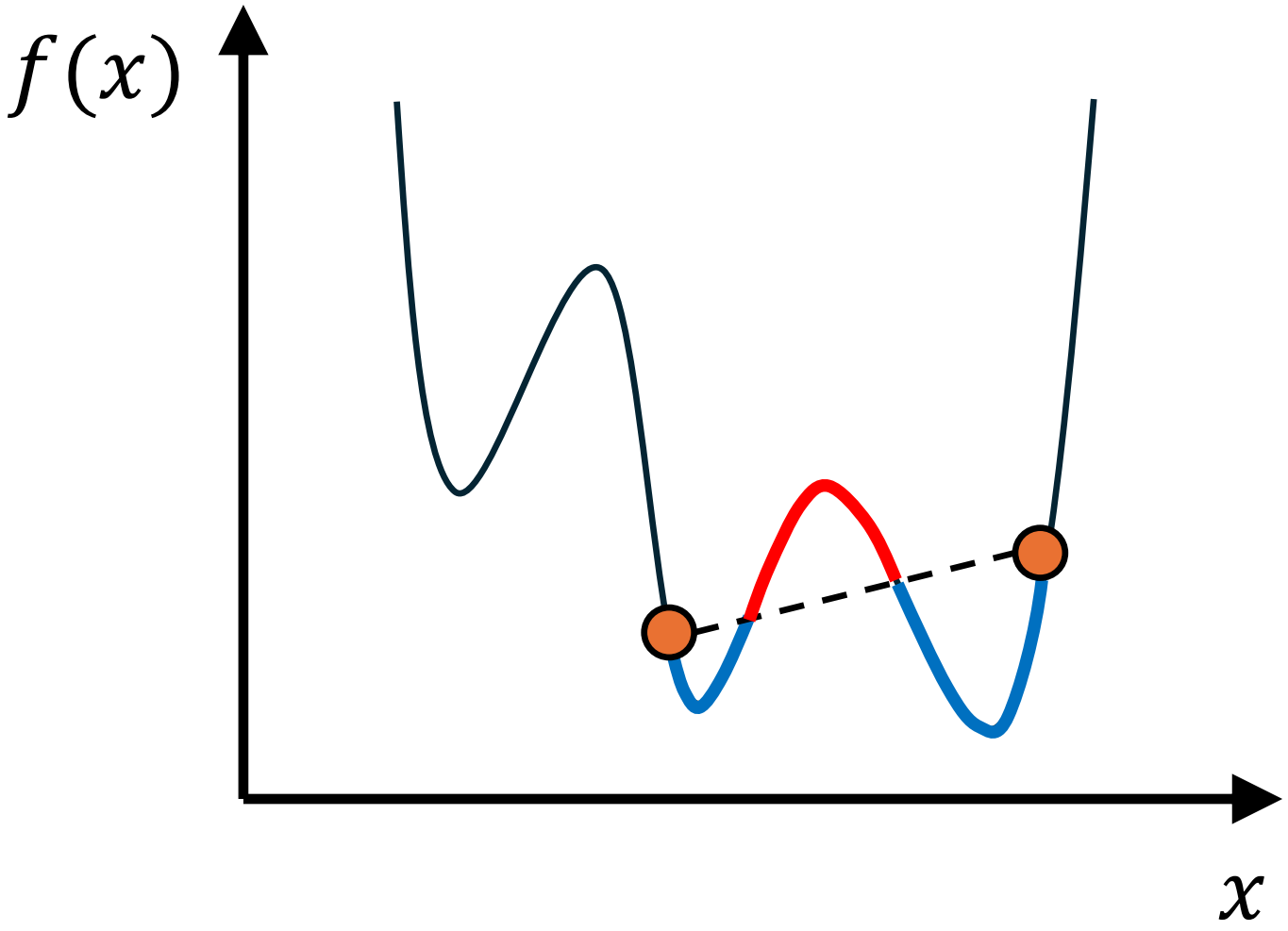


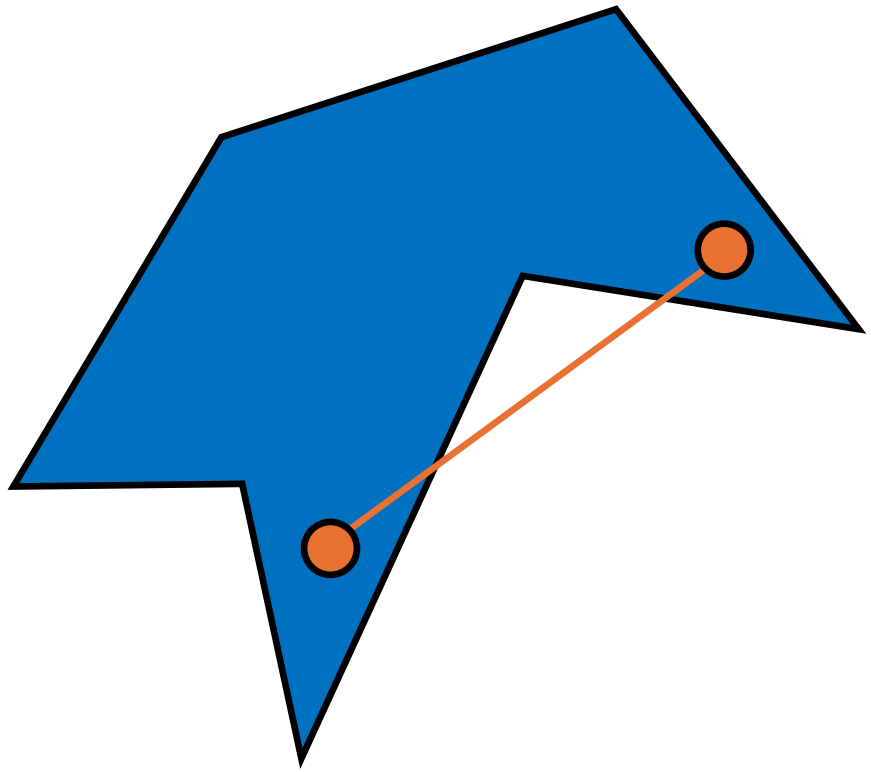
Convex function and set





Nonconvex function and set





Convexity

Convex set and convex function

https://www.youtube.com/watch?v=a_gRfwHUlhQ

Convexity and the principle of duality

https://www.youtube.com/watch?v=d0CF3d5aEGc

Convexity

LP → Convex

MILP → Nonconvex

NLP → Convex or Nonconvex

MINLP → Nonconvex

Convexity

Integer variables → Nonconvex

BE CAREFULL INCLUDING INTEGER VARIABLES

High computational times

Sometimes the optimization problems cannot be directly solved due to the high computational burden.

We can:

- Use approximation techniques (lecture 3).
- Use decomposition techniques (lecture 4).

3. Linear programming (LP)

Example 1

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$$0 \leq x \leq 5$$

$$0 \leq y \leq 3$$

Example 1

$$\max_{x,y} \quad z = 2x + y \longleftarrow$$ Objective function

s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$

Example 1

$$\max_{x,y} \quad z = 2x + y$$

Variables (output data)

s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$

Example 1

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$$\mathbf{0 \leq x \leq 5}$$

$$\mathbf{0 \leq y \leq 3}$$

} Constraints

Example 1

$$\max_{x,y} \quad z = \mathbf{2}x + \mathbf{1}y$$

s.t.:

$$\mathbf{0} \leq \mathbf{1}x \leq \mathbf{5}$$

$$\mathbf{0} \leq \mathbf{1}y \leq \mathbf{3}$$

Parameters (input data)

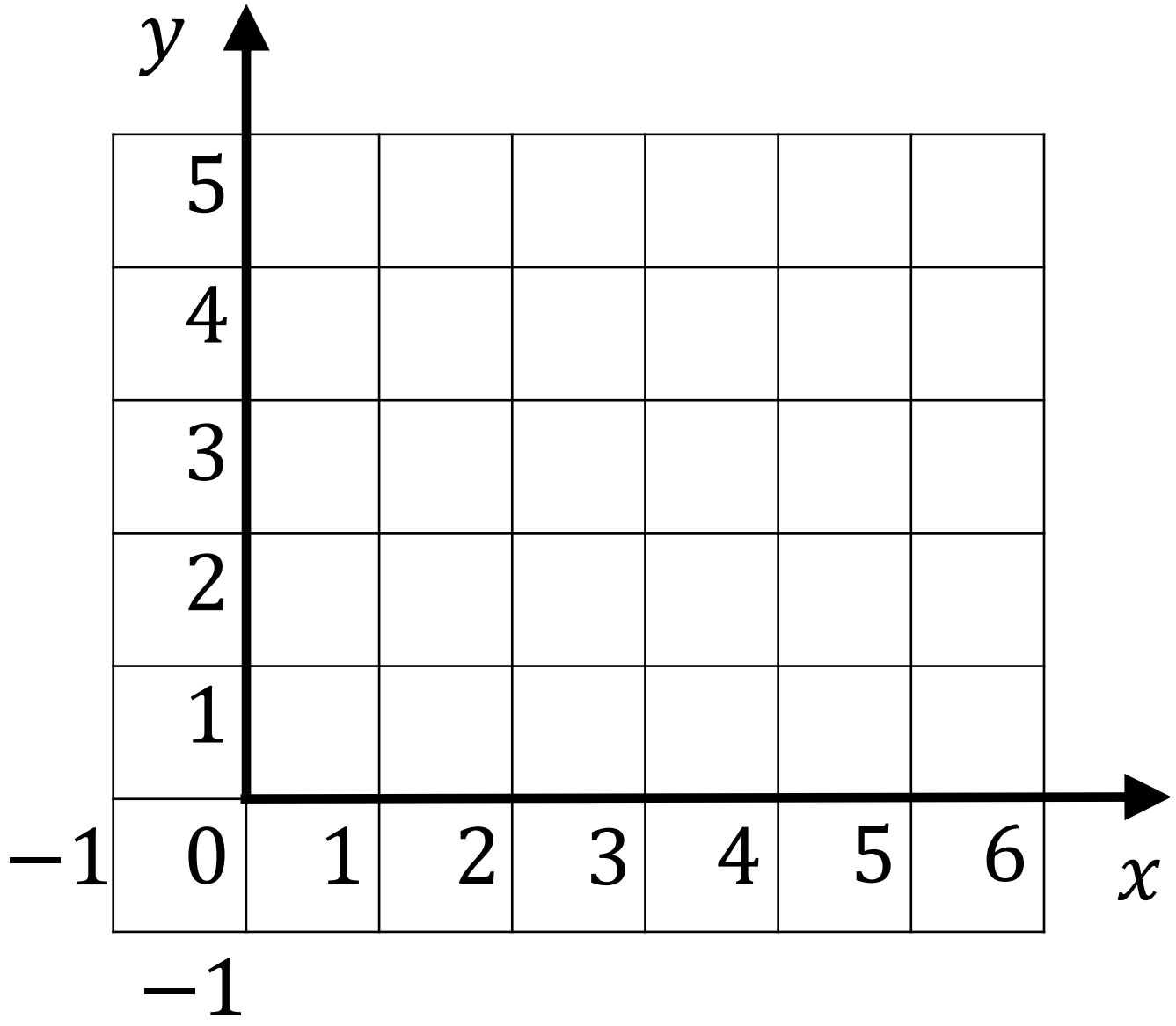
Example 1

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$



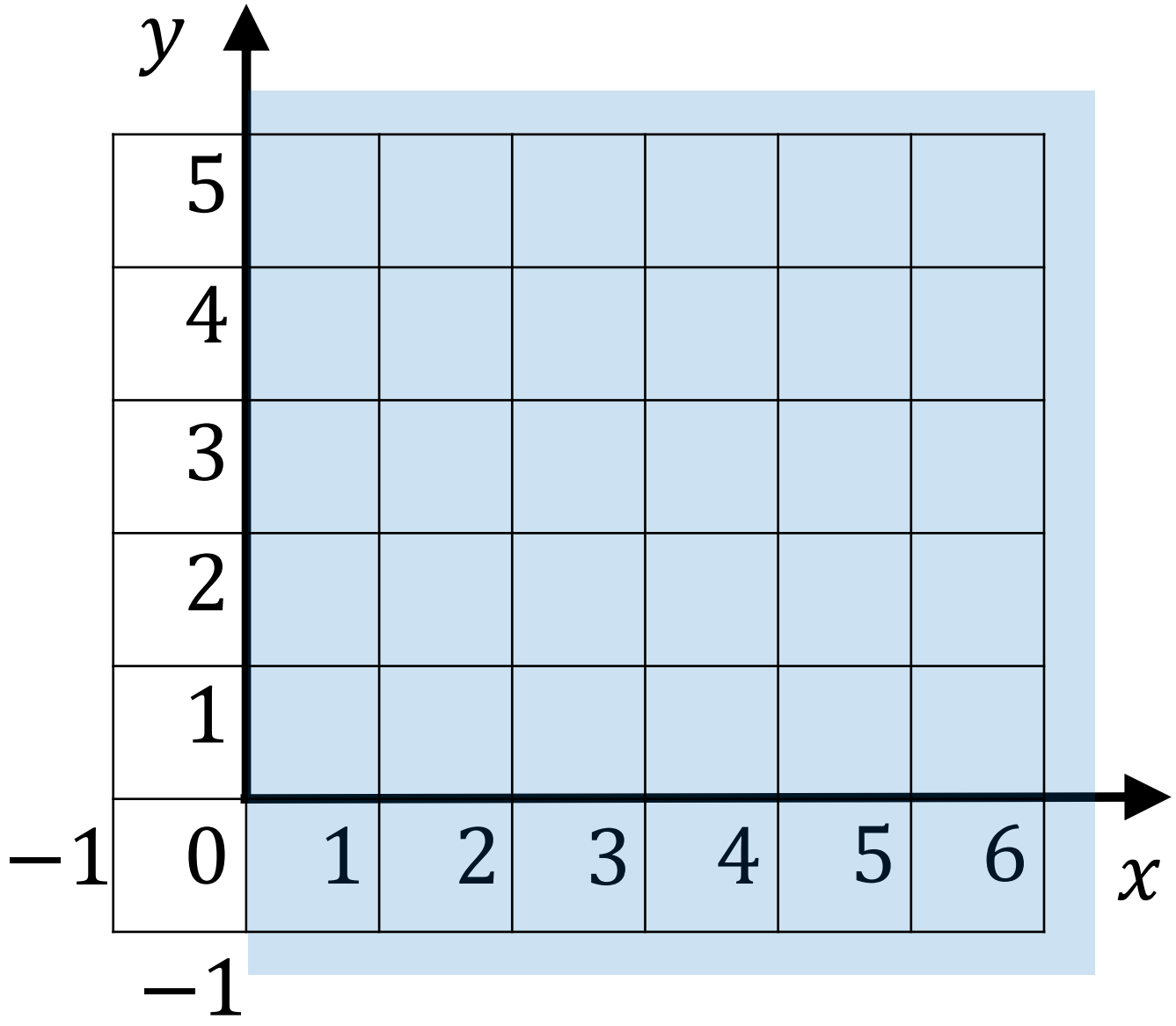
Example 1

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$\mathbf{0 \leq x} \leq 5$

$0 \leq y \leq 3$



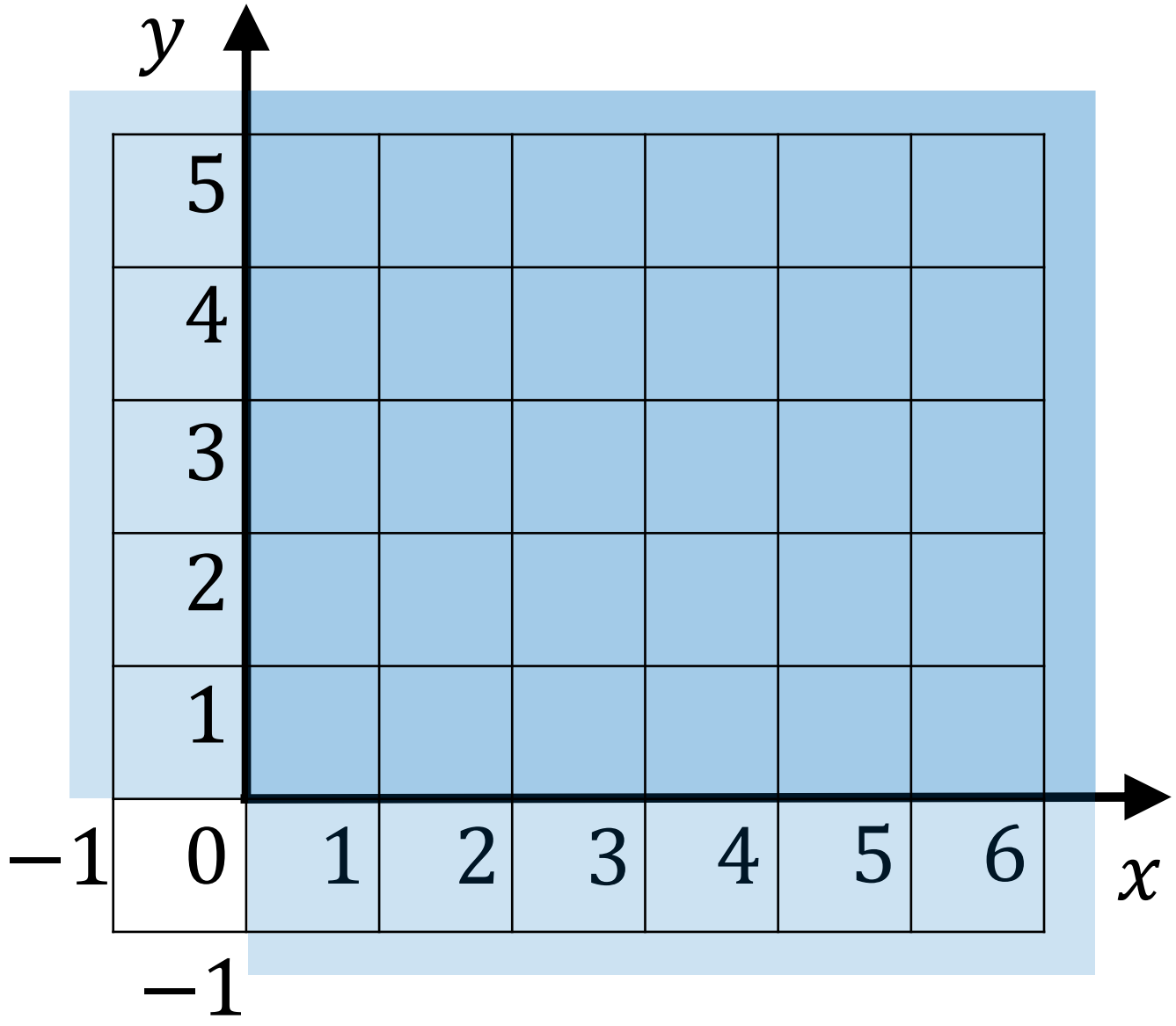
Example 1

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$$0 \leq x \leq 5$$

$$\mathbf{0 \leq y} \leq 3$$



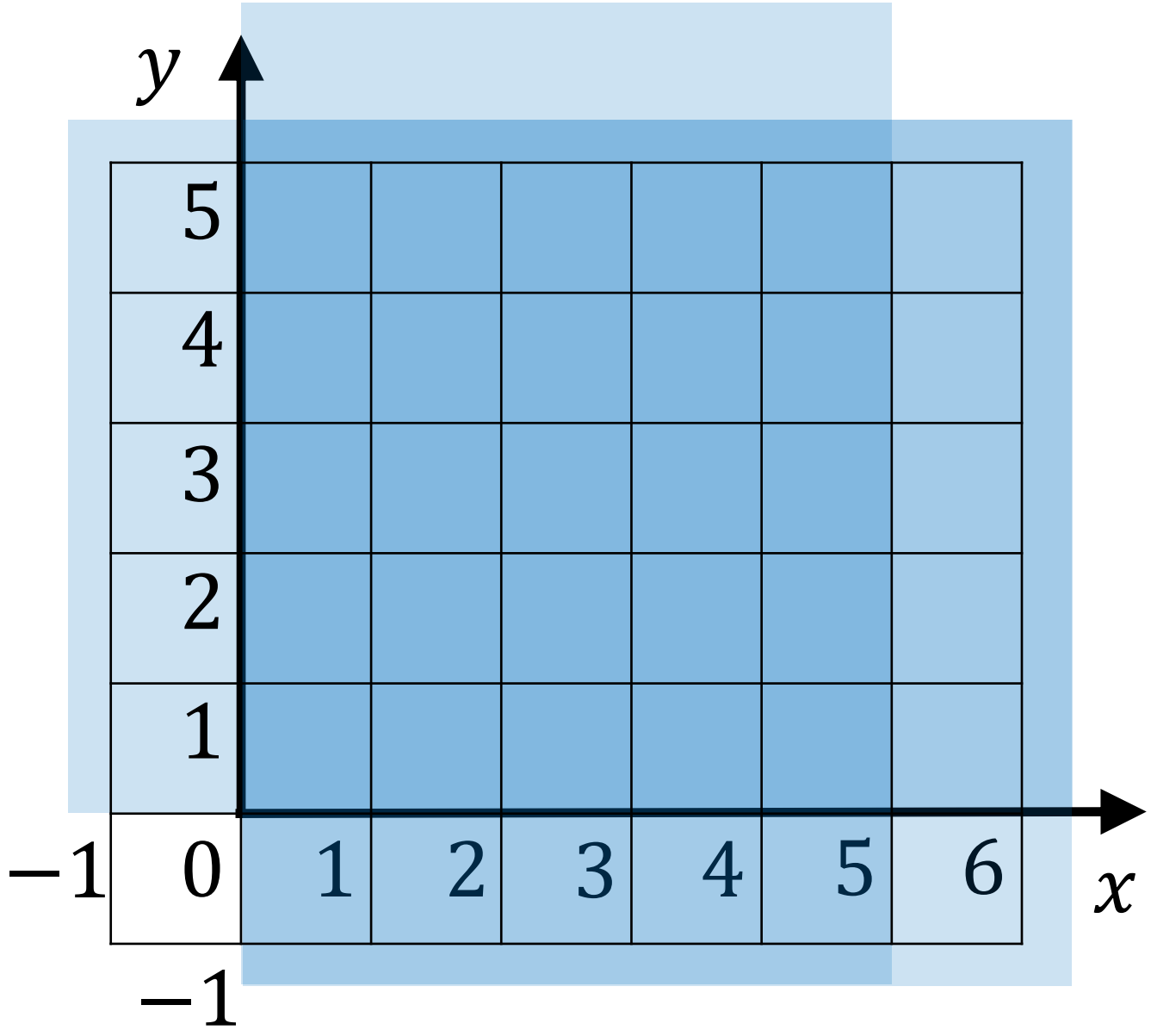
Example 1

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq \boldsymbol{x \leq 5}$

$0 \leq y \leq 3$



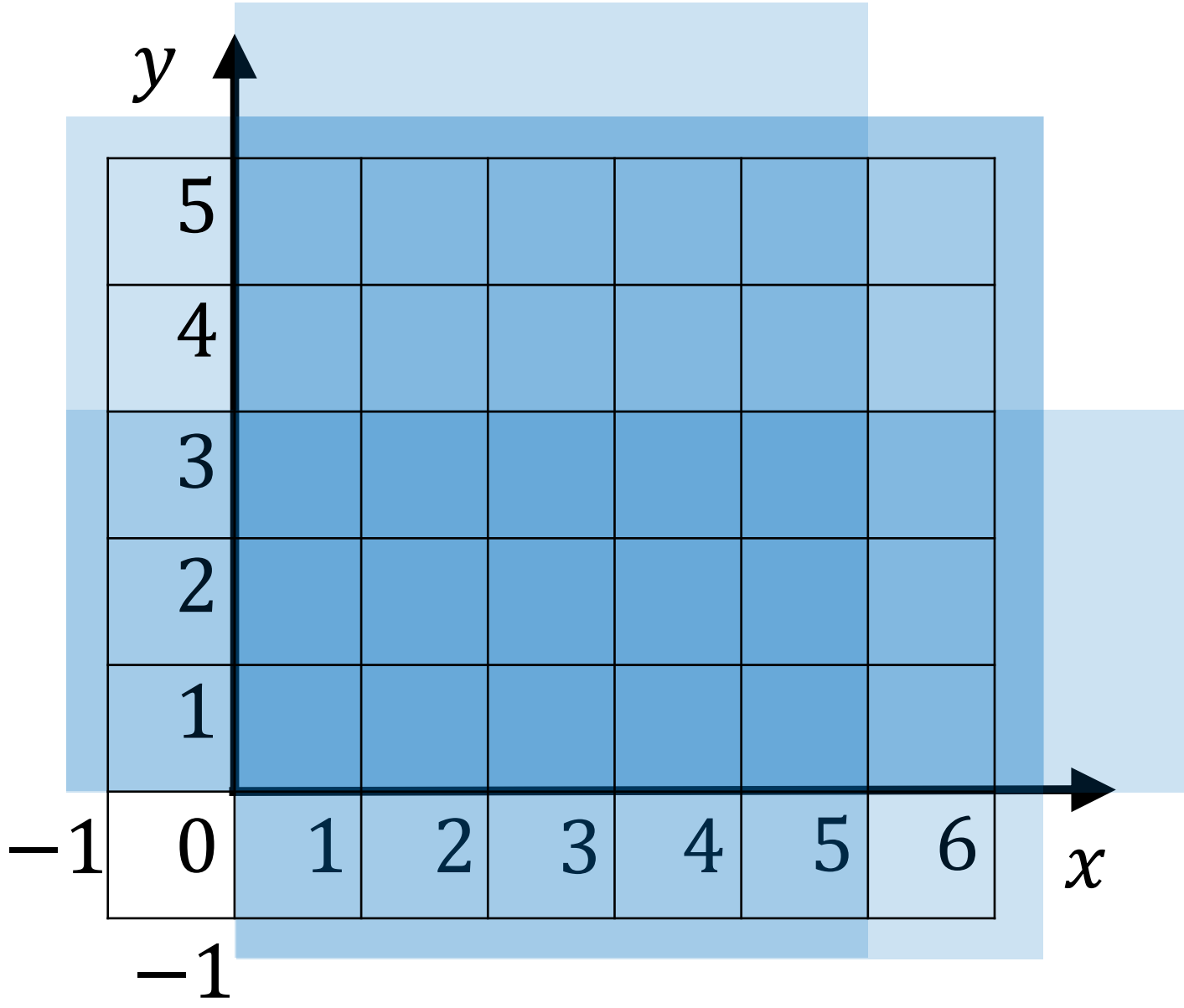
Example 1

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$$0 \leq x \leq 5$$

$$0 \leq \boldsymbol{y \leq 3}$$



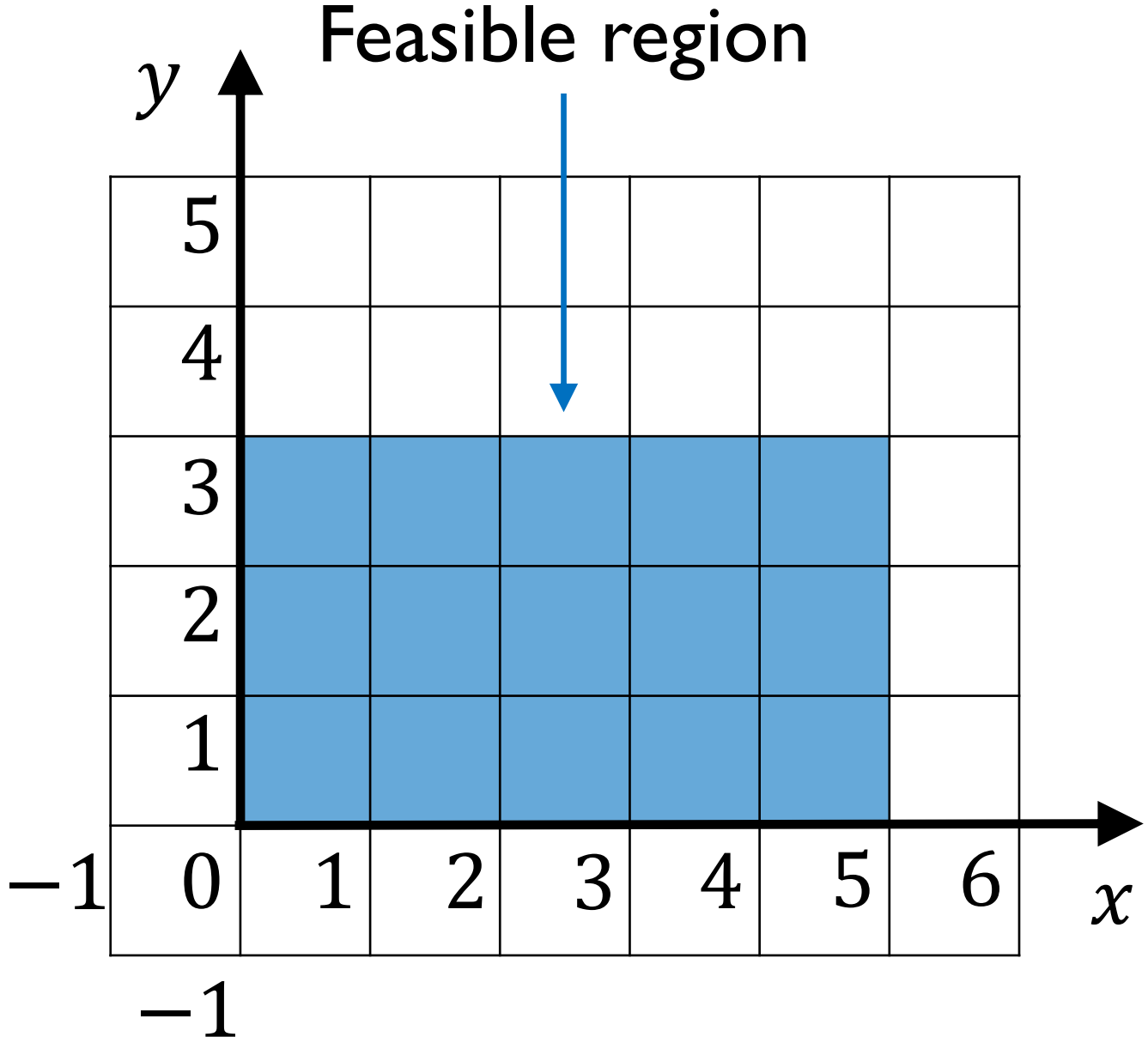
Example 1

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$



Example 1

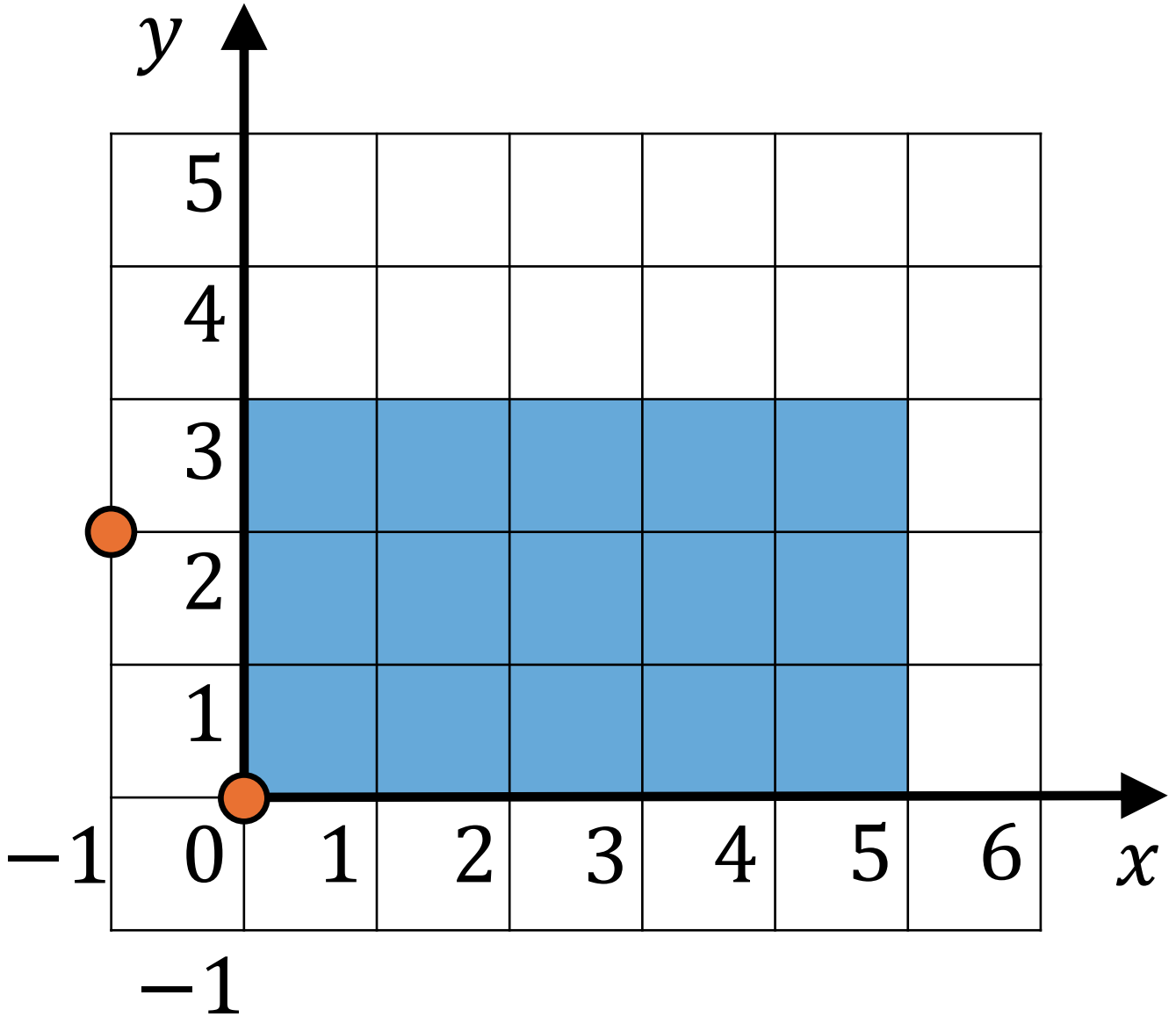
$$\max_{x,y} \quad z = 2x + y$$

$$2x + y = 0$$

Some possible solutions:

1) $x^* = 0, y^* = 0.$

2) $x^* = -1, y^* = 2.$



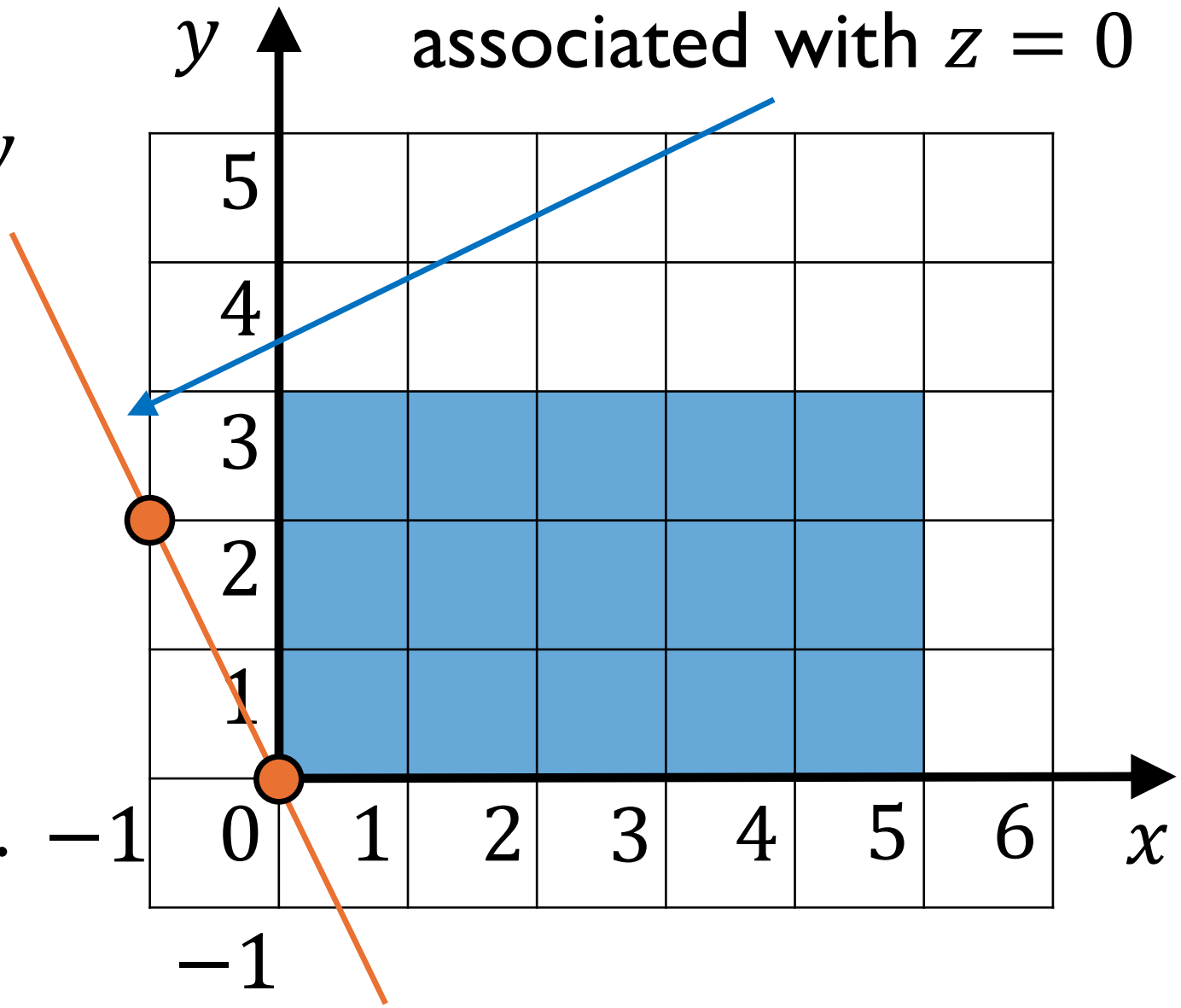
Example 1

$$\max_{x,y} \quad z = 2x + y$$

Some possible solutions of $2x + y = 0$:

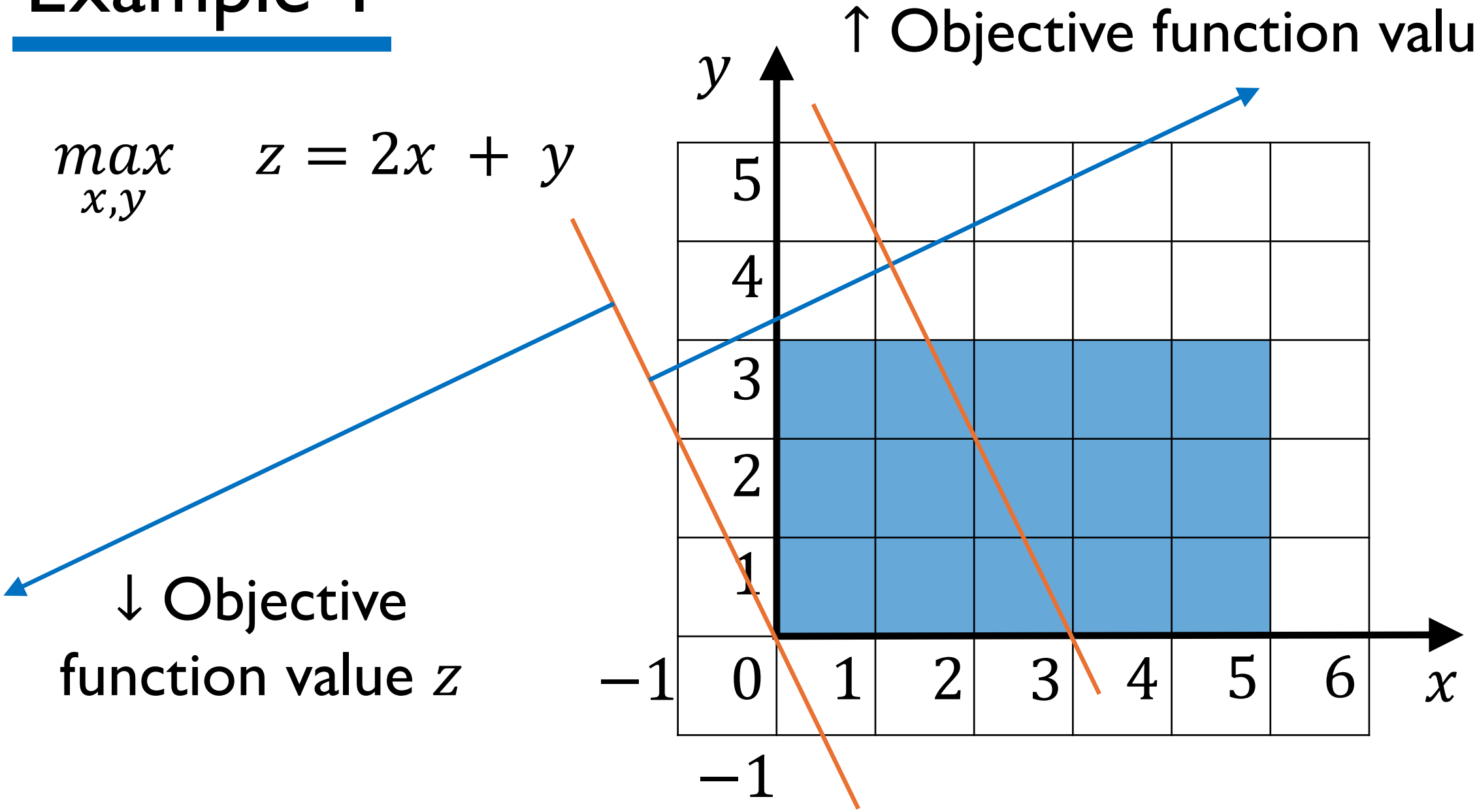
1) $x^* = 0, y^* = 0.$

2) $x^* = -1, y^* = 2.$



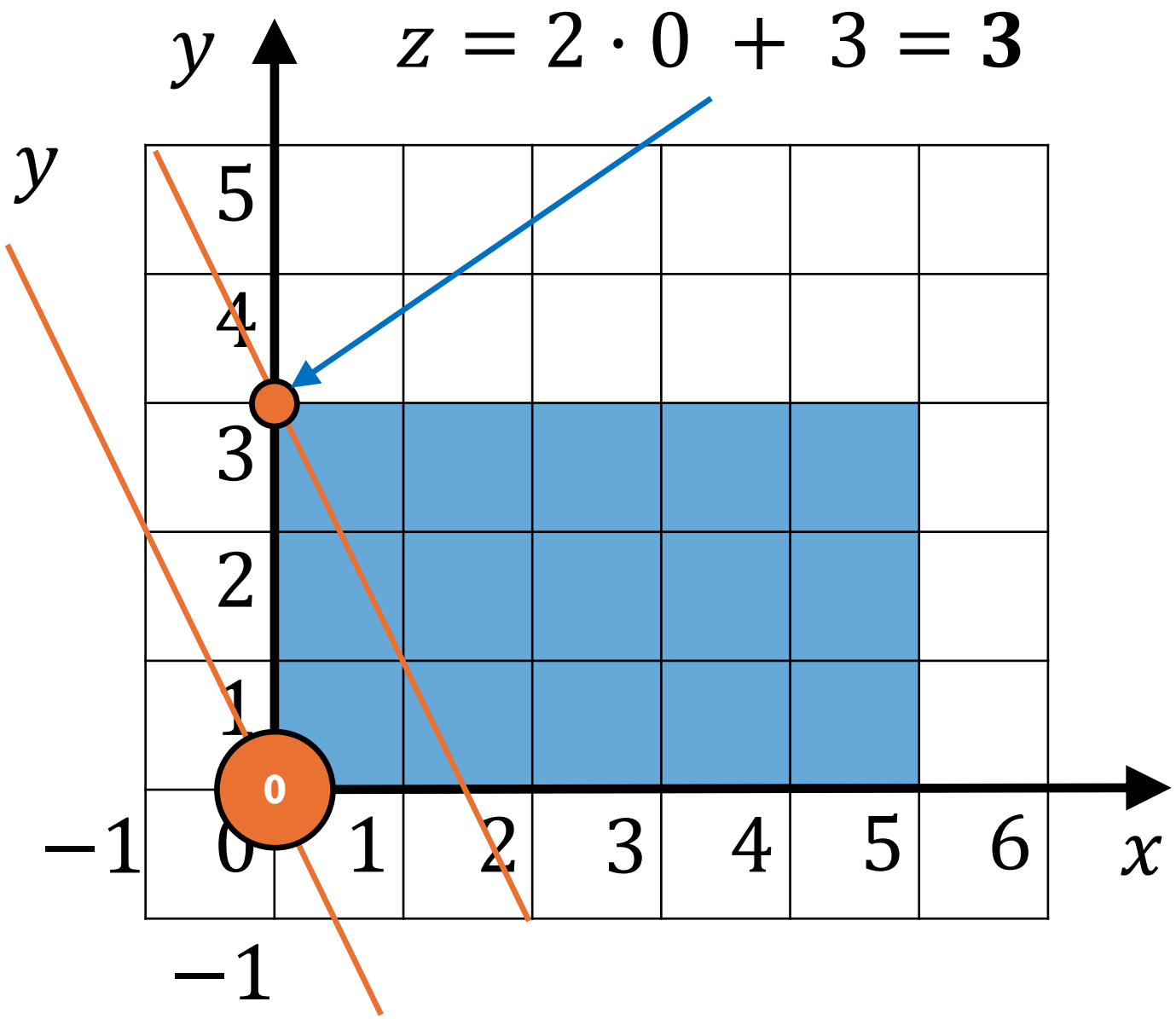
Example 1

$$\max_{x,y} \quad z = 2x + y$$



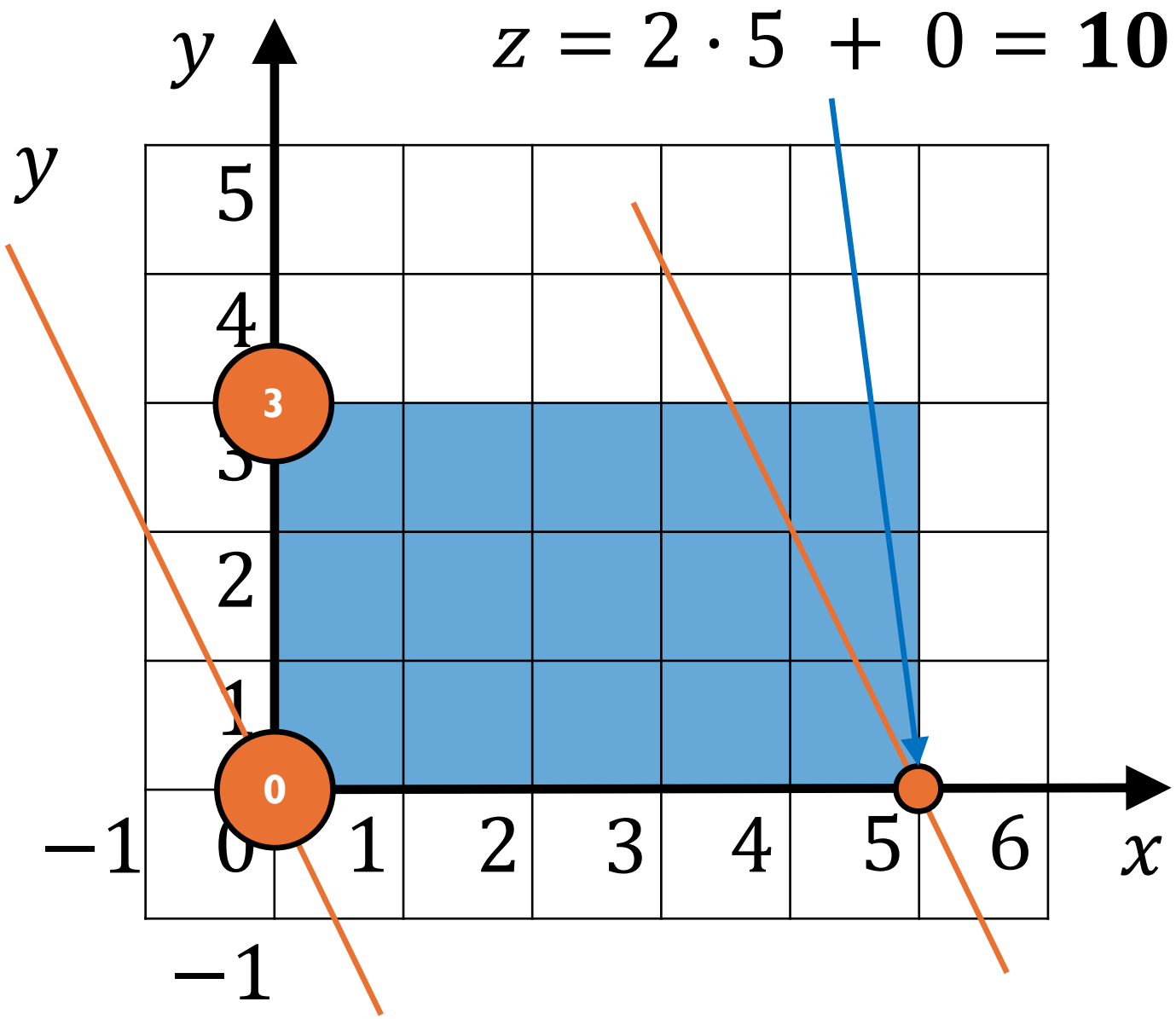
Example 1

$$\max_{x,y} \quad z = 2x + y$$



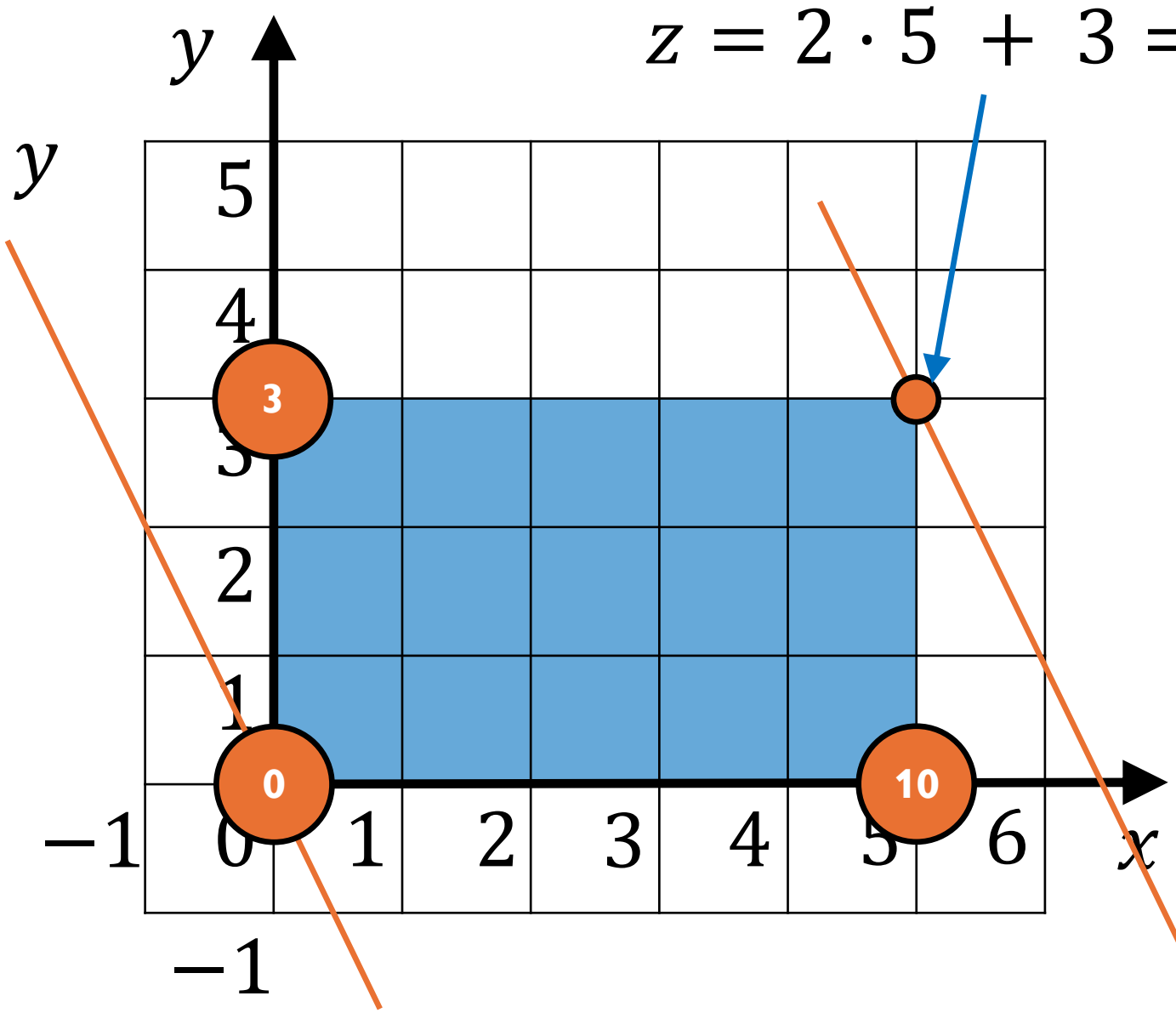
Example 1

$$\max_{x,y} \quad z = 2x + y$$



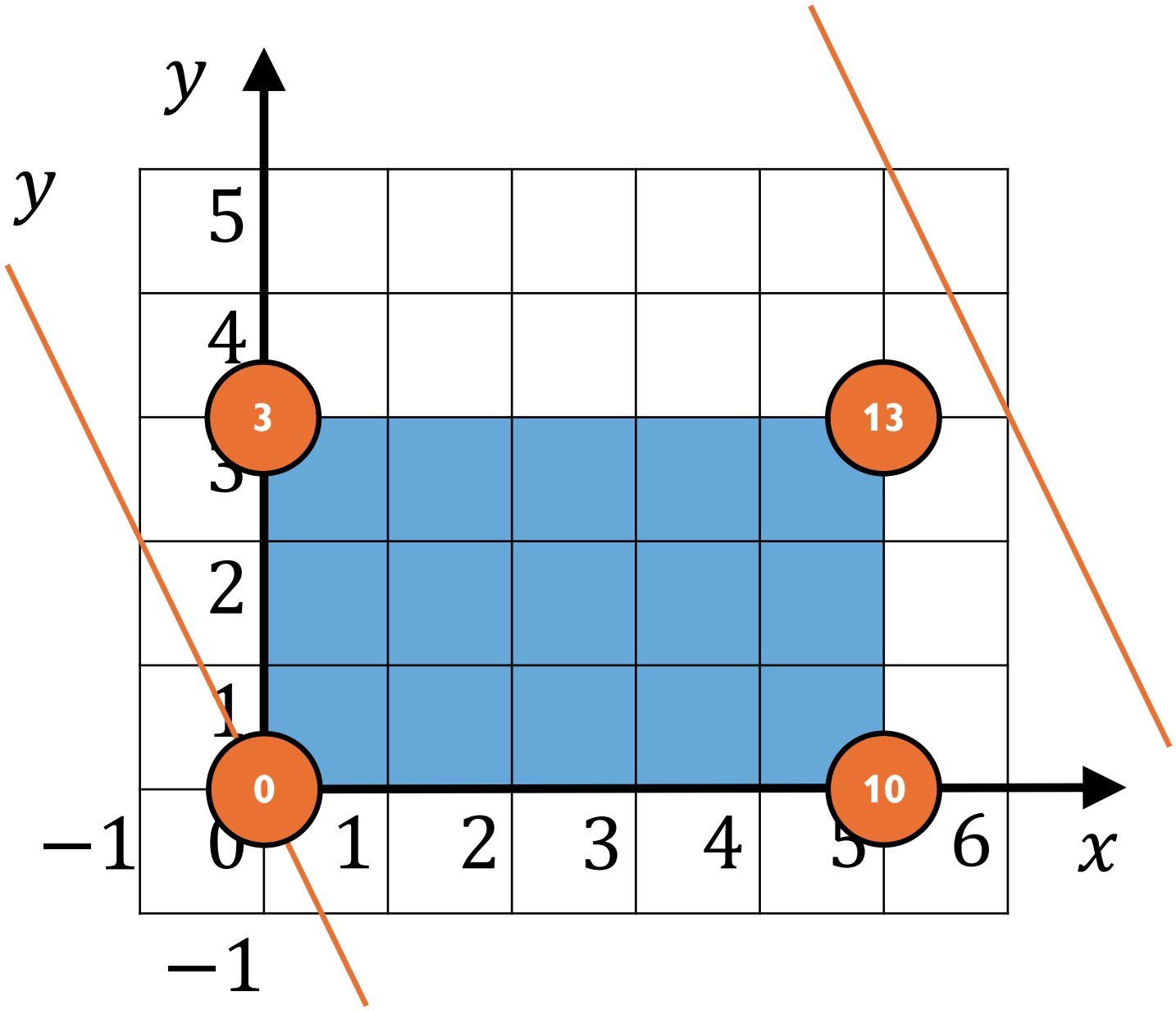
Example 1

$$\max_{x,y} \quad z = 2x + y$$



Example 1

$$\max_{x,y} \quad z = 2x + y$$

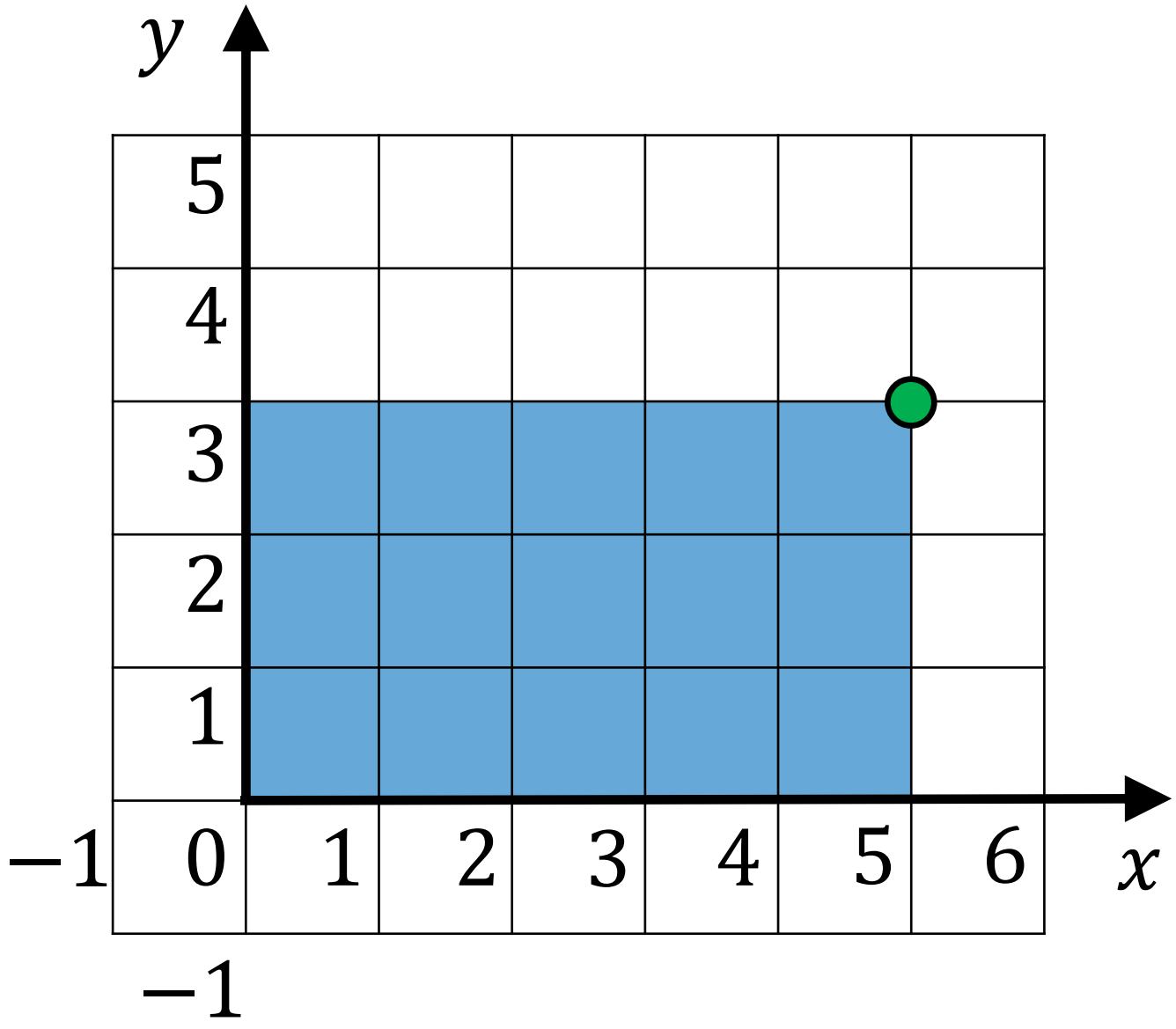


Example 1

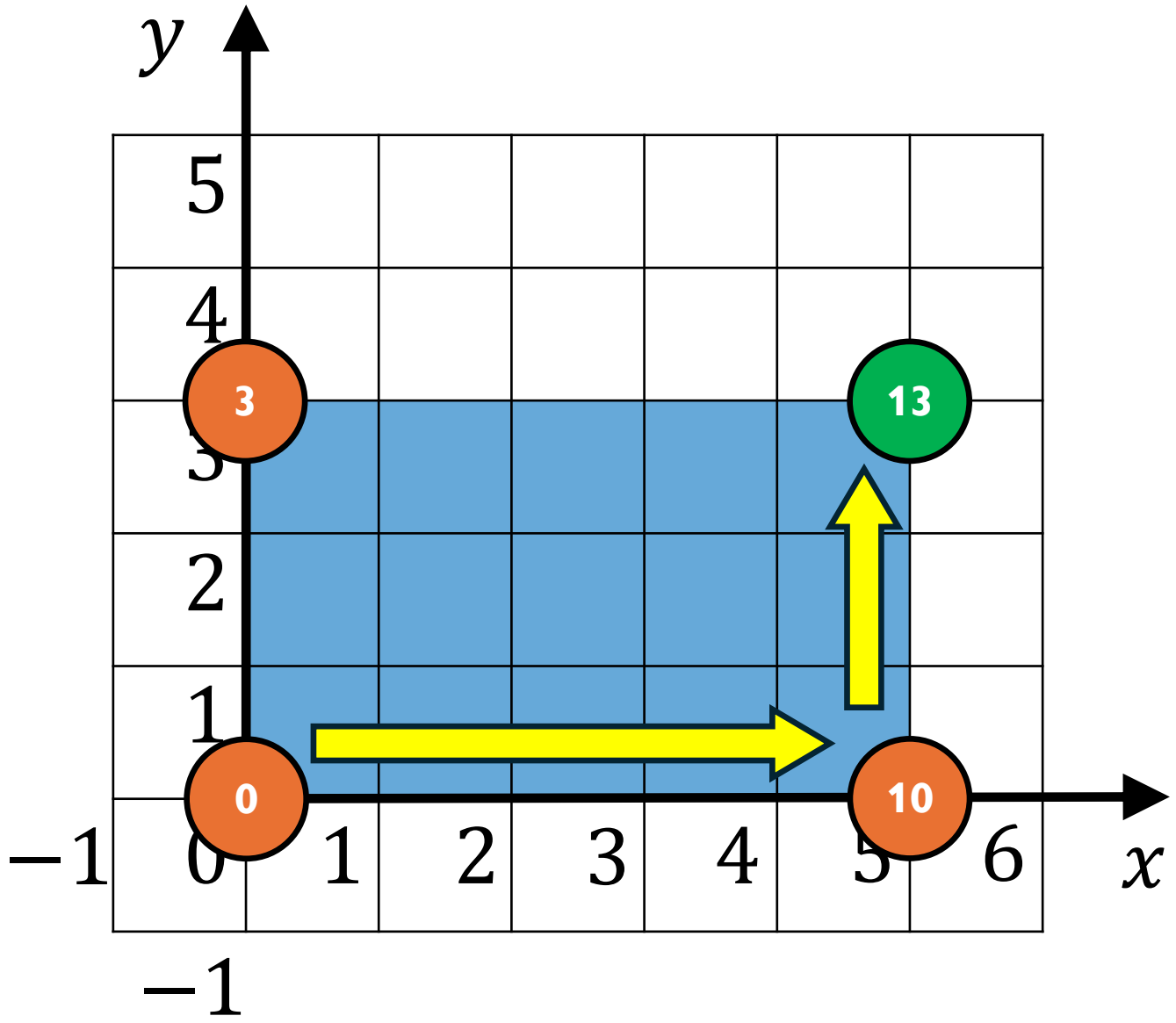
$$\max_{x,y} \quad z = 2x + y$$

Optimal solution:

$x^* = 5$
$y^* = 3$
$z^* = 13$



Simplex method



Simplex method

Channel of Prof. Yong Wang (Binghamton University):

Basic Feasible Solution:

https://www.youtube.com/watch?v=0P5OAcXdUec

Graphical Explanation:

https://www.youtube.com/watch?v=Mz__0uBb2-U

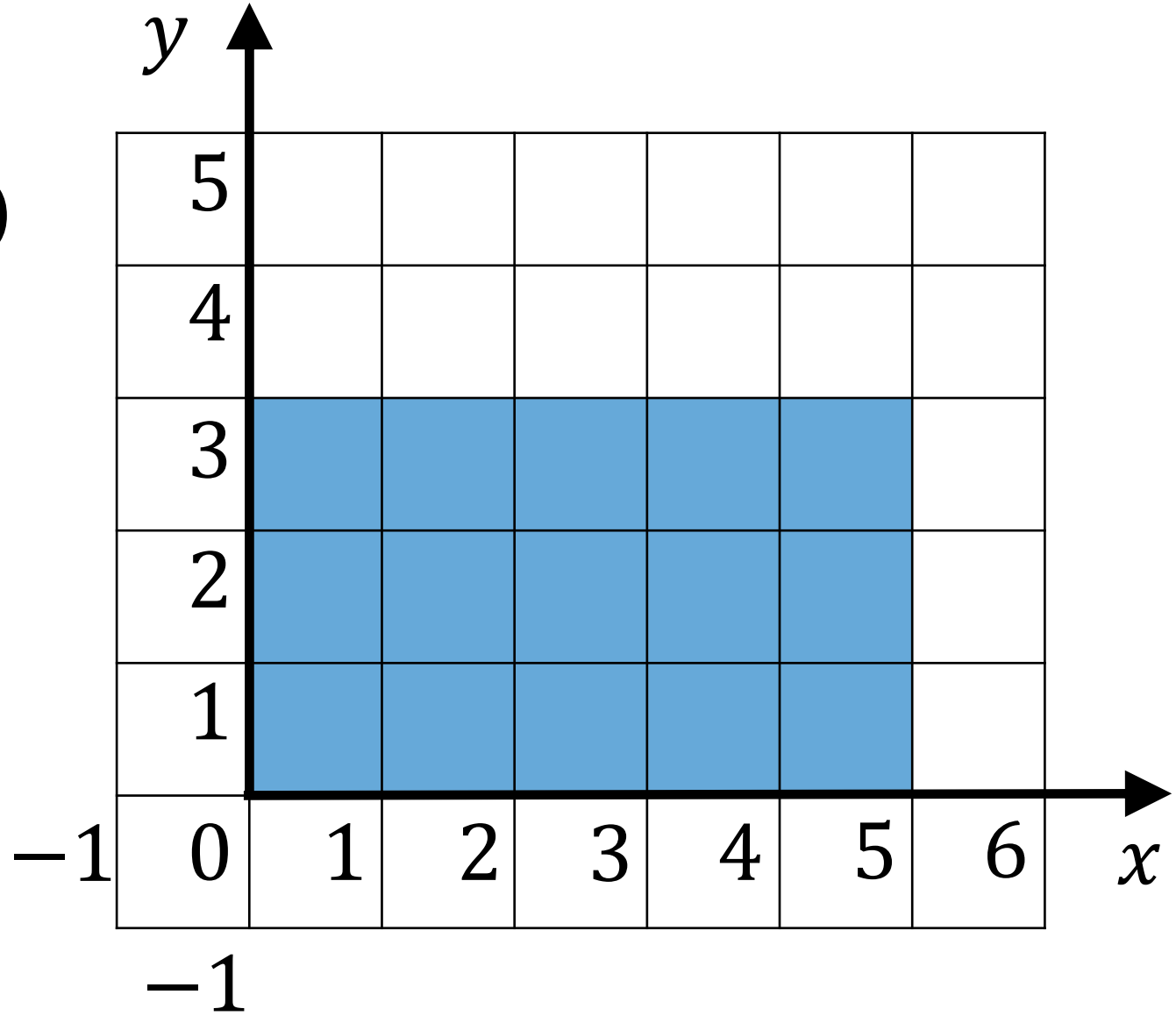
Example 2

$$\min_{x,y} \quad z = -(2x + y)$$

s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$



Example 2

$$\min_{x,y} \quad z = -(2x + y)$$

s.t.:

$0 \leq x \leq 5$

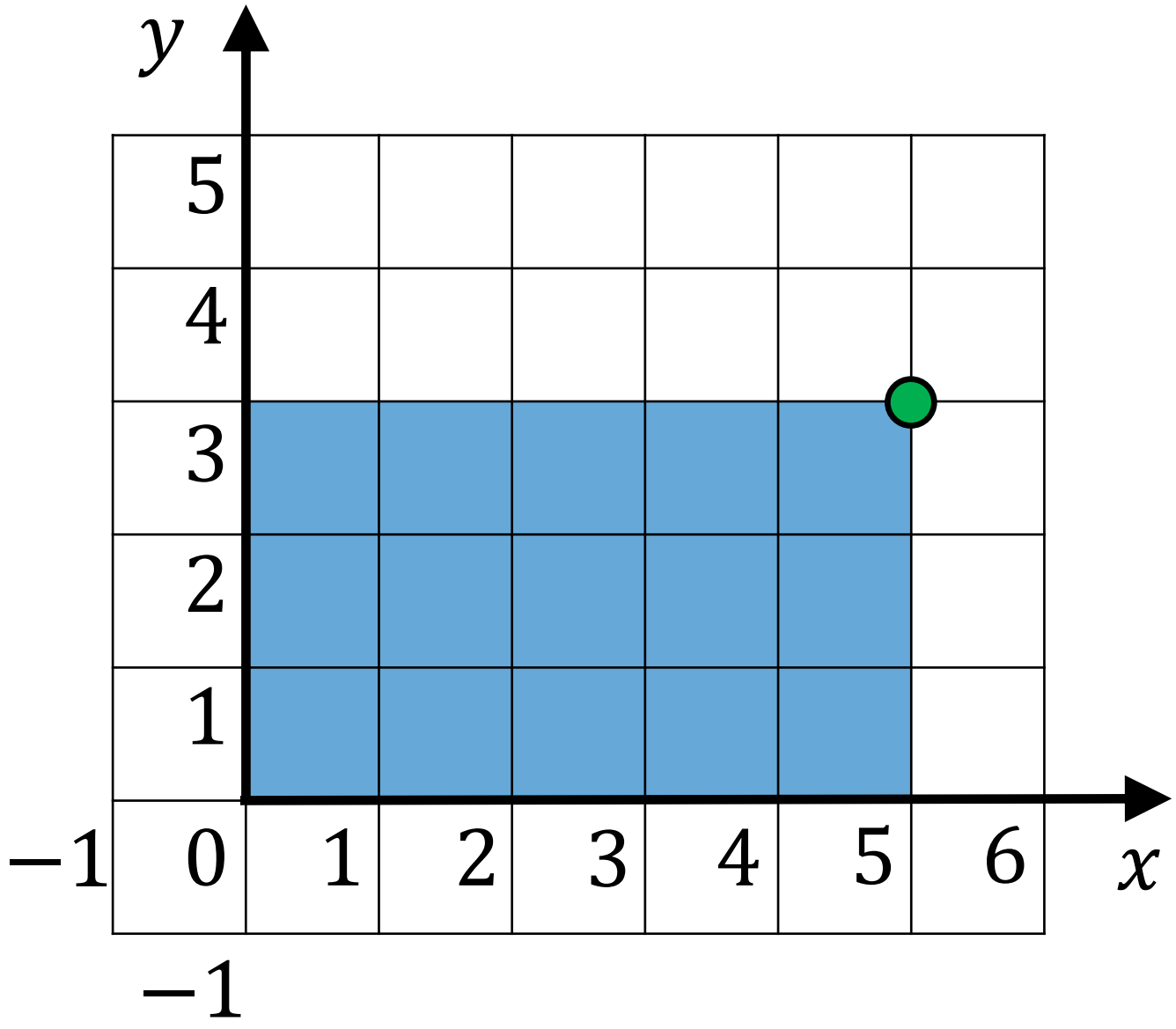
$0 \leq y \leq 3$

Optimal solution:

$x^* = 5$

$y^* = 3$

$z^* = -13$



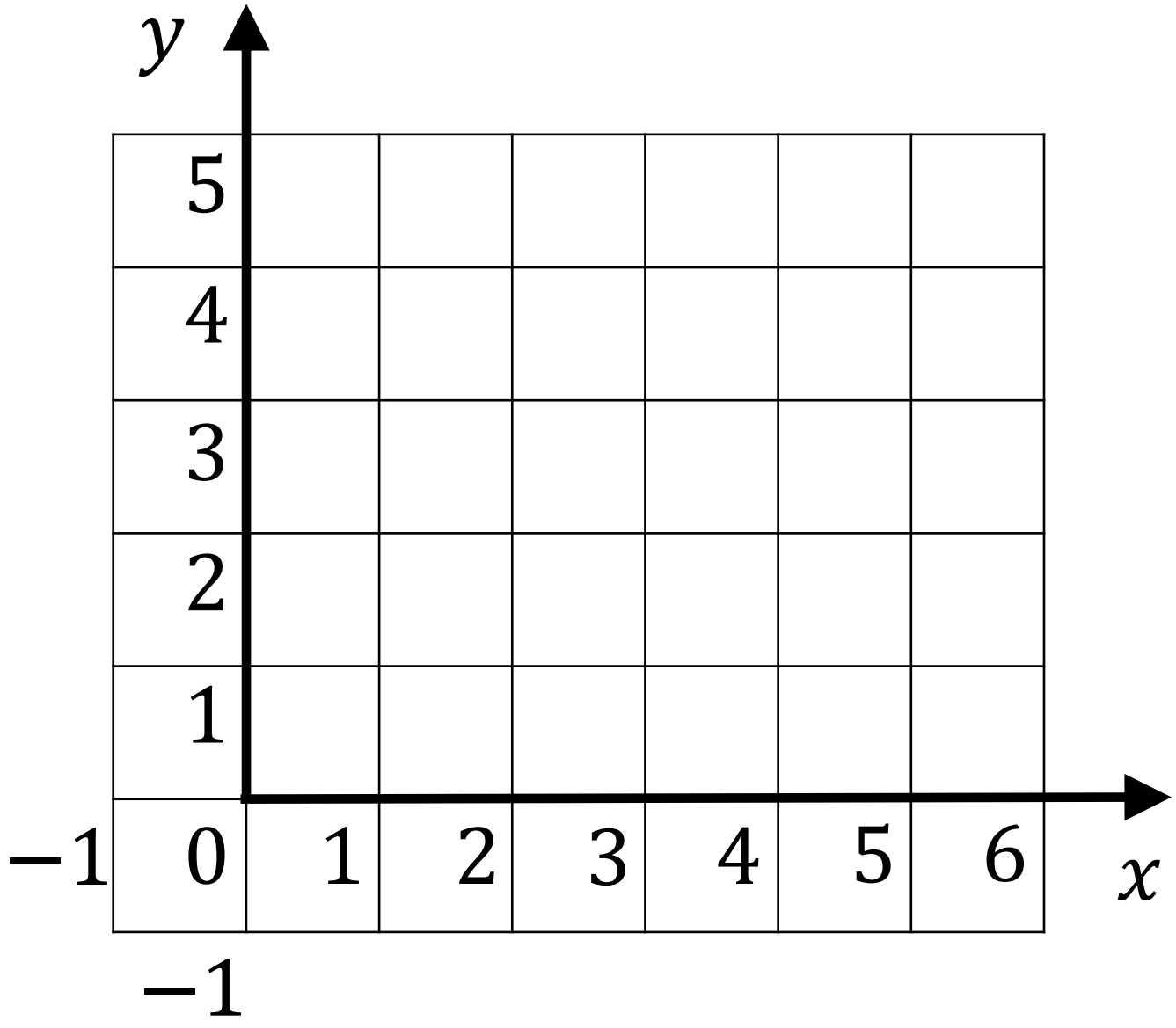
Example 3

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq x$

$0 \leq y$



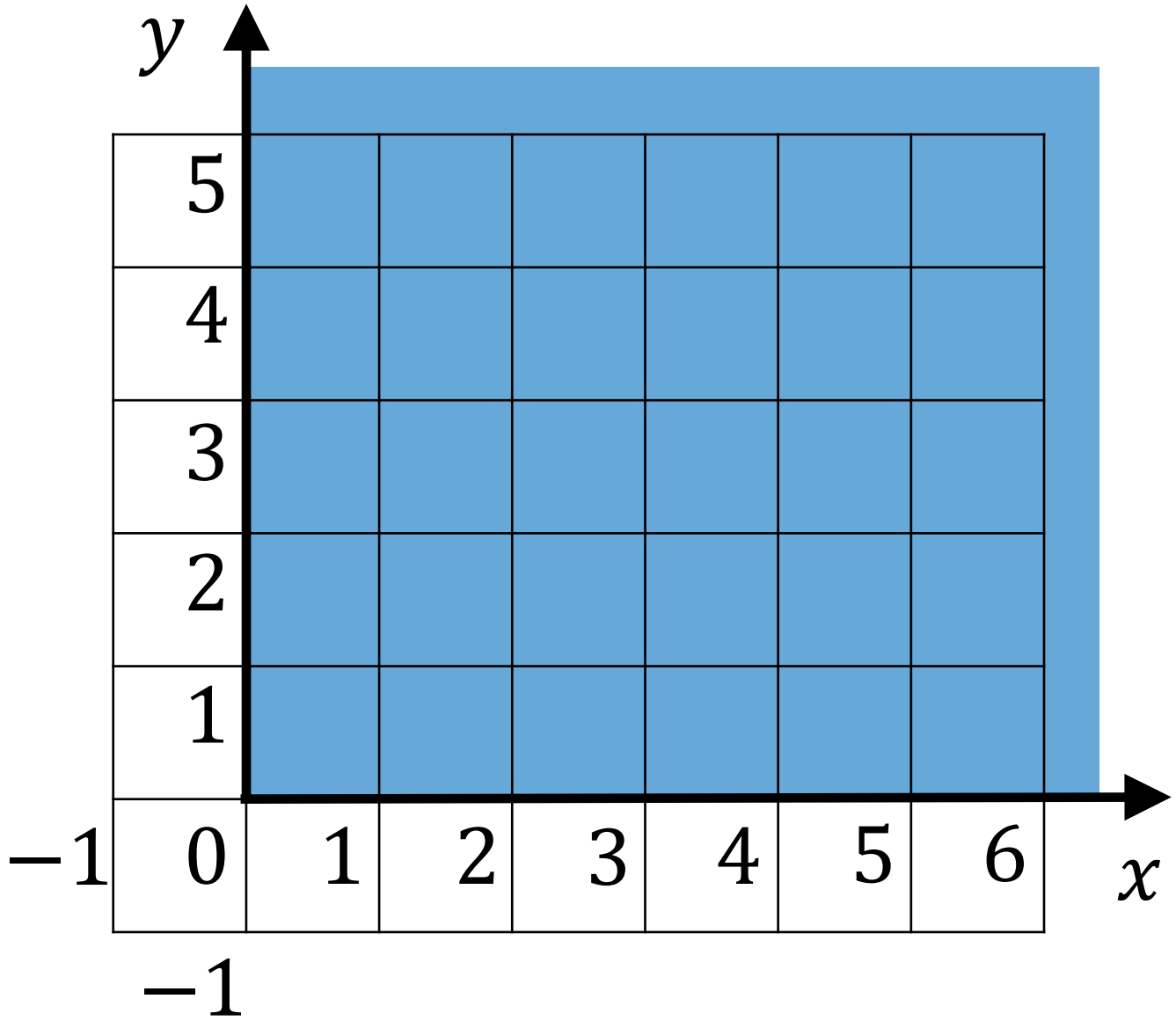
Example 3

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq x$

$0 \leq y$



Example 3

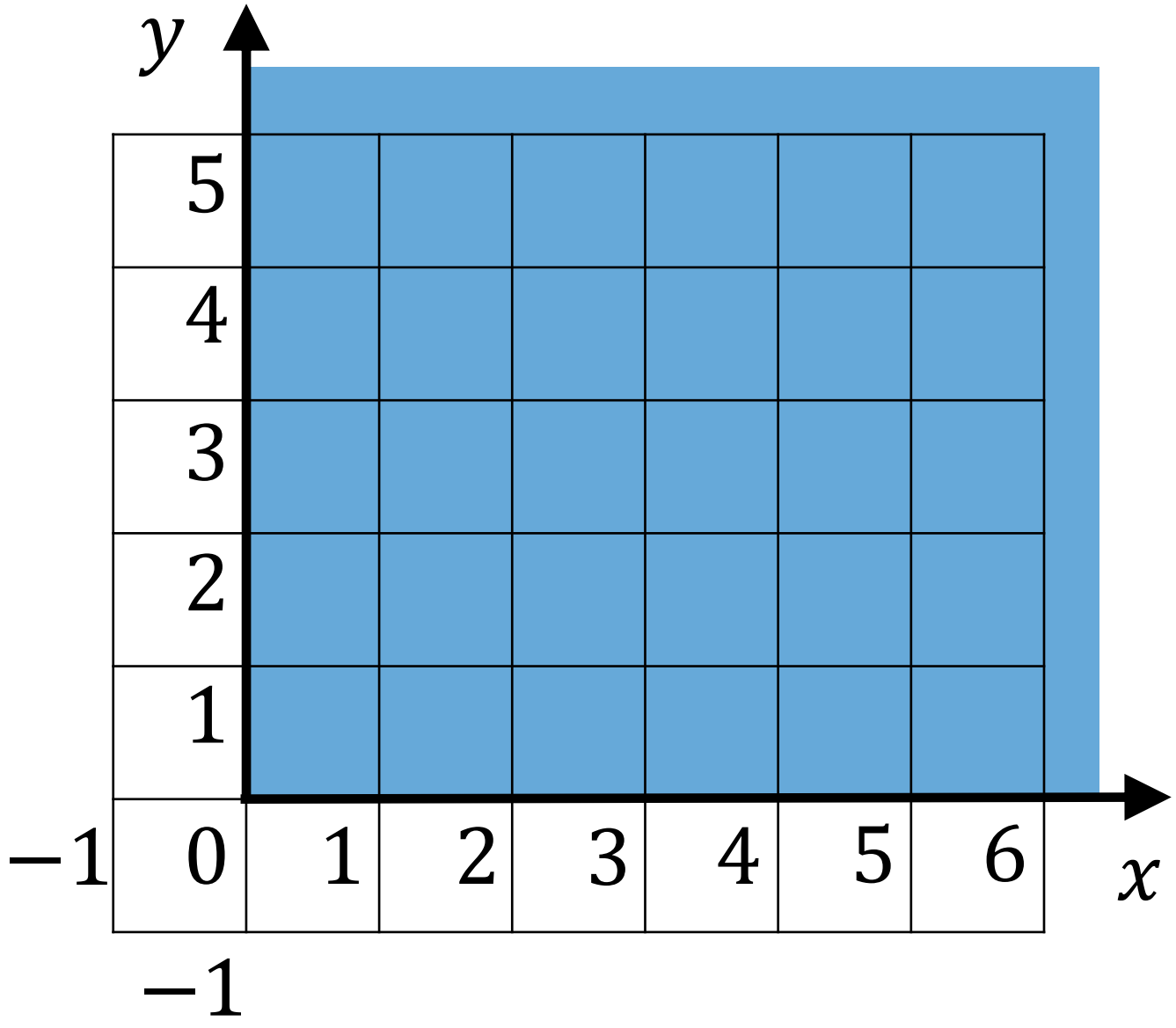
$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq x$

$0 \leq y$

Unbounded problem



Example 4

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$

$4 \leq y$

Example 4

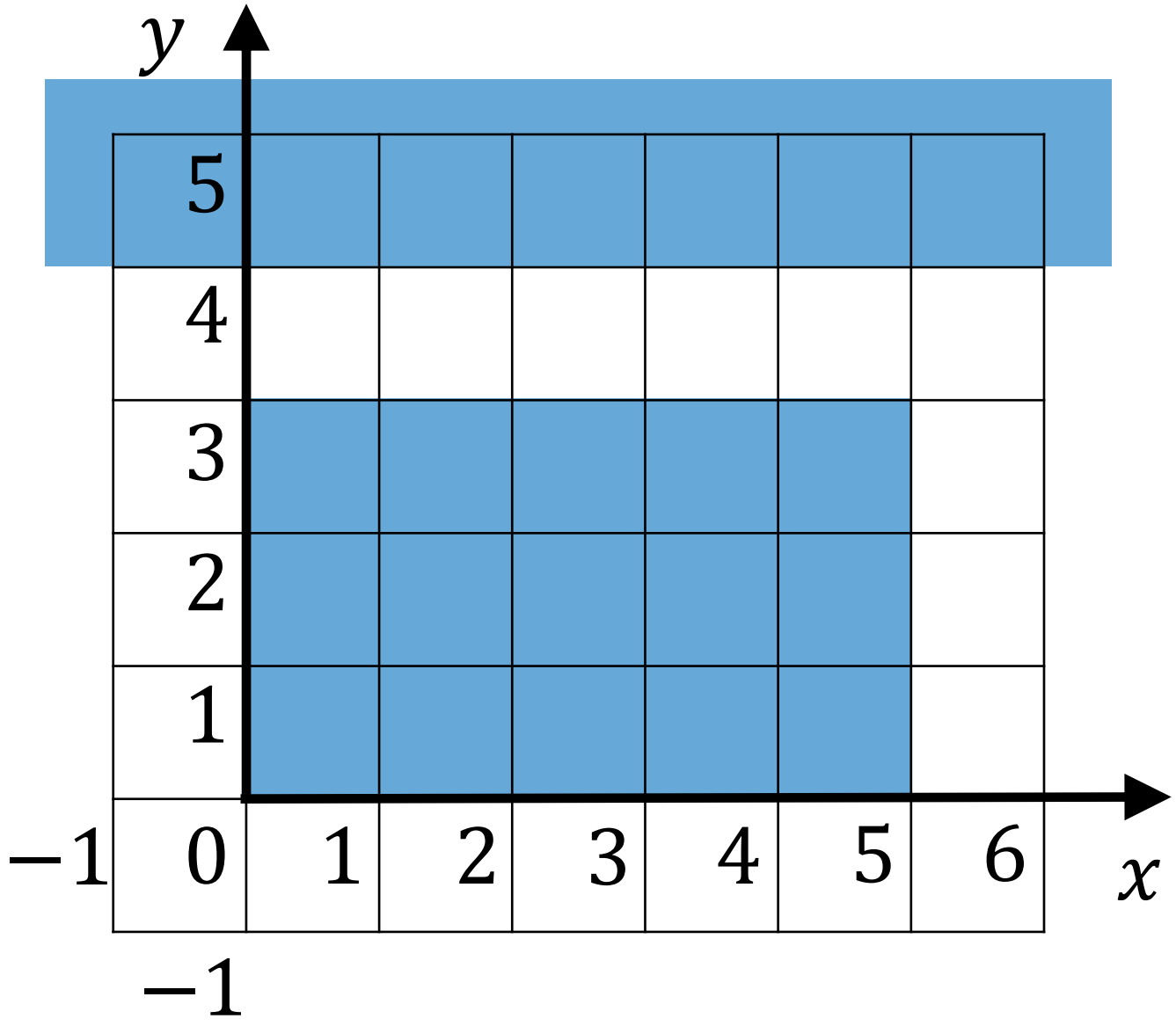
$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$

$4 \leq y$



Example 4

$$\max_{x,y} \quad z = 2x + y$$

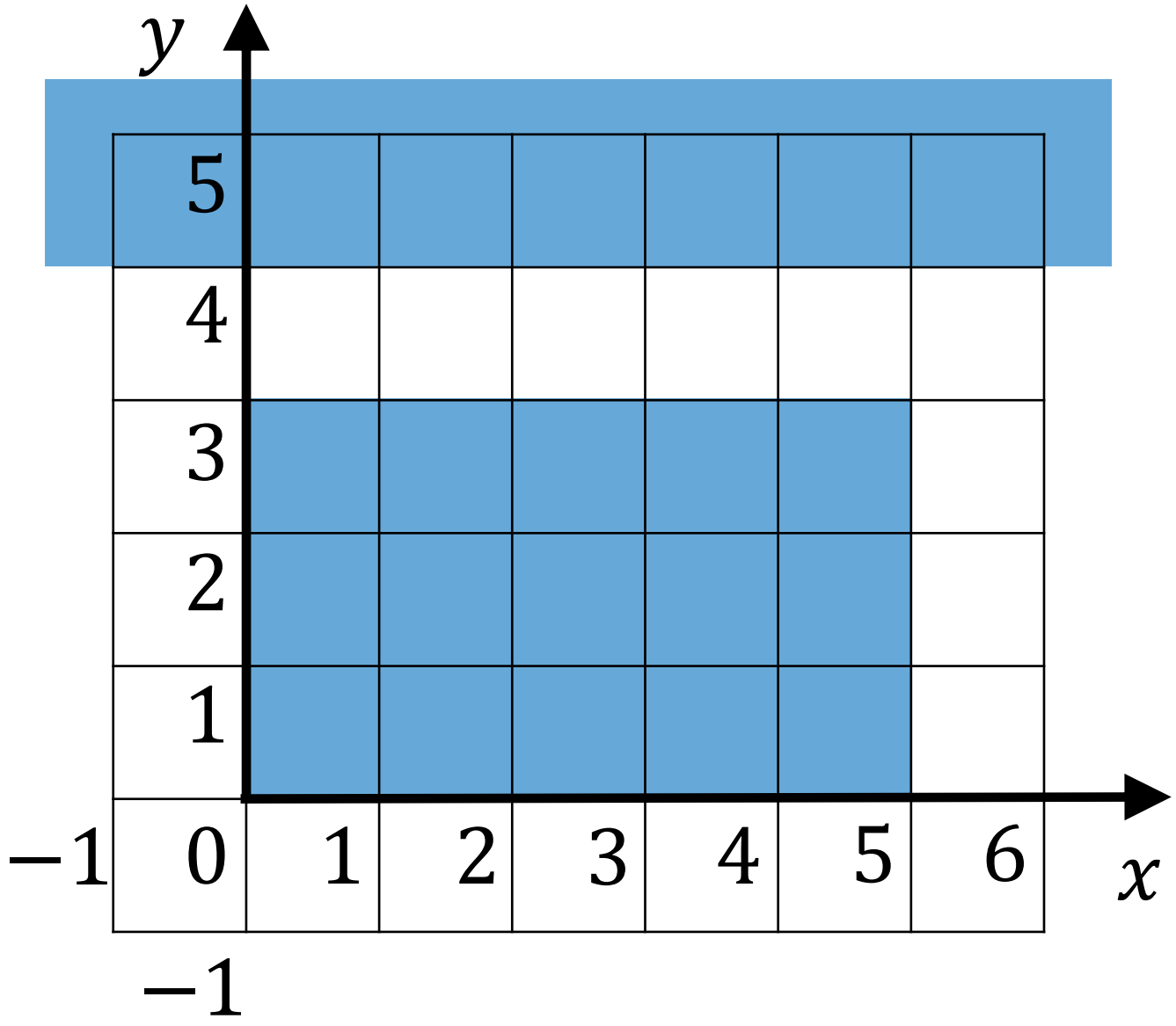
s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$

$4 \leq y$

Infeasible problem



Example 5

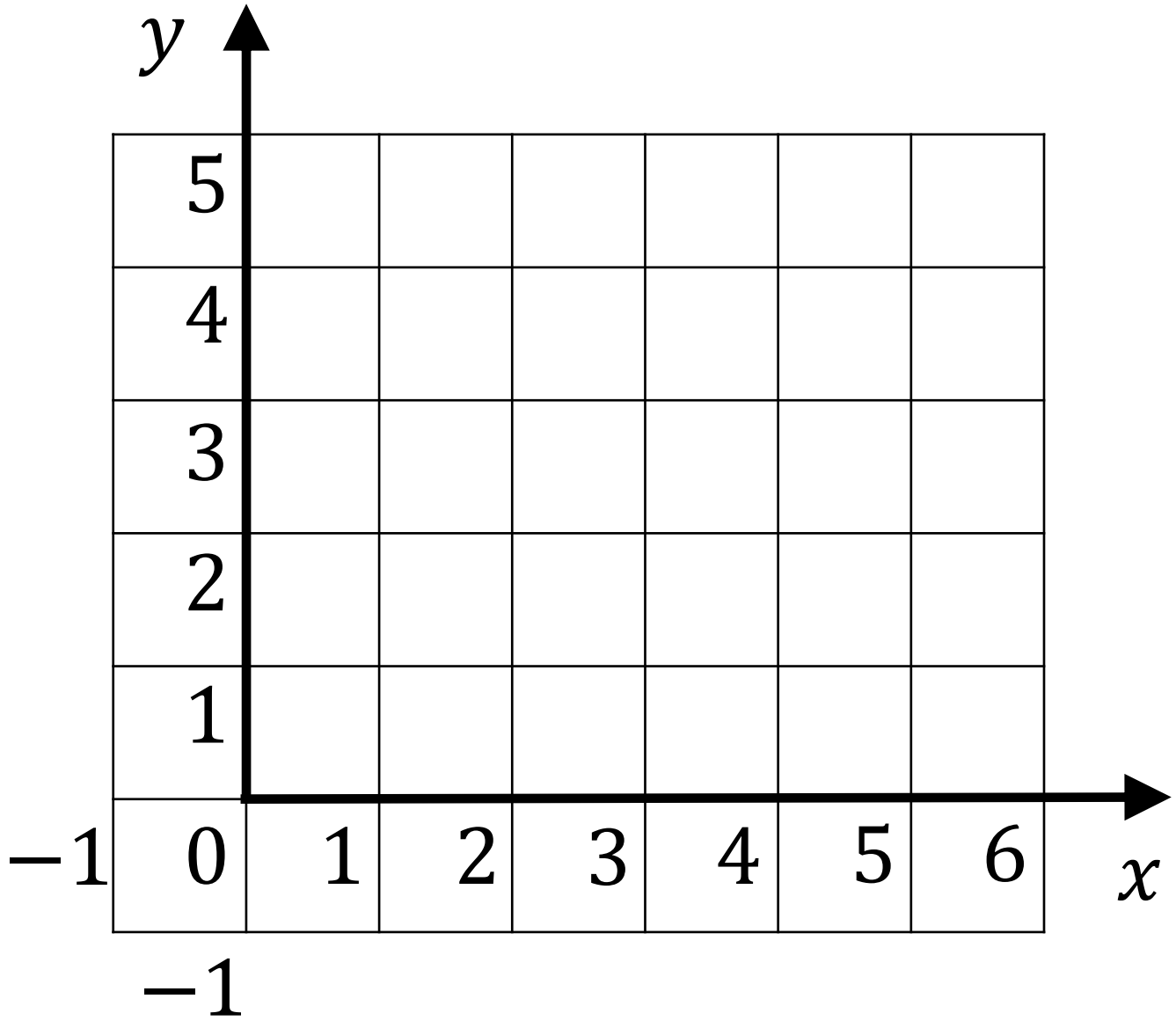
$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$

$x + y \leq 4$



Example 5

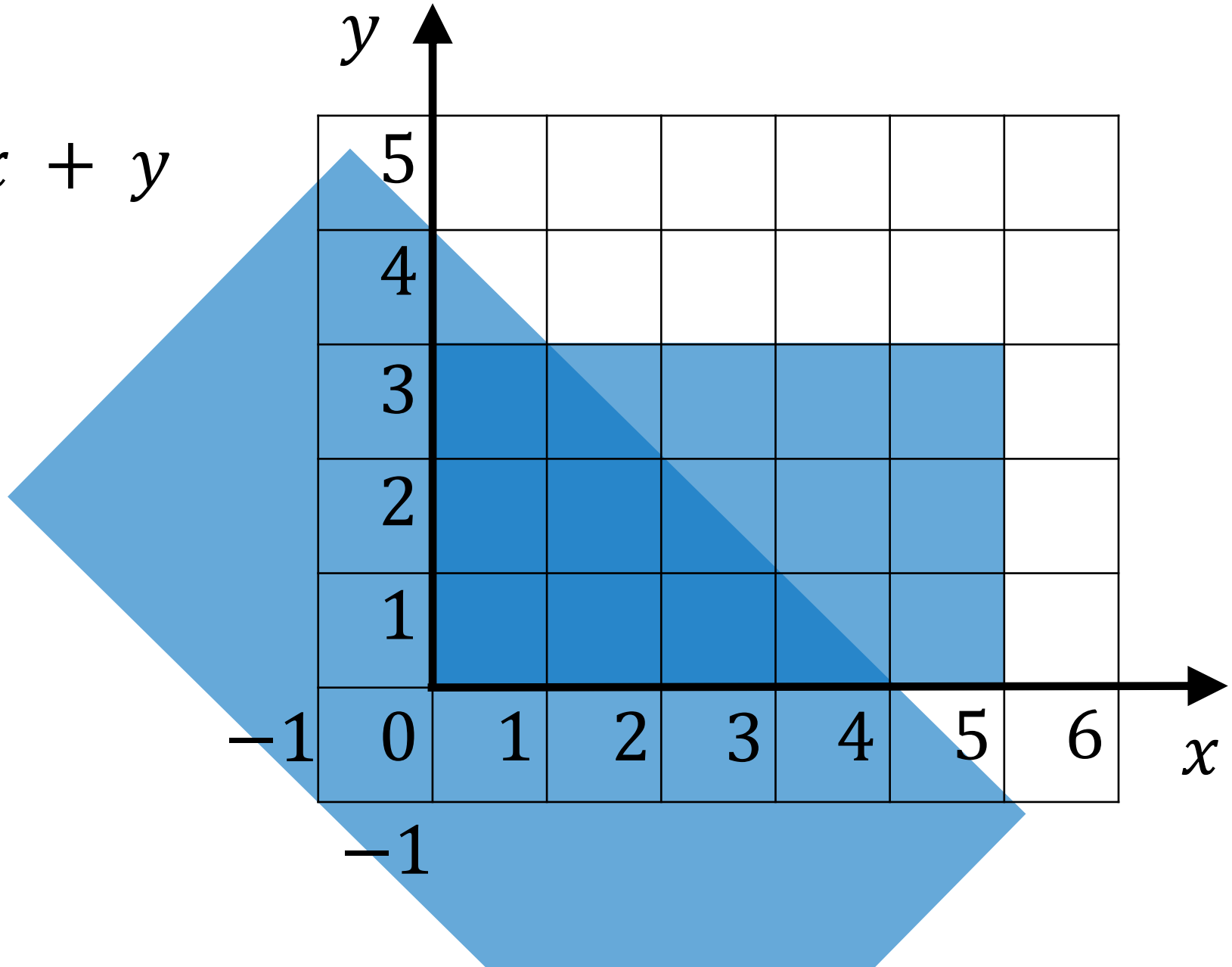
$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$

$x + y \leq 4$



Example 5

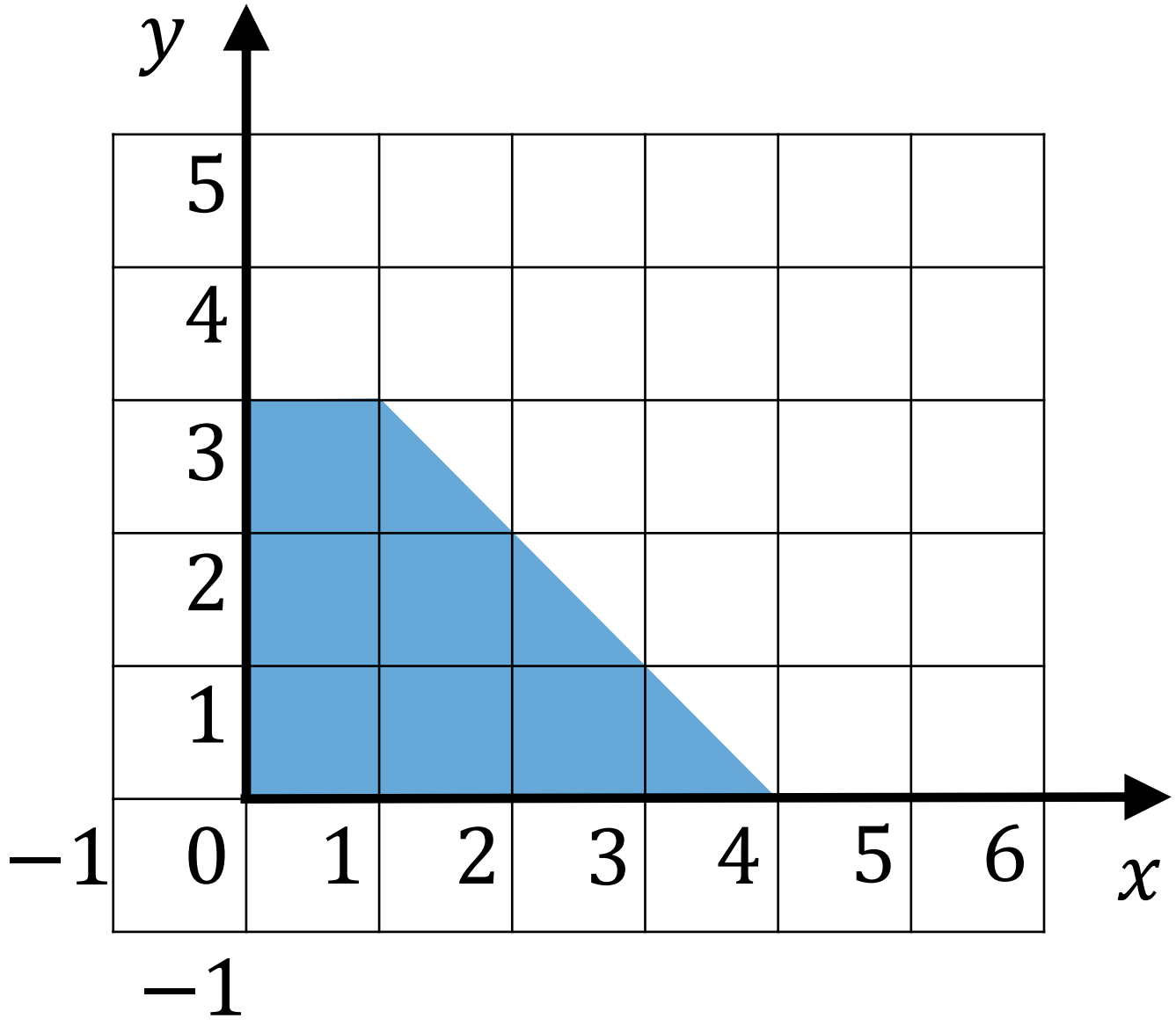
$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$

$x + y \leq 4$



Example 5

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$

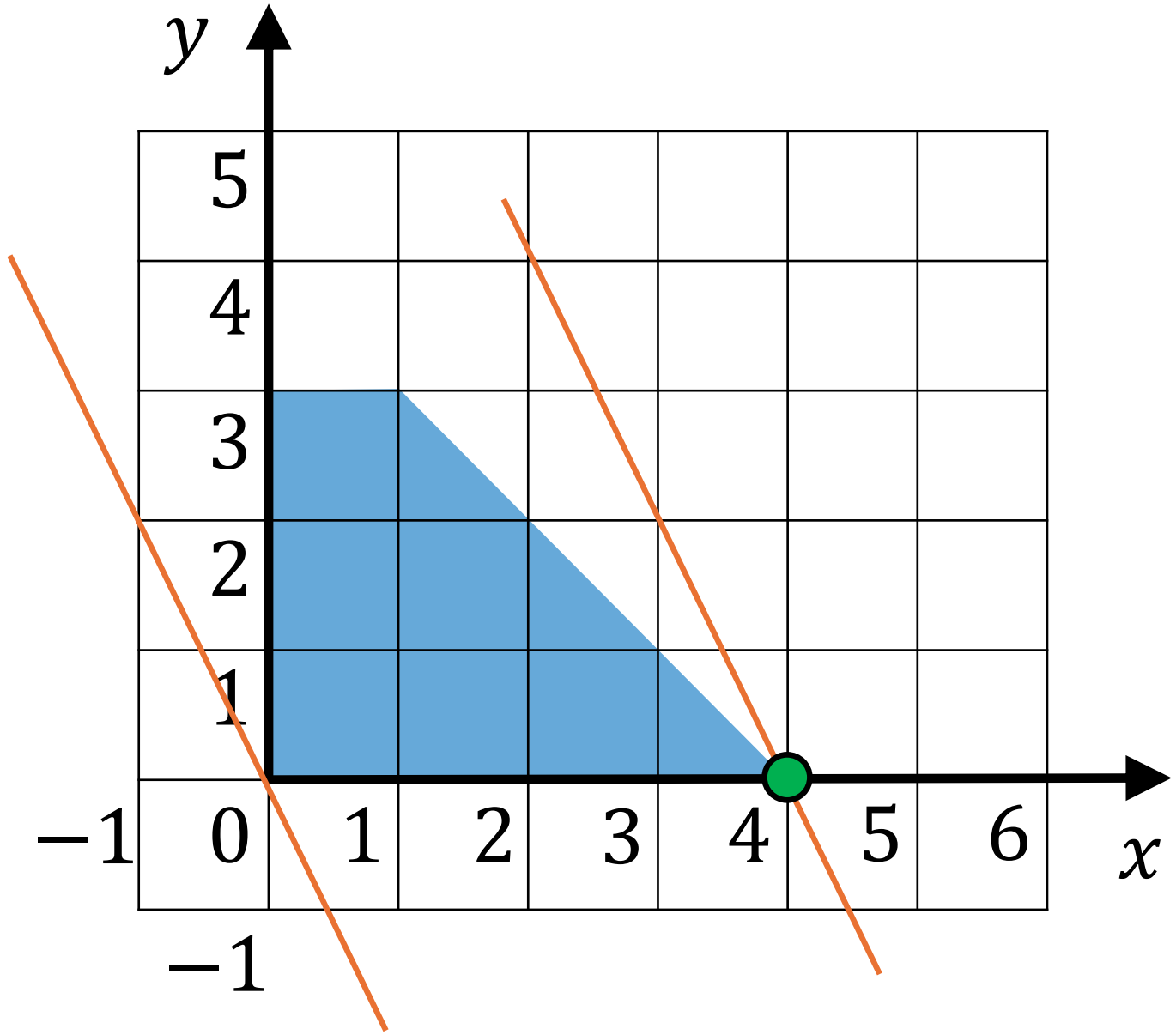
$x + y \leq 4$

Optimal solution:

$\boldsymbol{x^* = 4}$

$\boldsymbol{y^* = 0}$

$\boldsymbol{z^* = 8}$



Example 5

$$\max_{x,y} \quad z = 2x + y$$

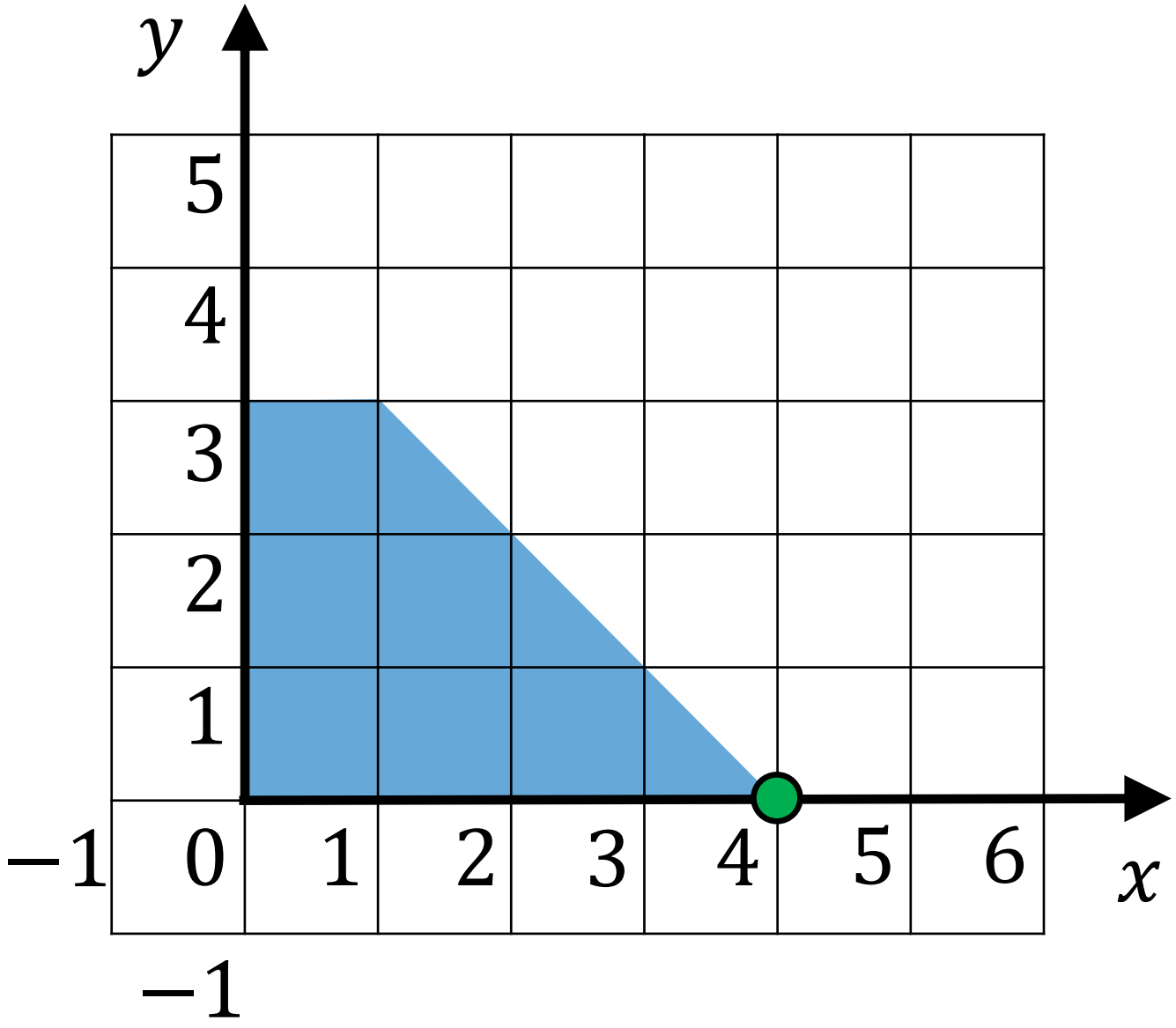
s.t.:

$0 \leq \boldsymbol{x} \leq \mathbf{5}$

$0 \leq y \leq 3$

$x + y \leq 4$

Inactive constraints
(they do not influence the optimal solution)



Example 5

Example 1: $z = 13$.

Example 5: $z = 8$.

In a maximization/minimization problem, adding constraints leads to an objective function value lower/greater than or equal to the objective function value obtained without the additional constraints.

Example 6

$$\max_{x,y} \quad z = 2x + y$$

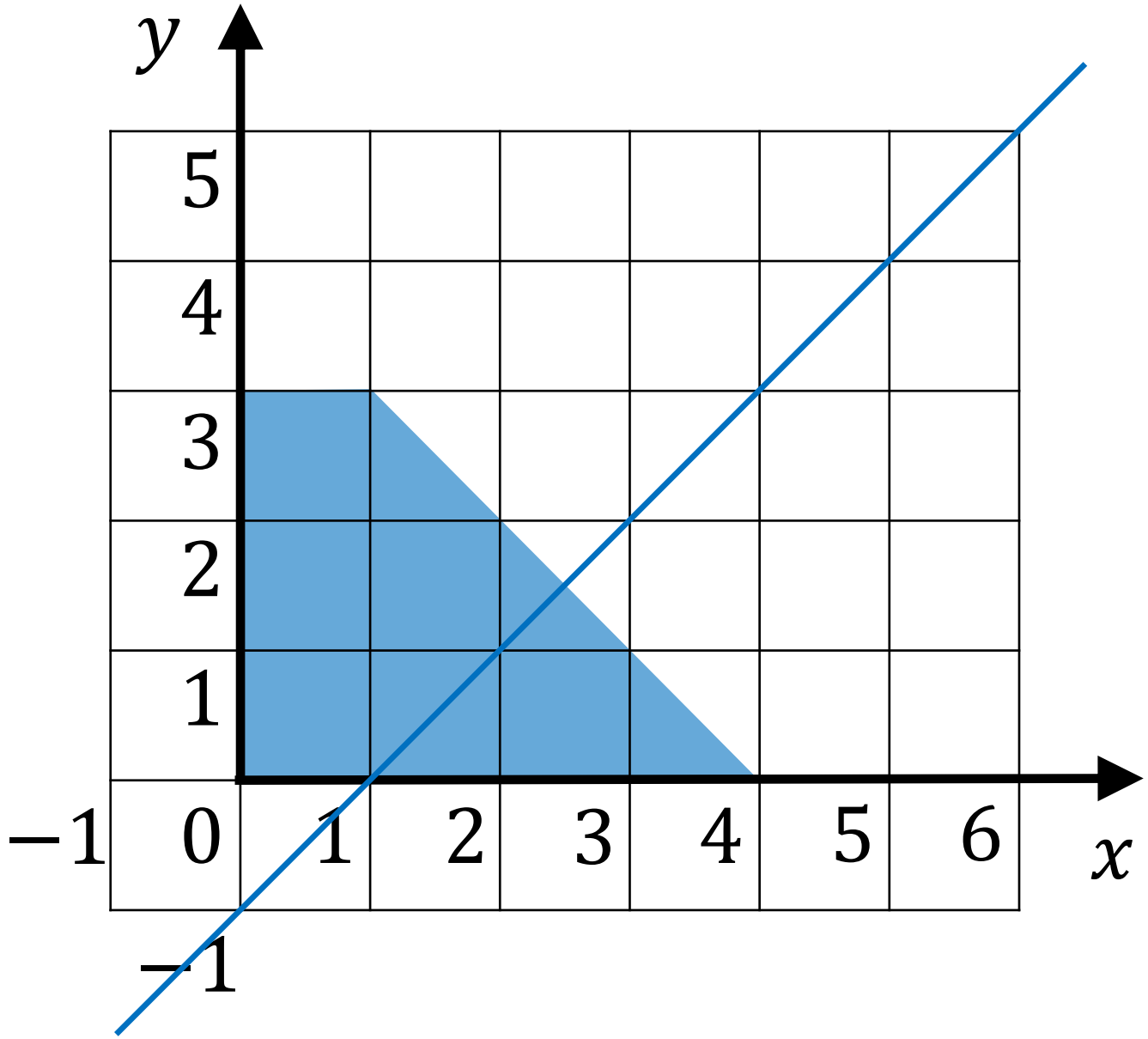
s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$

$x + y \leq 4$

$\boldsymbol{x - y = 1}$



Example 6

$$\max_{x,y} \quad z = 2x + y$$

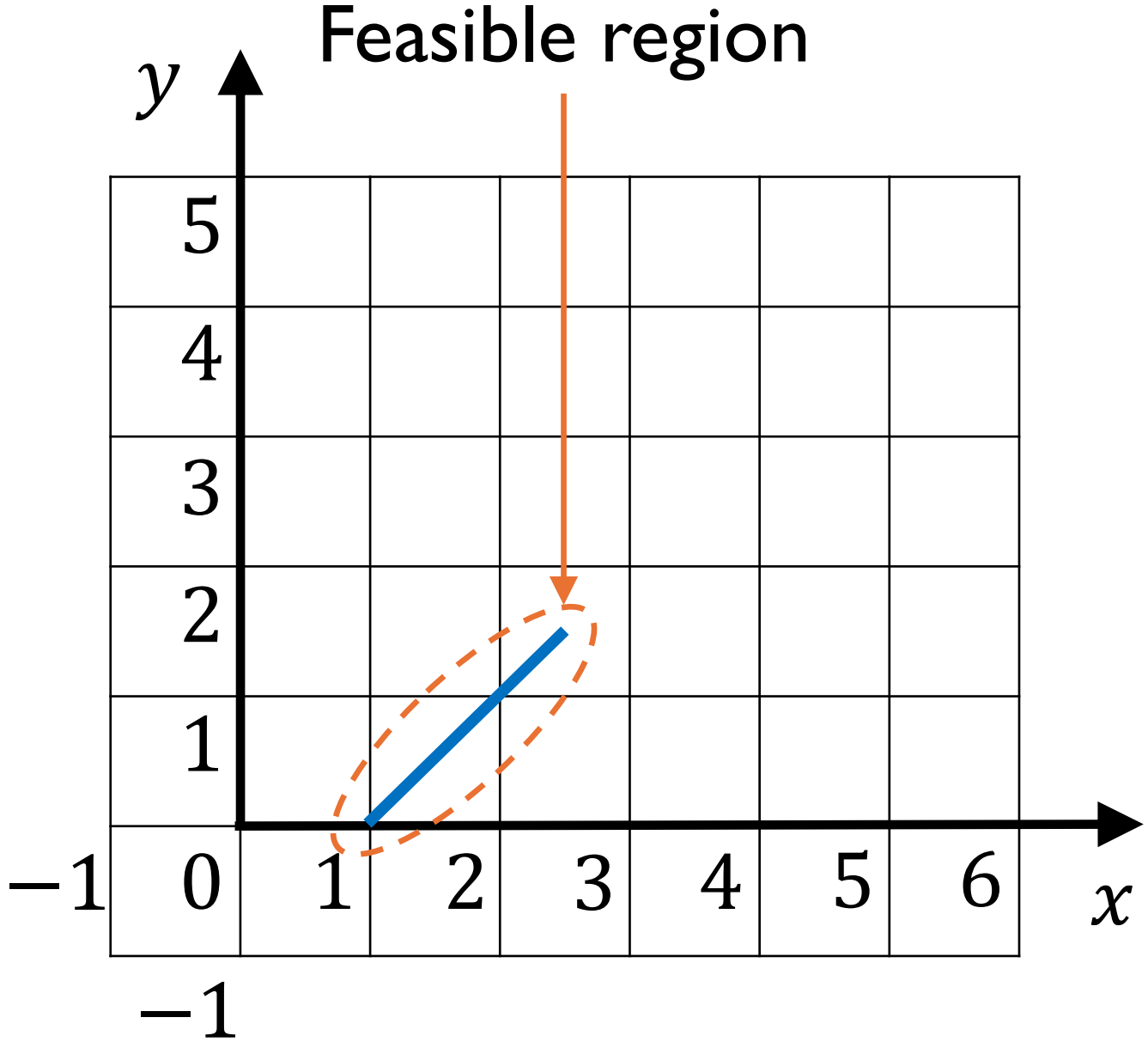
s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$

$x + y \leq 4$

$x - y = 1$



Example 6

$$\max_{x,y} \quad z = 2x + y$$

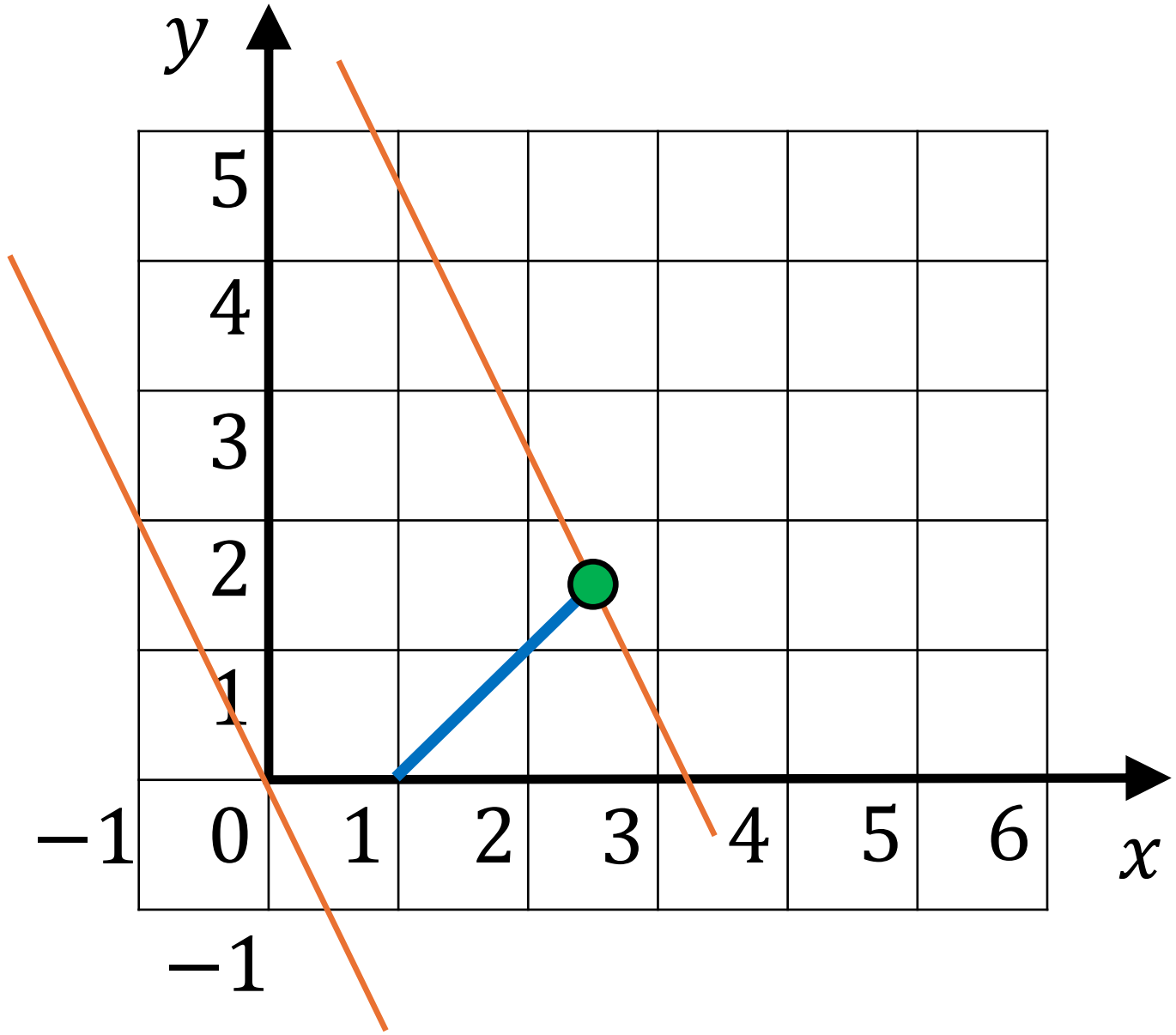
s.t.:

$0 \leq x \leq 5$

$0 \leq y \leq 3$

$x + y \leq 4$

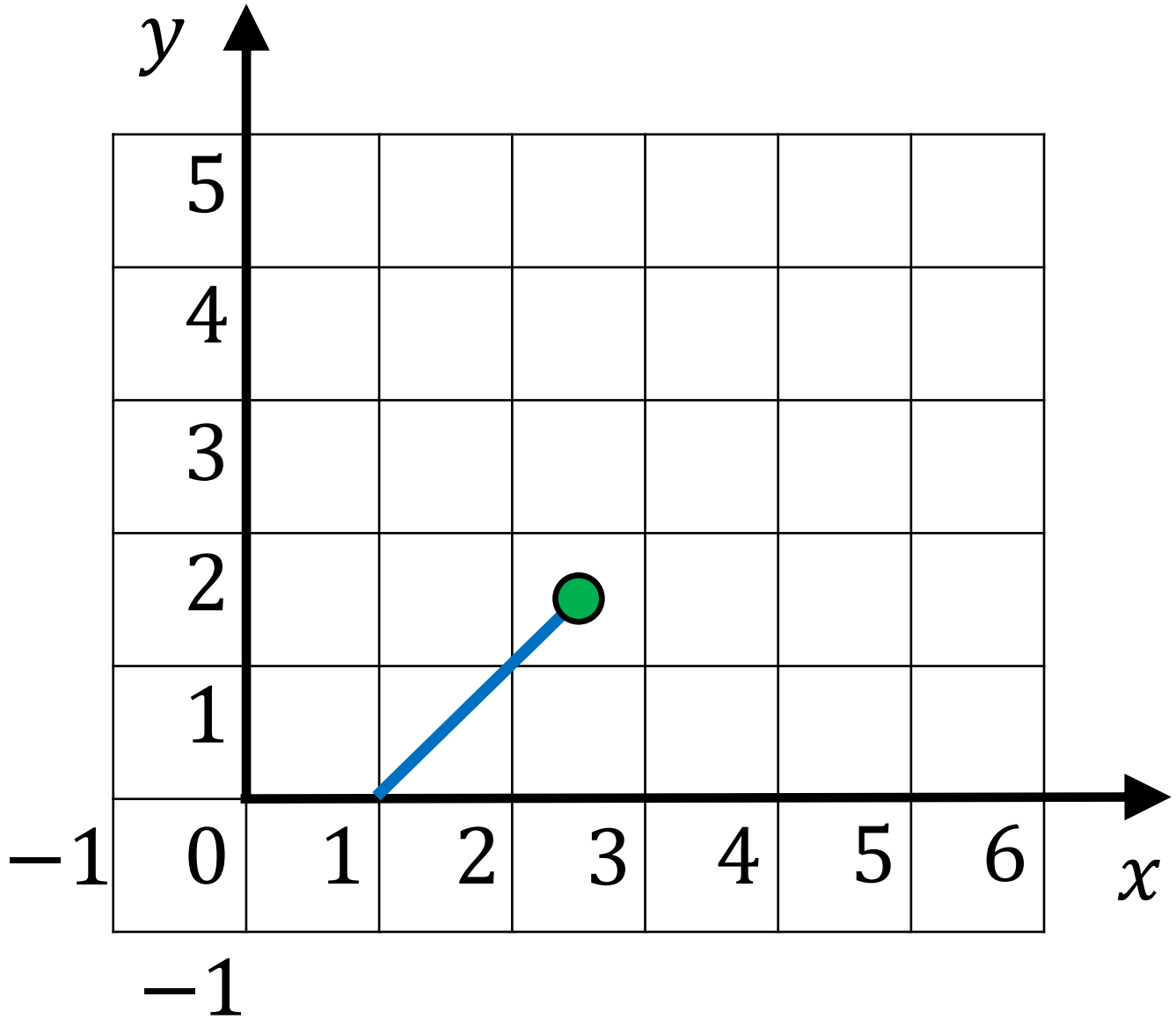
$x - y = 1$



Example 6

Optimal solution:

$$x^* = 2.5$$
$$y^* = 1.5$$
$$z^* = 6.5$$



4. Mixed-integer linear programming (MILP)

Example 7

$$\max_{x,y} \quad z = 2x + y$$

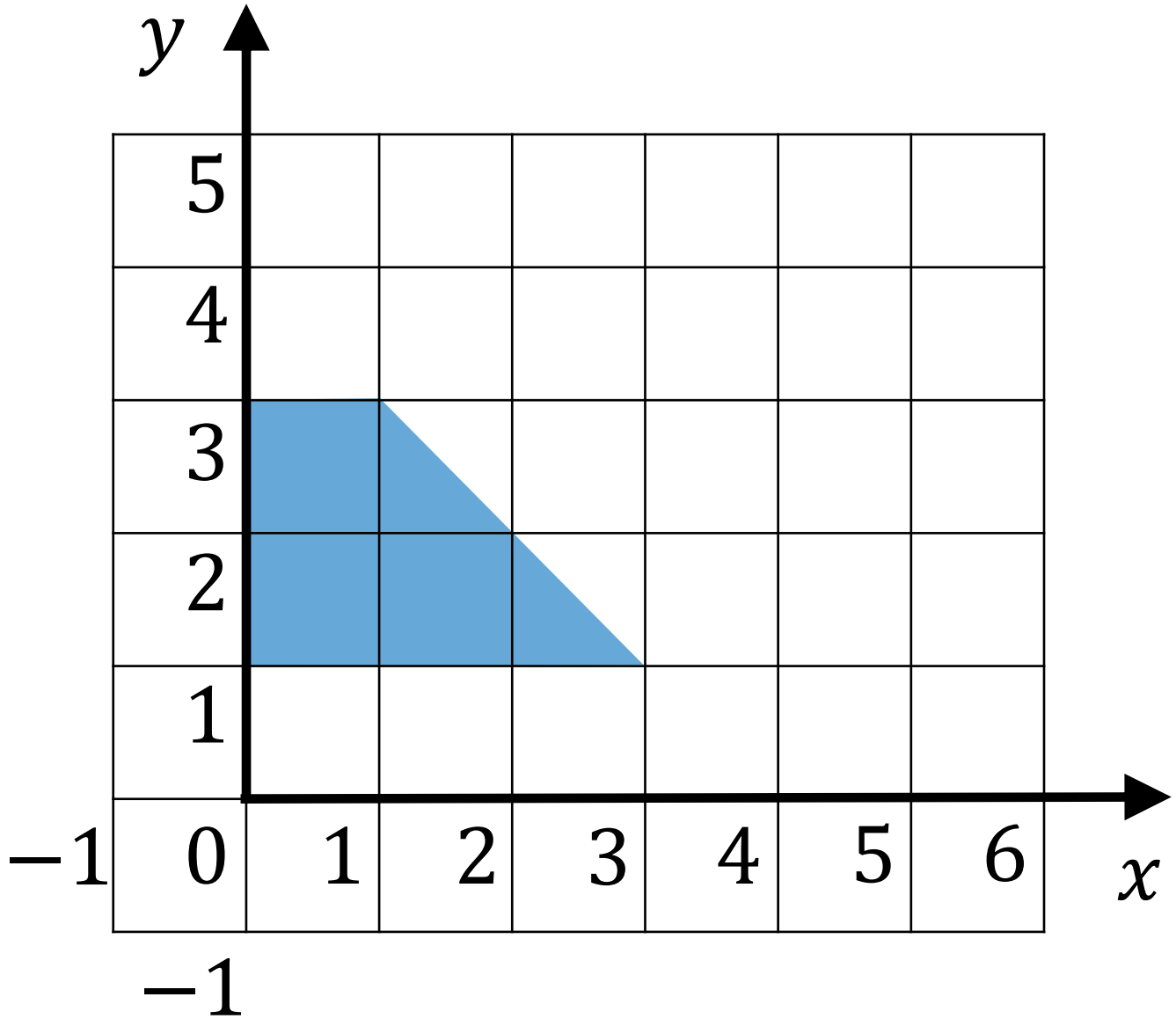
s.t.:

$\mathbf{0 \leq x \leq 5}$

$\mathbf{1 \leq y \leq 3}$

$\mathbf{x + y \leq 4}$

$y \in \mathbb{N}$



Example 7

$$\max_{x,y} \quad z = 2x + y$$

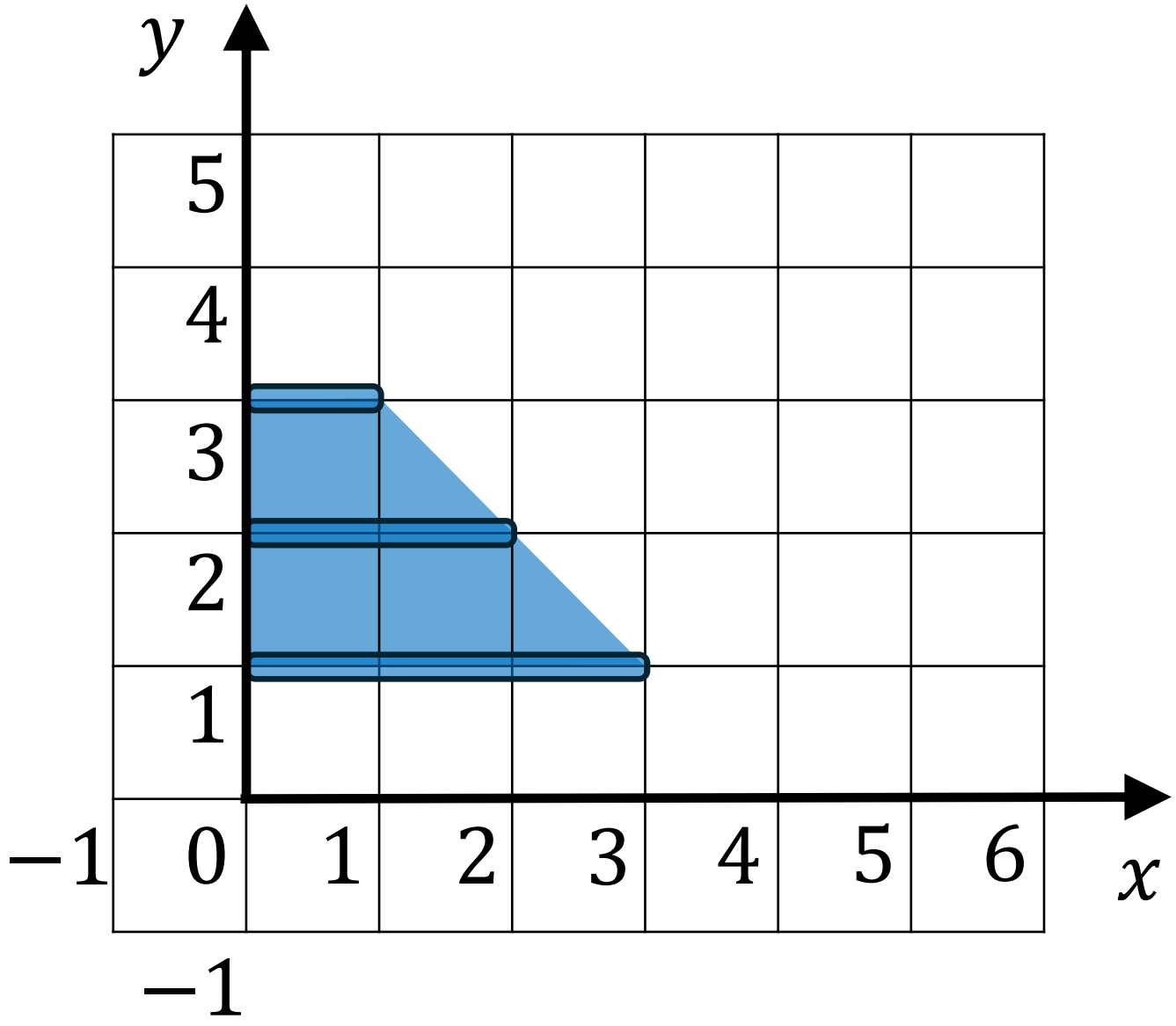
s.t.:

$0 \leq x \leq 5$

$1 \leq y \leq 3$

$x + y \leq 4$

$y \in \mathbb{N}$



Example 7

$$\max_{x,y} \quad z = 2x + y$$

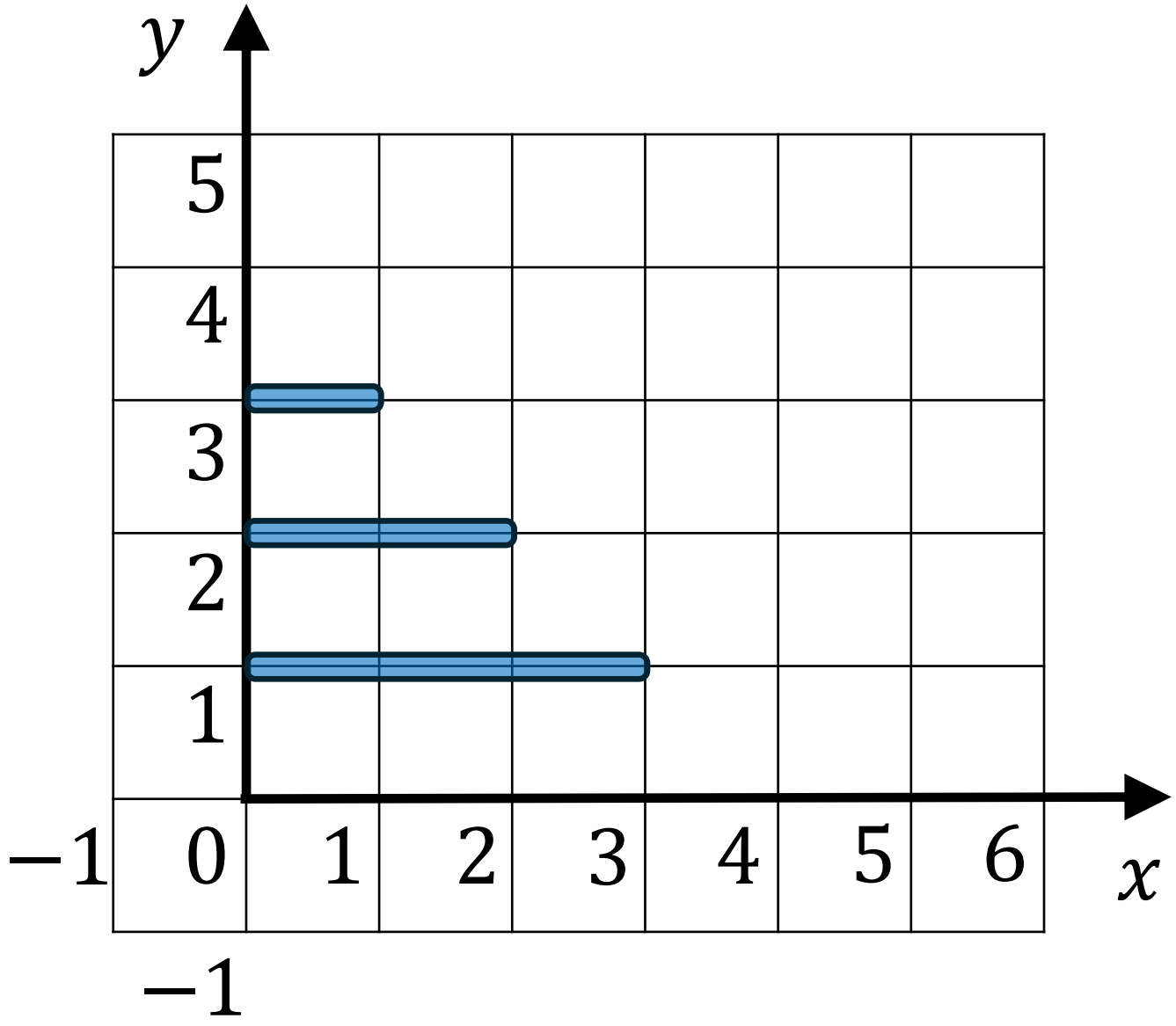
s.t.:

$0 \leq x \leq 5$

$1 \leq y \leq 3$

$x + y \leq 4$

$y \in \mathbb{N}$



Example 7

$$\max_{x,y} \quad z = 2x + y$$

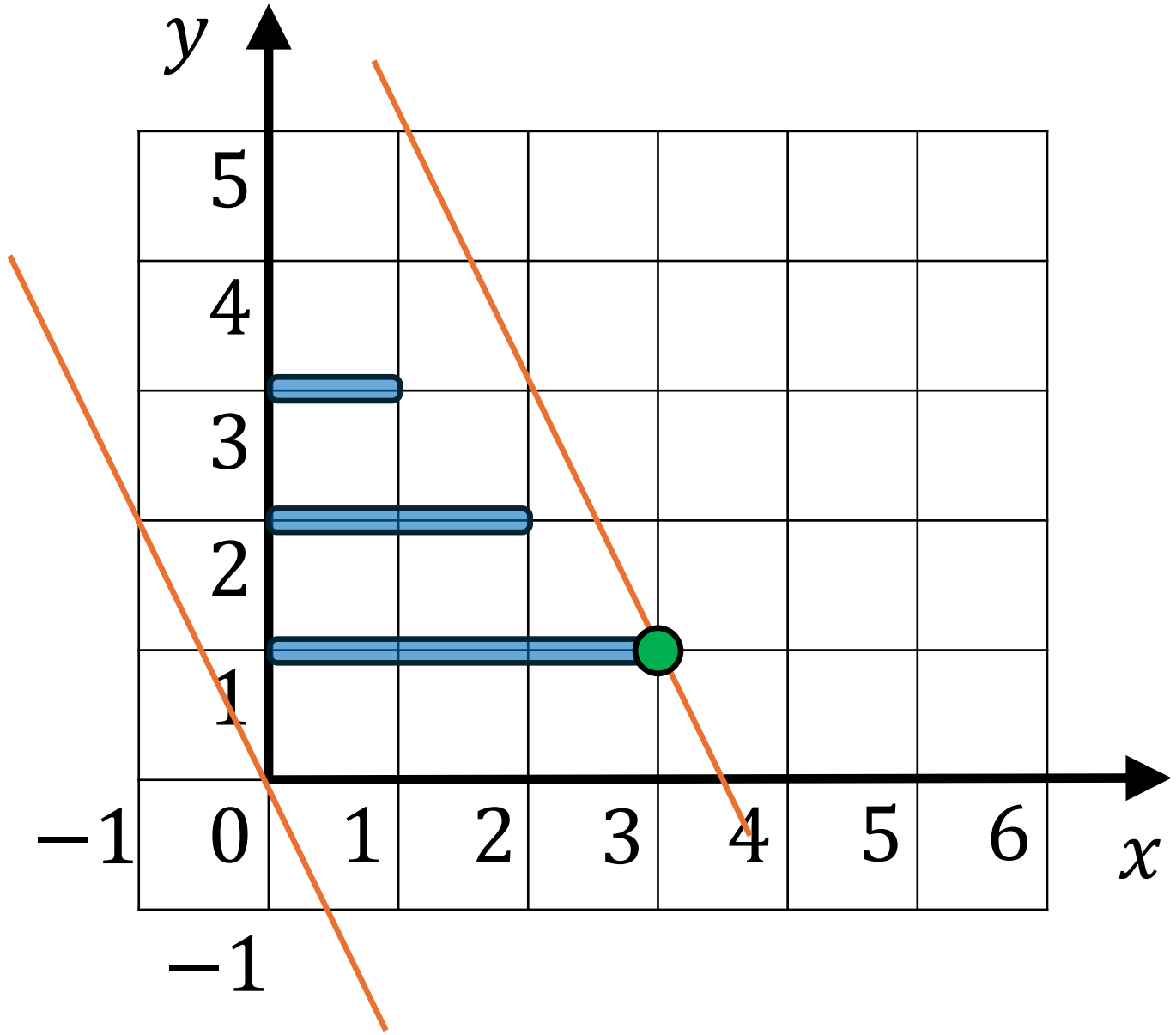
s.t.:

$0 \leq x \leq 5$

$1 \leq y \leq 3$

$x + y \leq 4$

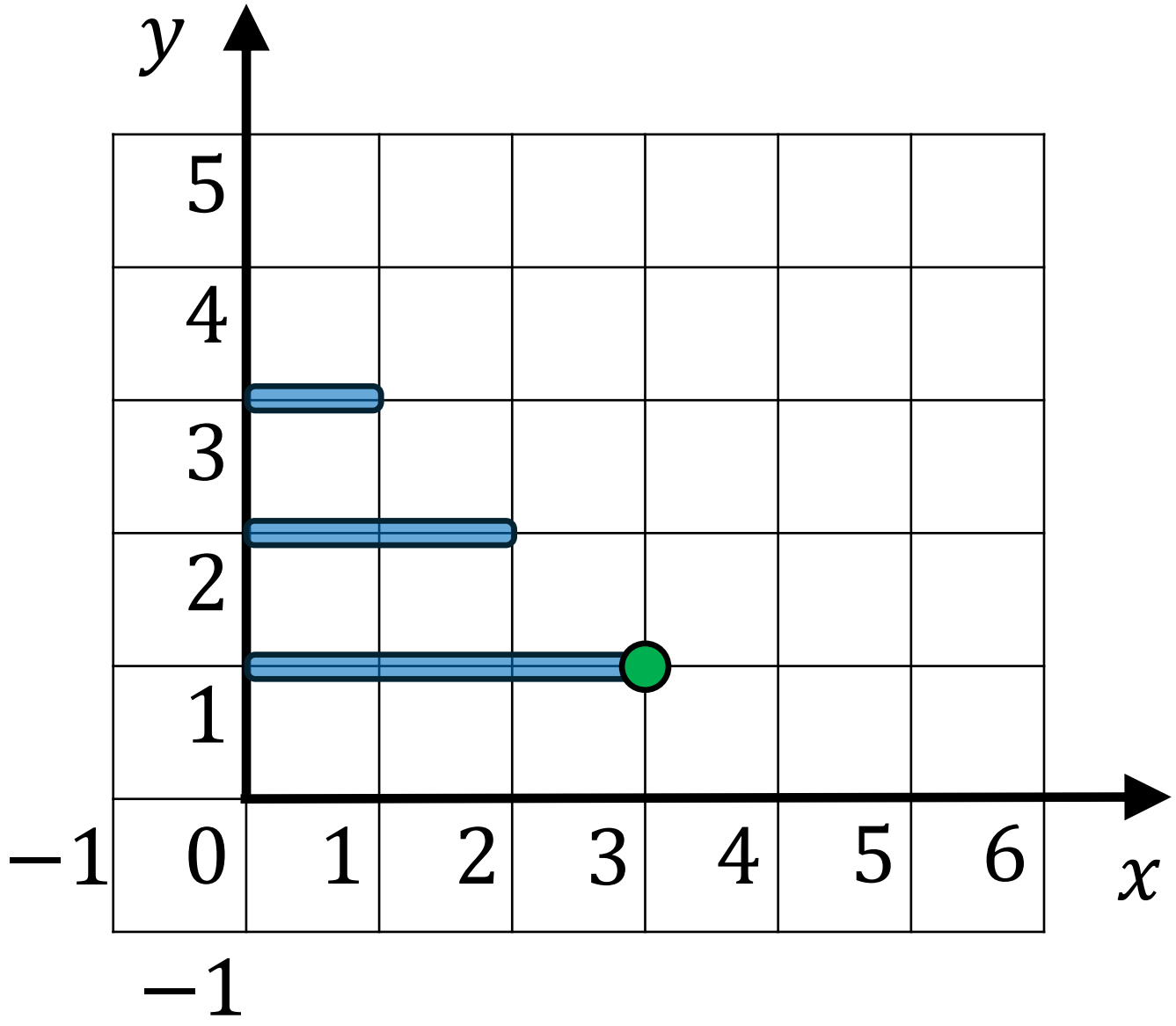
$y \in \mathbb{N}$



Example 7

Optimal solution:

$$x^* = 3$$
$$y^* = 1$$
$$z^* = 7$$



Example 8

$$\max_{x,y} \quad z = 2x + y$$

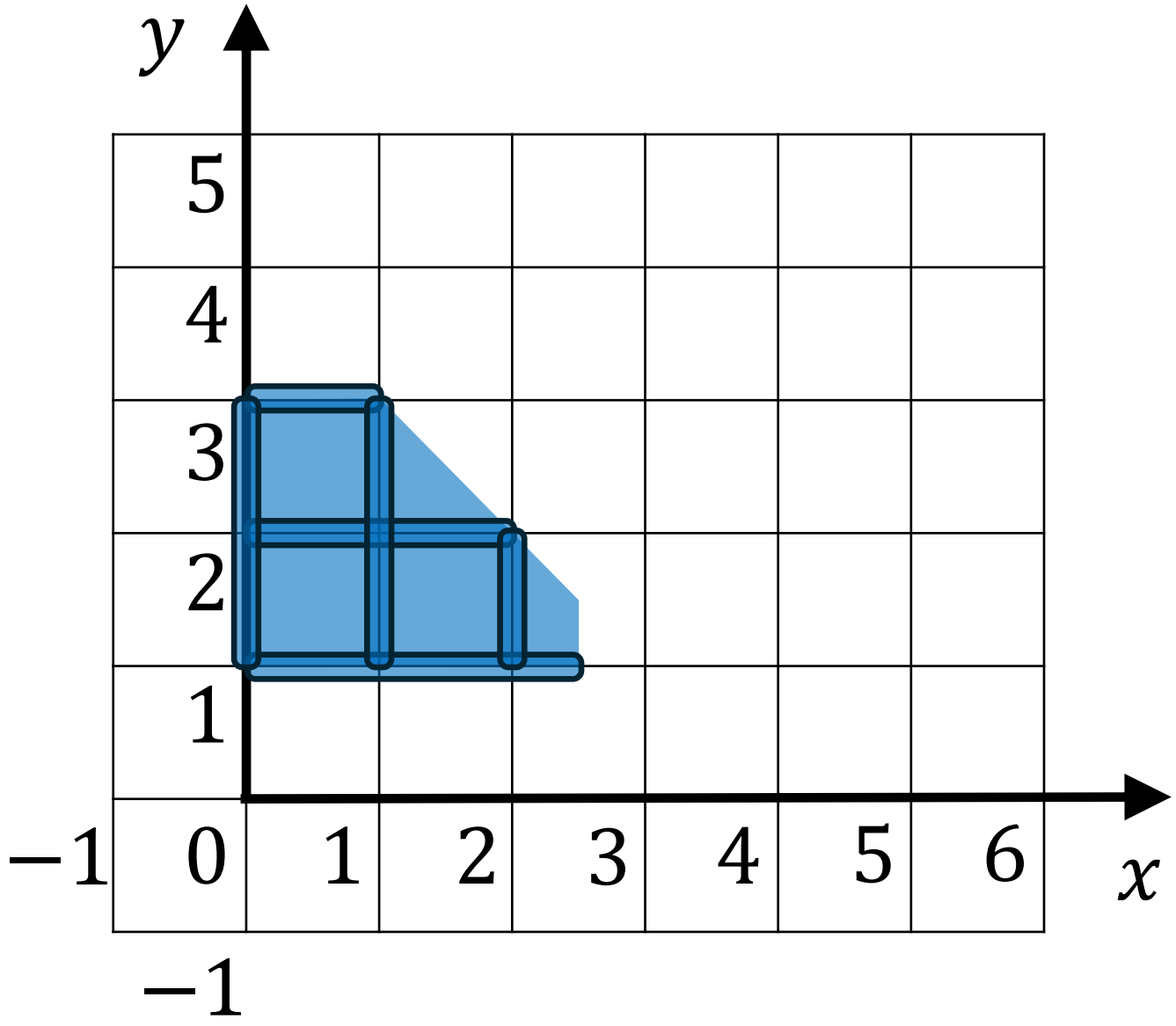
s.t.:

$0 \leq \boldsymbol{x \leq 2.5}$

$1 \leq y \leq 3$

$x + y \leq 4$

$\boldsymbol{x}, y \in \mathbb{N}$



Example 8

$$\max_{x,y} \quad z = 2x + y$$

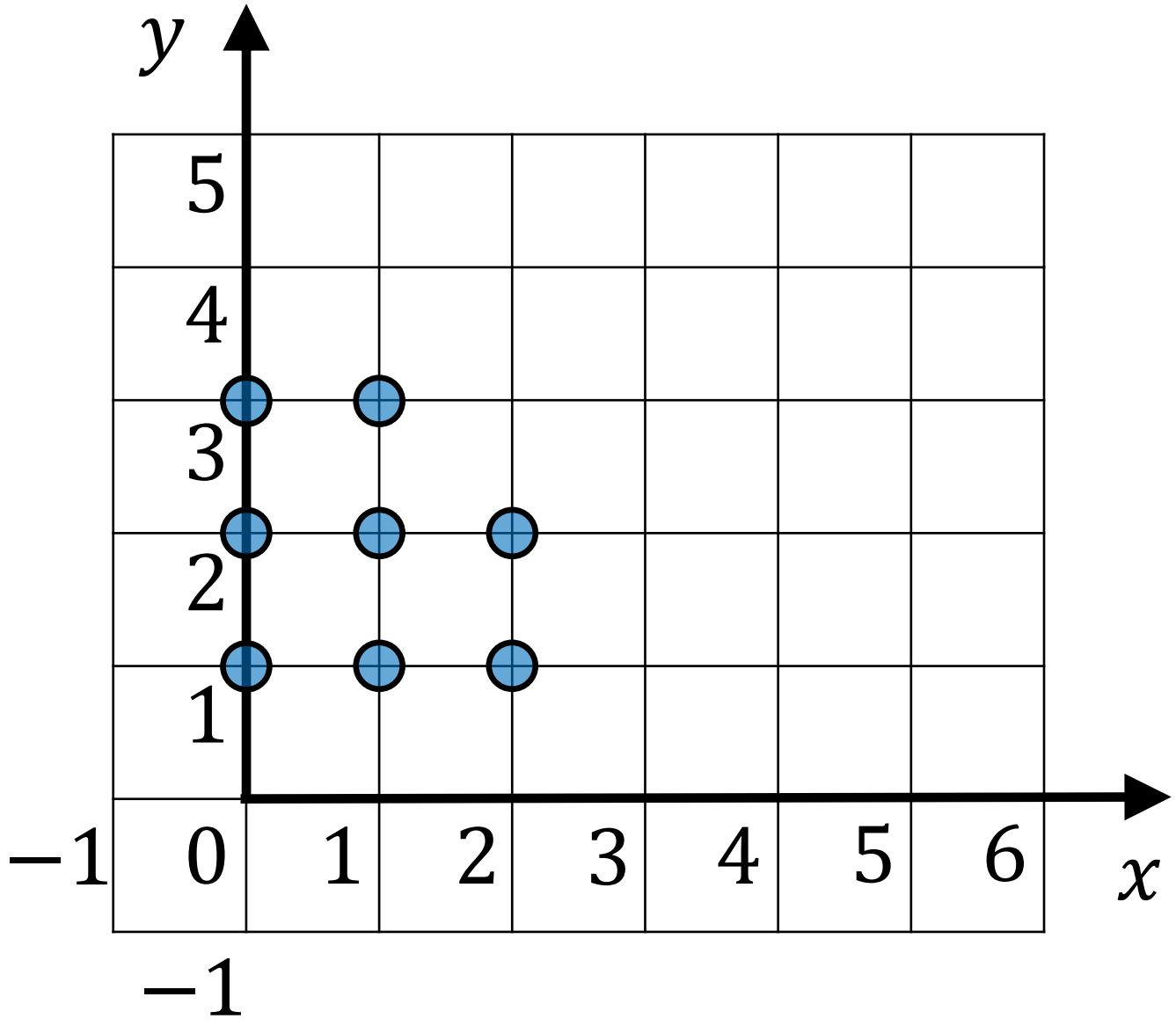
s.t.:

$0 \leq x \leq 2.5$

$1 \leq y \leq 3$

$x + y \leq 4$

$x, y \in \mathbb{N}$



Example 8

$$\max_{x,y} \quad z = 2x + y$$

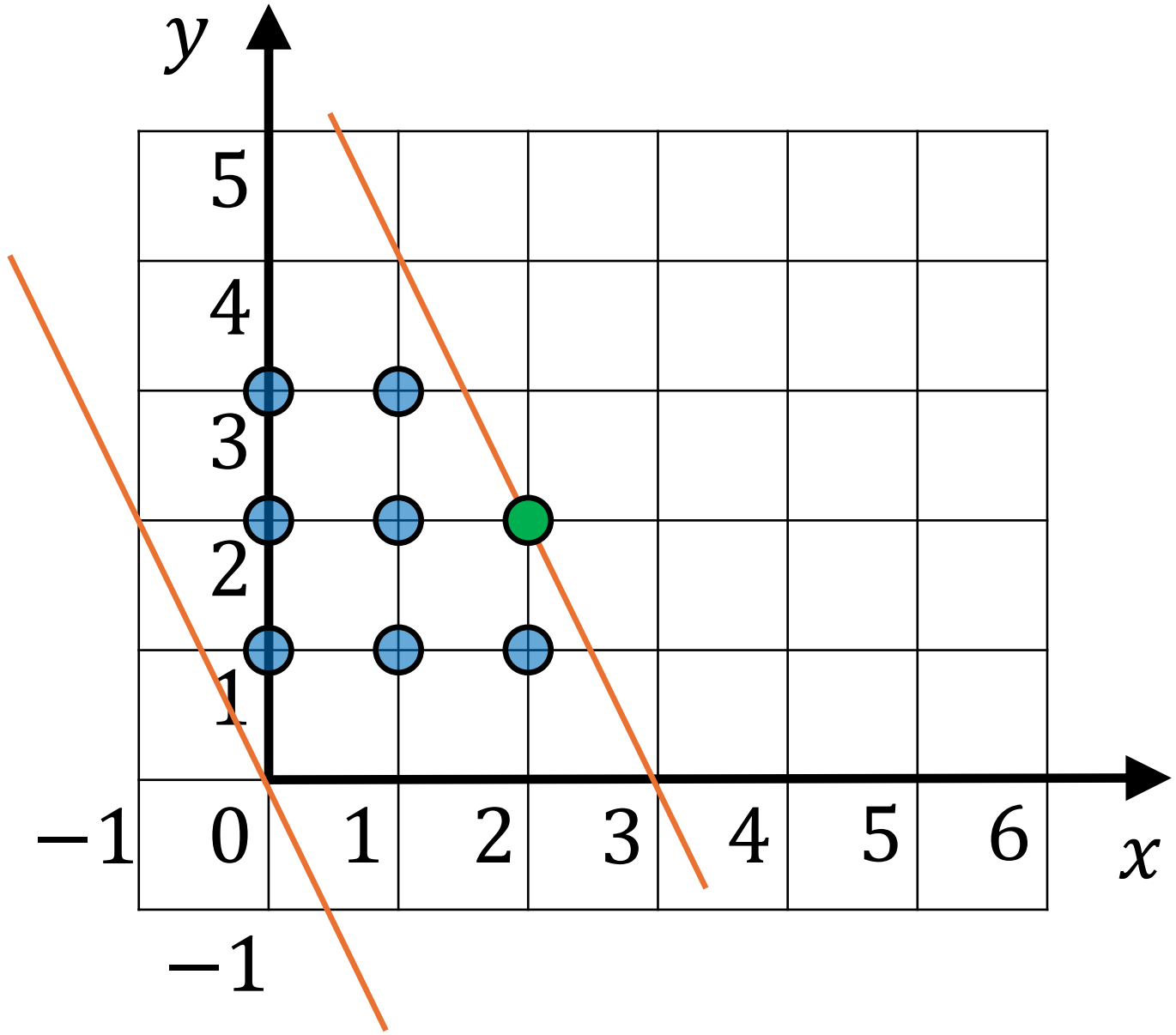
s.t.:

$0 \leq x \leq 2.5$

$1 \leq y \leq 3$

$x + y \leq 4$

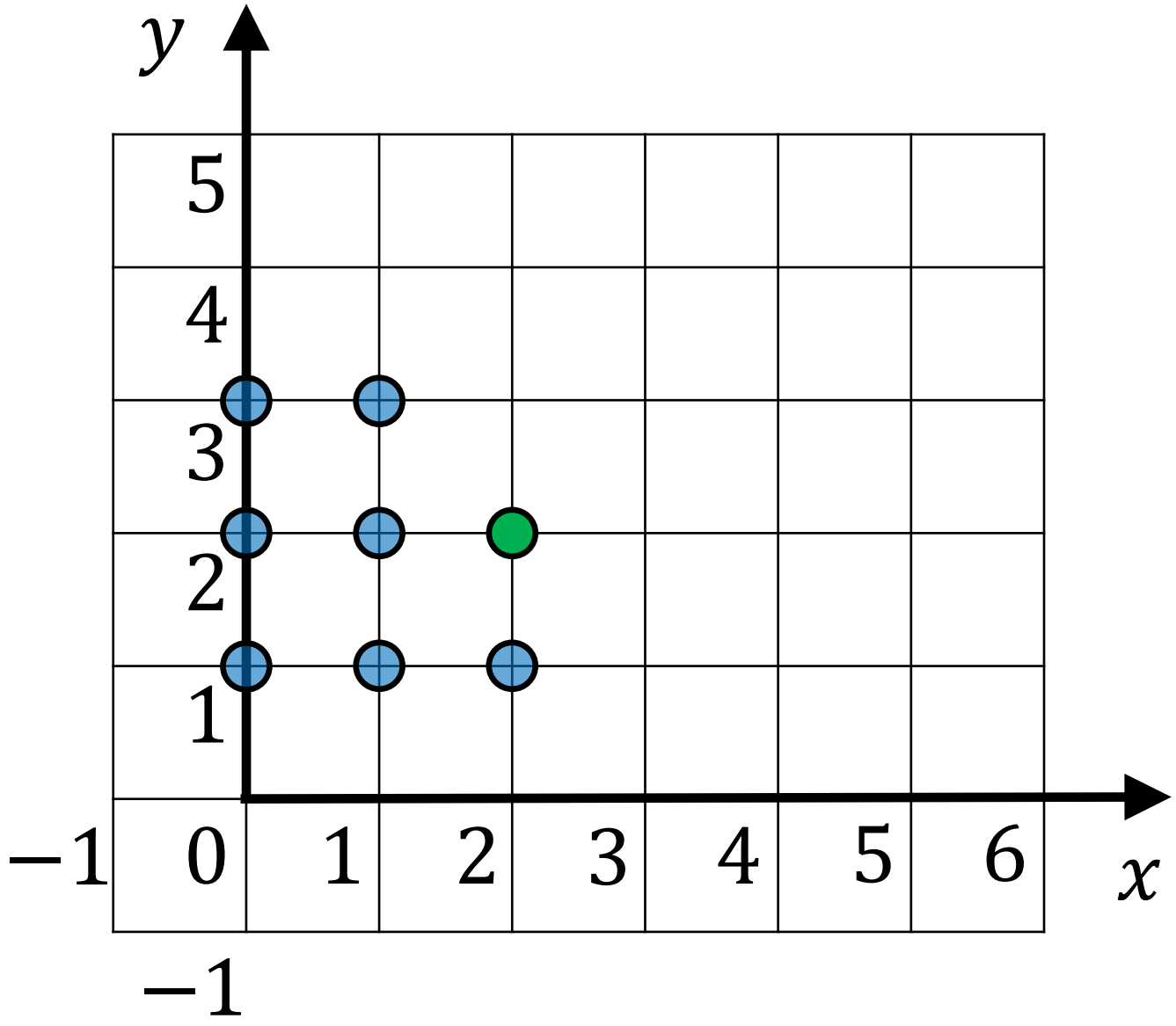
$x, y \in \mathbb{N}$



Example 8

Optimal solution:

$$x^* = 2$$
$$y^* = 2$$
$$z^* = 6$$



Branch and bound

- Algorithm based on the use of a relaxed formulation.
- Integer variables → Continuous variables.
- Iterative procedure → All the search tree is analyzed, or the optimality GAP condition is fulfilled: if $|UB - LB| > GAP$, continue; otherwise, stop.
- A relative optimality GAP condition can also be considered: if $\frac{|UB-LB|}{|LB|} > GAP$, continue; otherwise, stop.

Branch and bound

- If $GAP > 0$, we cannot claim that the solution found is the global optimum.
- However, the solution may be good enough for the decision maker.
- https://www.youtube.com/watch?v=tBUfL3O_Nzg

5. Nonlinear programming (NLP)

Example 9

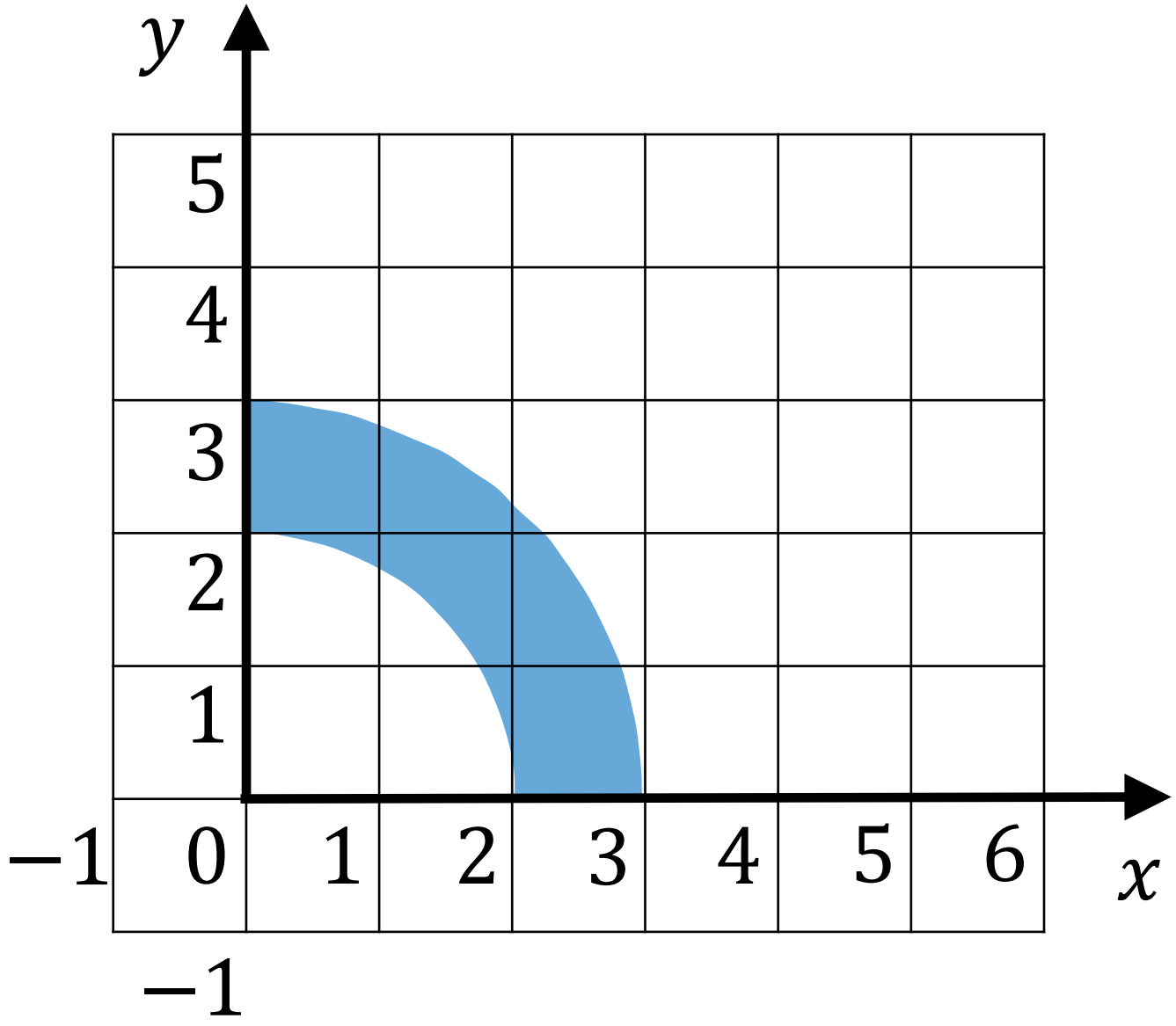
$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq x$

$0 \leq y$

$\mathbf{4 \leq x^2 + y^2 \leq 9}$



Example 9

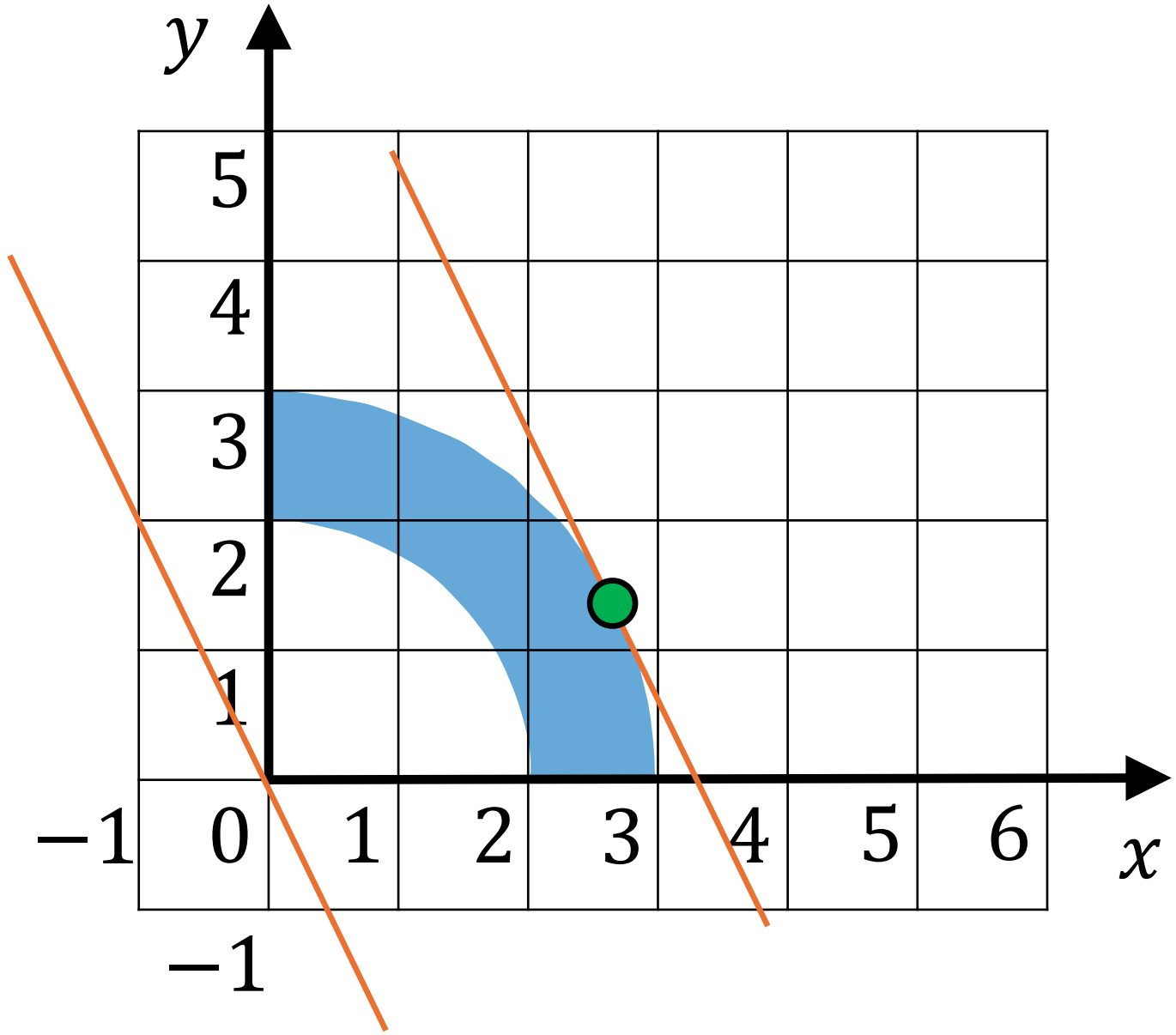
$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq x$

$0 \leq y$

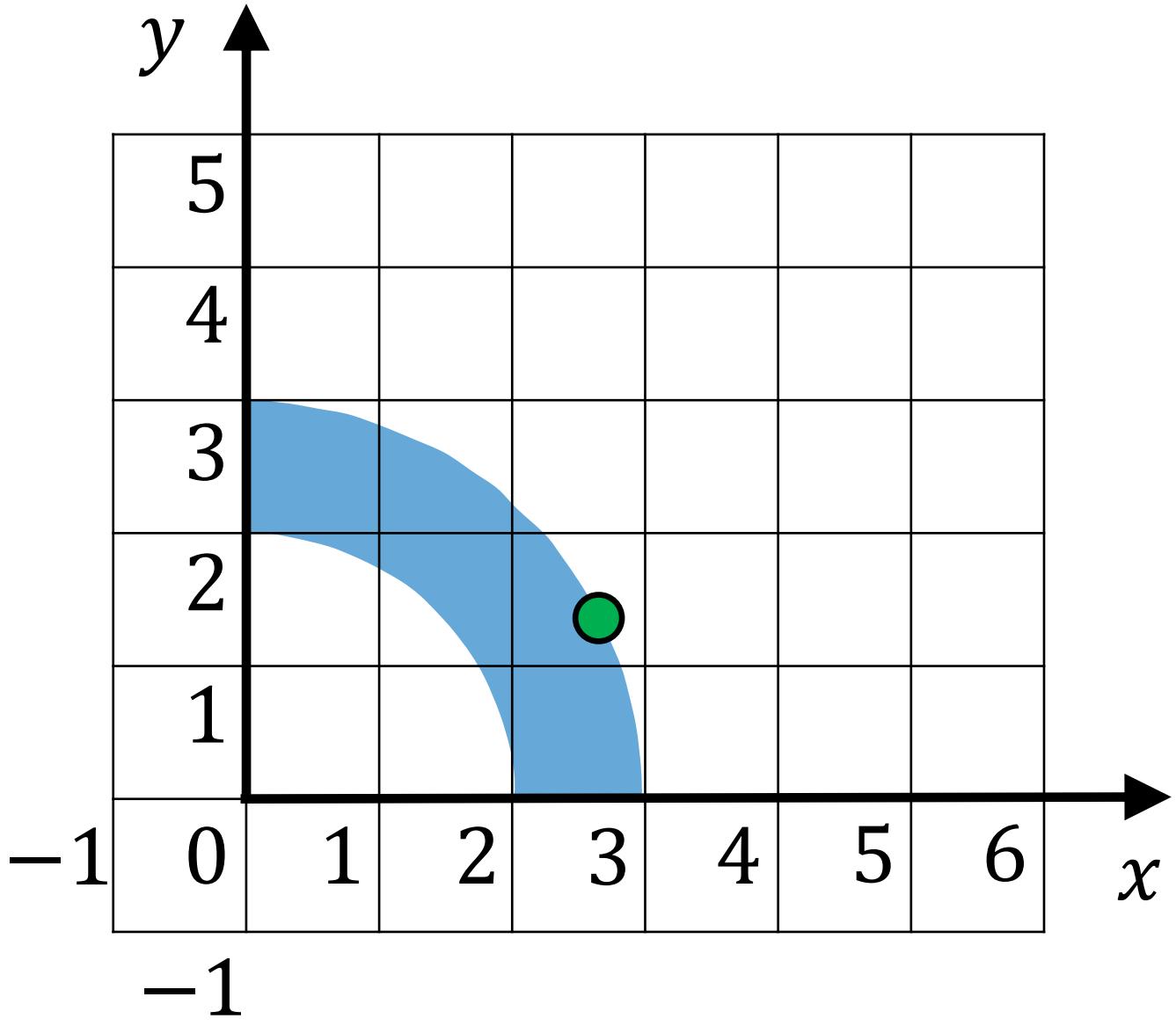
$4 \leq x^2 + y^2 \leq 9$



Example 9

Optimal solution:

$$x^* = 2.683$$
$$y^* = 1.3462$$
$$z^* = 6.708$$



Example 10

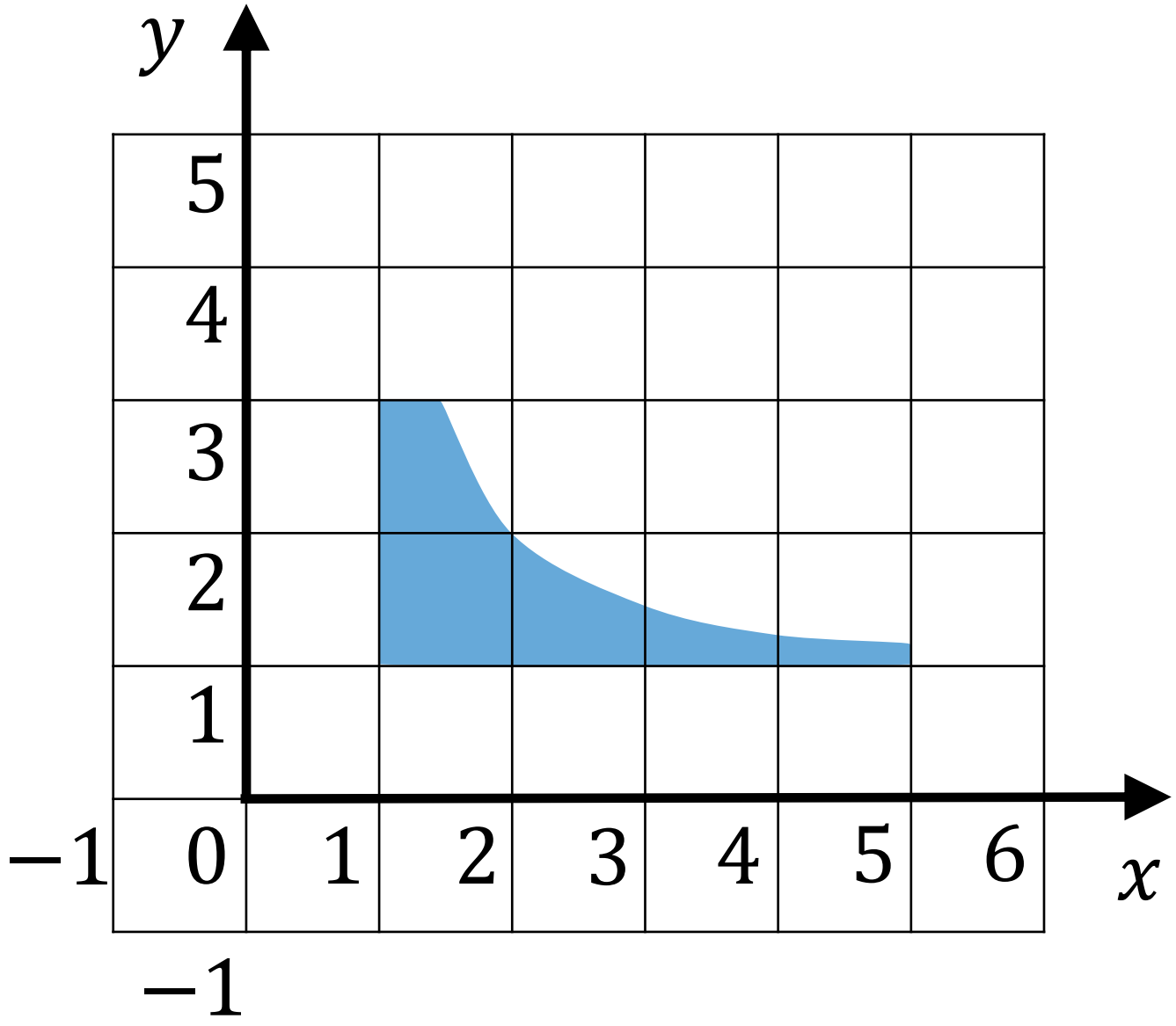
$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$1 \leq x \leq 5$

$1 \leq y \leq 3$

$y(x-1) \leq x$



Example 10

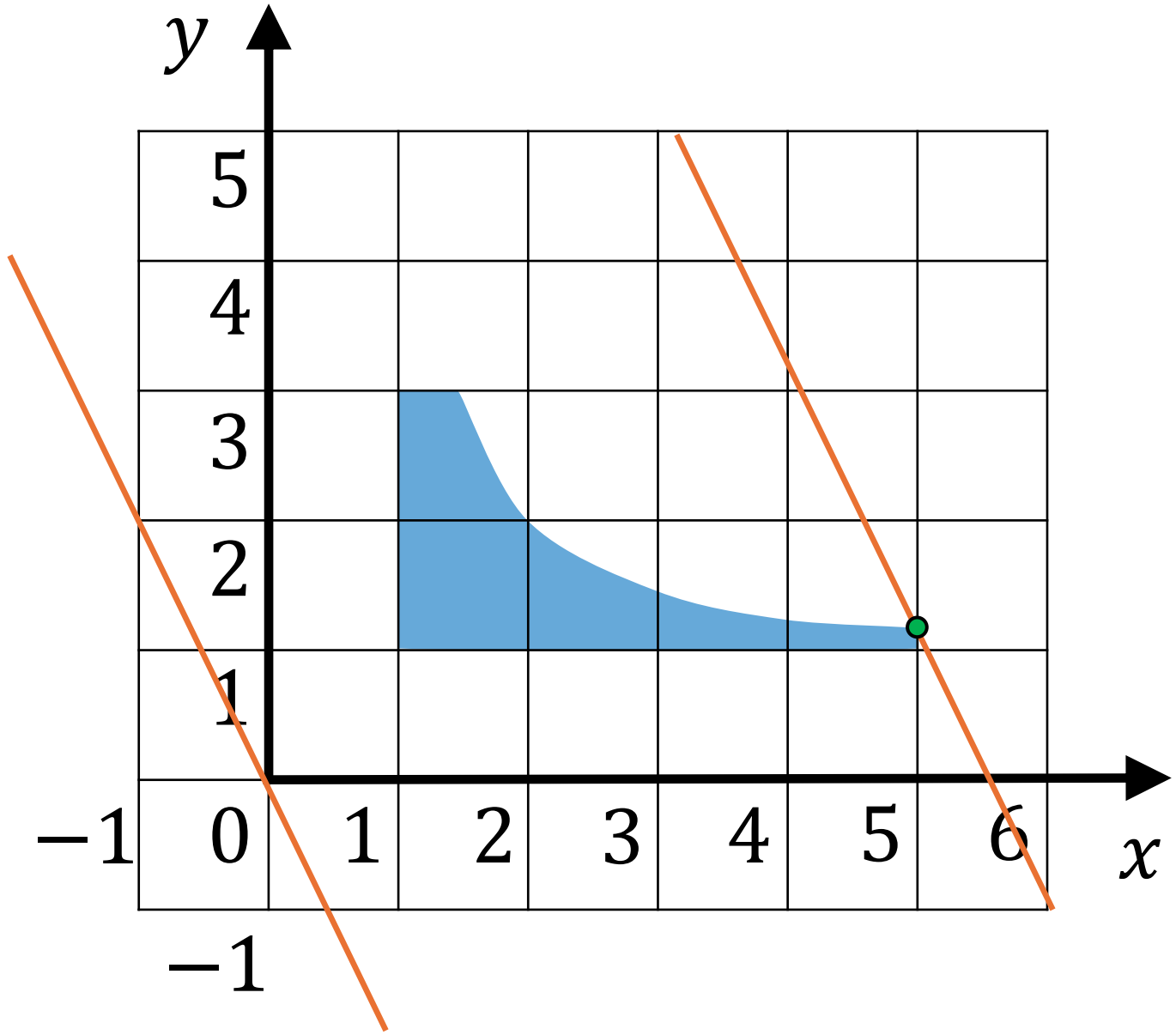
$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$1 \leq x \leq 5$

$1 \leq y \leq 3$

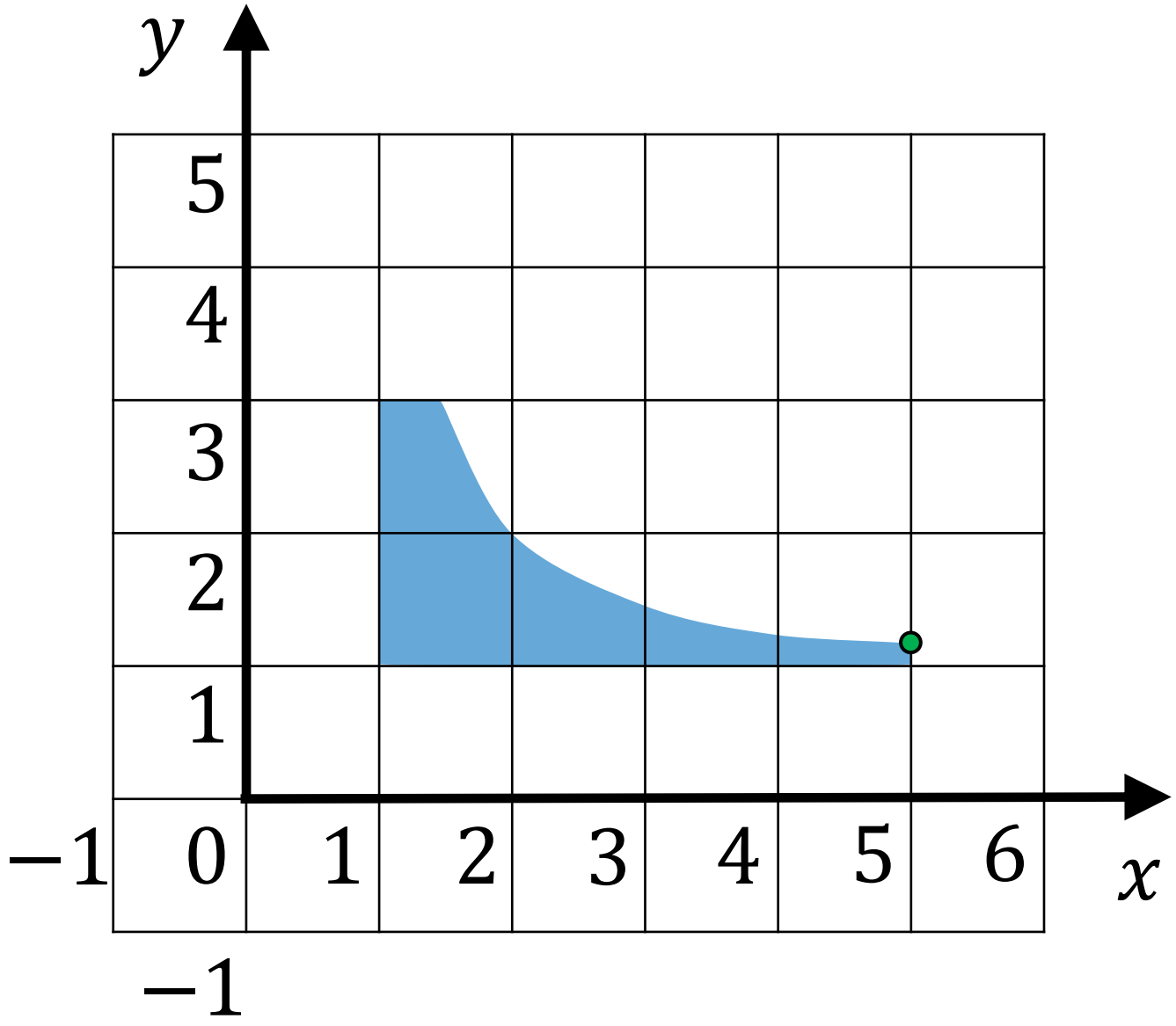
$y(x-1) \leq x$



Example 10

Optimal solution:

$$x^* = 5$$
$$y^* = 1.25$$
$$z^* = 11.25$$



6. Optimization software

Optimization Software

- **GAMS**: General Algebraic Modeling System.
- Pyomo.
- AMPL.
- AIMMS.
- JuMP.
- …

GAMS



Get started:

https://www.gams.com/free-licenses/#get-started

Download:

https://www.gams.com/download/

Documentation:

https://www.gams.com/latest/docs/index.html

Solvers

- Gurobi.
- IBM CPLEX.
- CONOPT.
- MOSEK.
- KNITRO.
- …

Exercise 1

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$$0 \leq x \leq 5$$
$$0 \leq y \leq 3$$
$$x + y \leq 4$$

Optimal solution:

$$x^* = 4$$
$$y^* = 0$$
$$z^* = 8$$

Exercise 2

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$0 \leq x \leq 2.5$

$1 \leq y \leq 3$

$x + y \leq 4.7$

$x, y \in \mathbb{N}$

Optimal solution:

$$x^* = 2$$
$$y^* = 2$$
$$z^* = 6$$

Exercise 3

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$$0 \leq x$$
$$0 \leq y$$
$$1 \leq x^2 + y^2 \leq 9$$

Optimal solution:

$$x^* = 2.683$$
$$y^* = 1.342$$
$$z^* = 6.708$$

Exercise 4

$$\max_{x,y} \quad z = 2x + y$$

s.t.:

$$1 \leq x \leq 5$$
$$1 \leq y \leq 3$$
$$y(x-1) \leq x$$

Optimal solution:

$$x^* = 5$$
$$y^* = 1.25$$
$$z^* = 11.25$$

Exercise 5

$$\min_{x} \quad z = -2x_1 - x_2 + 4x_3 + x_4$$

s.t.:

$$x_1, x_2, x_3, x_4 \geq 0$$
$$x_1 \leq 4$$
$$x_2 \leq 3$$
$$x_3 \leq 5$$
$$x_4 \leq 1$$
$$x_1 + x_2 + x_3 + x_4 = 10$$

Optimal solution:

$$x_1^* = 4$$
$$x_2^* = 3$$
$$x_3^* = 2$$
$$x_4^* = 1$$
$$z^* = -2$$

Notation

I will use the following notation in the course:

- Uppercase letters → Parameters (input data).
- Lowercase letters → Variables (output data).

Exercise 6 – Compact formulation

$$\min_{\boldsymbol{x}} \quad z = \sum_{i=1}^{I} A_i x_i$$

s.t.:

$$0 \leq x_i \leq \bar{X}_i,\ i = 1, \ldots, I$$

$$\sum_{i=1}^{I} x_i = B$$

$\boldsymbol{A} = [-2, -1, 4, 1].$
$\bar{\boldsymbol{X}} = [4, 3, 5, 1].$
$B = 10.$

Optimal solution:

$$x_1^* = 4$$
$$x_2^* = 3$$
$$x_3^* = 2$$
$$x_4^* = 1$$
$$z^* = -2$$

Exercise 7 – Sensitivity analysis

Solve exercise 6 for the following values of parameter B: 0, 3, 8, 12, and 21. Use the LOOP statement in GAMS.

Solution:

$B = 0 \rightarrow z^* = 0.$

$B = 3 \rightarrow z^* = -6.$

$B = 8 \rightarrow z^* = -10.$

$B = 12 \rightarrow z^* = 6.$

$B = 21 \rightarrow z^* = 6$ (INFEASIBLE) → Check the solution report.

7. Dual variables

Dual variables

$$\min_{x} \quad z = f(x)$$

s.t.:

$h(x) = 0 : \boldsymbol{\lambda}$

$g(x) \leq 0 : \boldsymbol{\mu}$

Dual variables

Dual variables

Primal

min primal objective function (variables x, y)
subject to primal constraints (variables x, y) : λ, μ

Dual

max dual objective function (variables λ, μ)
subject to dual constraints (variables λ, μ) : x, y

Dual variables

Primal

max primal objective function (variables x, y)
subject to primal constraints (variables x, y) : λ, μ

Dual

min dual objective function (variables λ, μ)
subject to dual constraints (variables λ, μ) : x, y

Dual variables

Channel of Prof. Michel Bierlaire (École Polytechnique Fédérale de Lausanne):

Duality: duality in linear optimization

https://www.youtube.com/watch?v=R3Z1fRXXxyg

Duality: Lagrangian and dual problem

https://www.youtube.com/watch?v=4OifjG2kIJQ

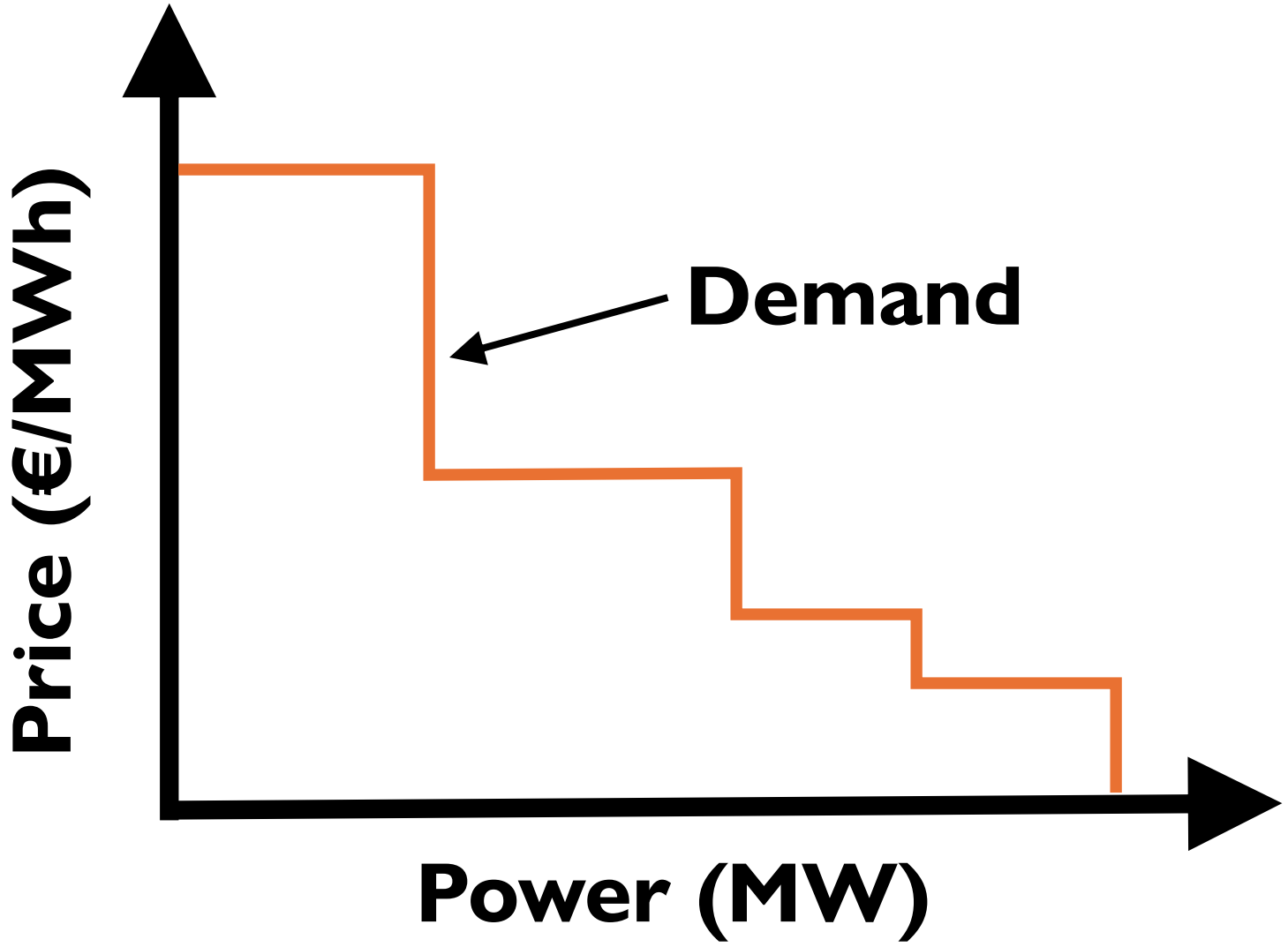
Formulation

I will consider hourly time periods for the sake of simplicity.

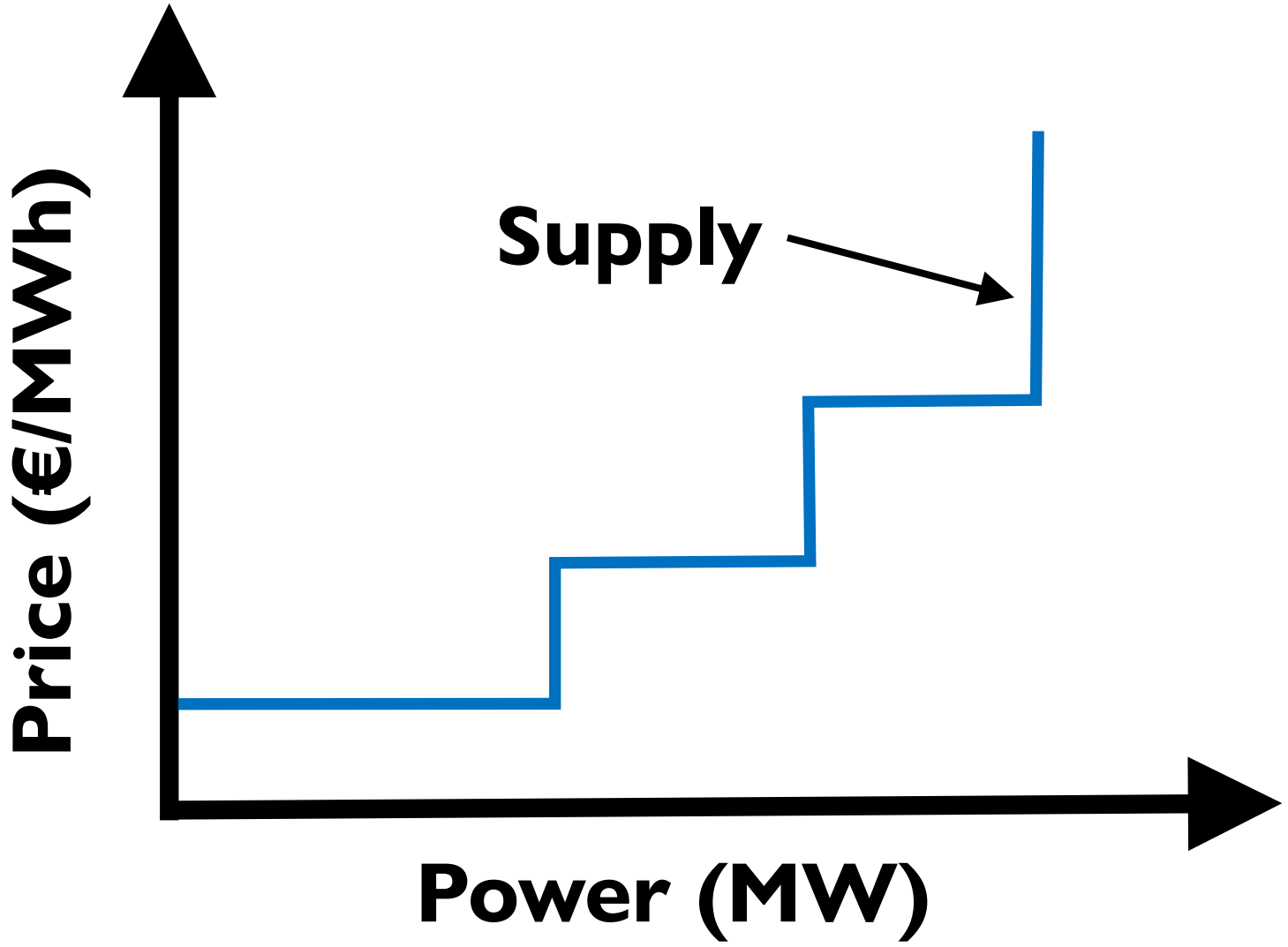
Hence, power (MW) and energy (MWh) units are equivalent.

If this is not the case, the duration of the time periods should be included in the formulation.

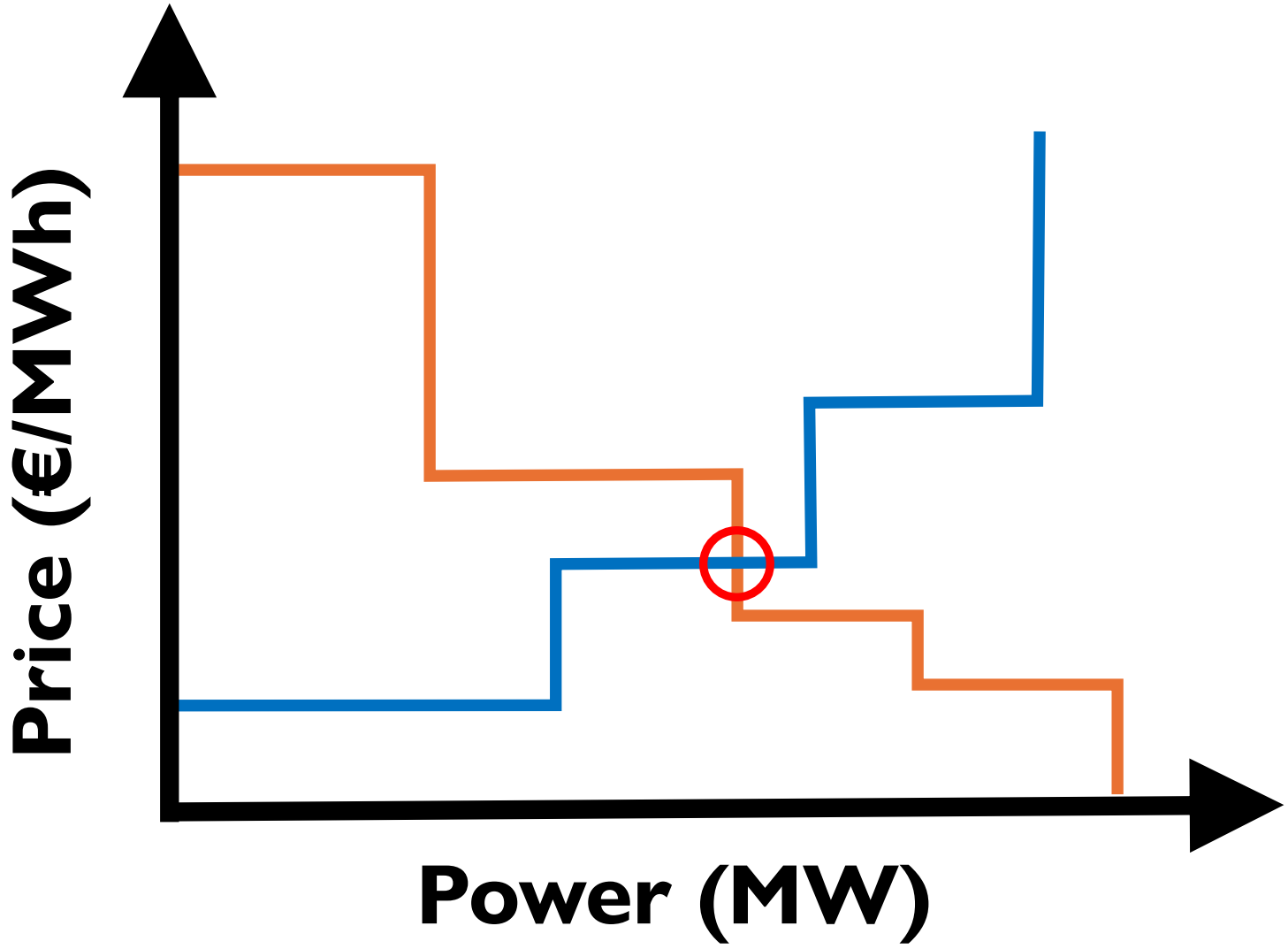
Day-ahead electricity market



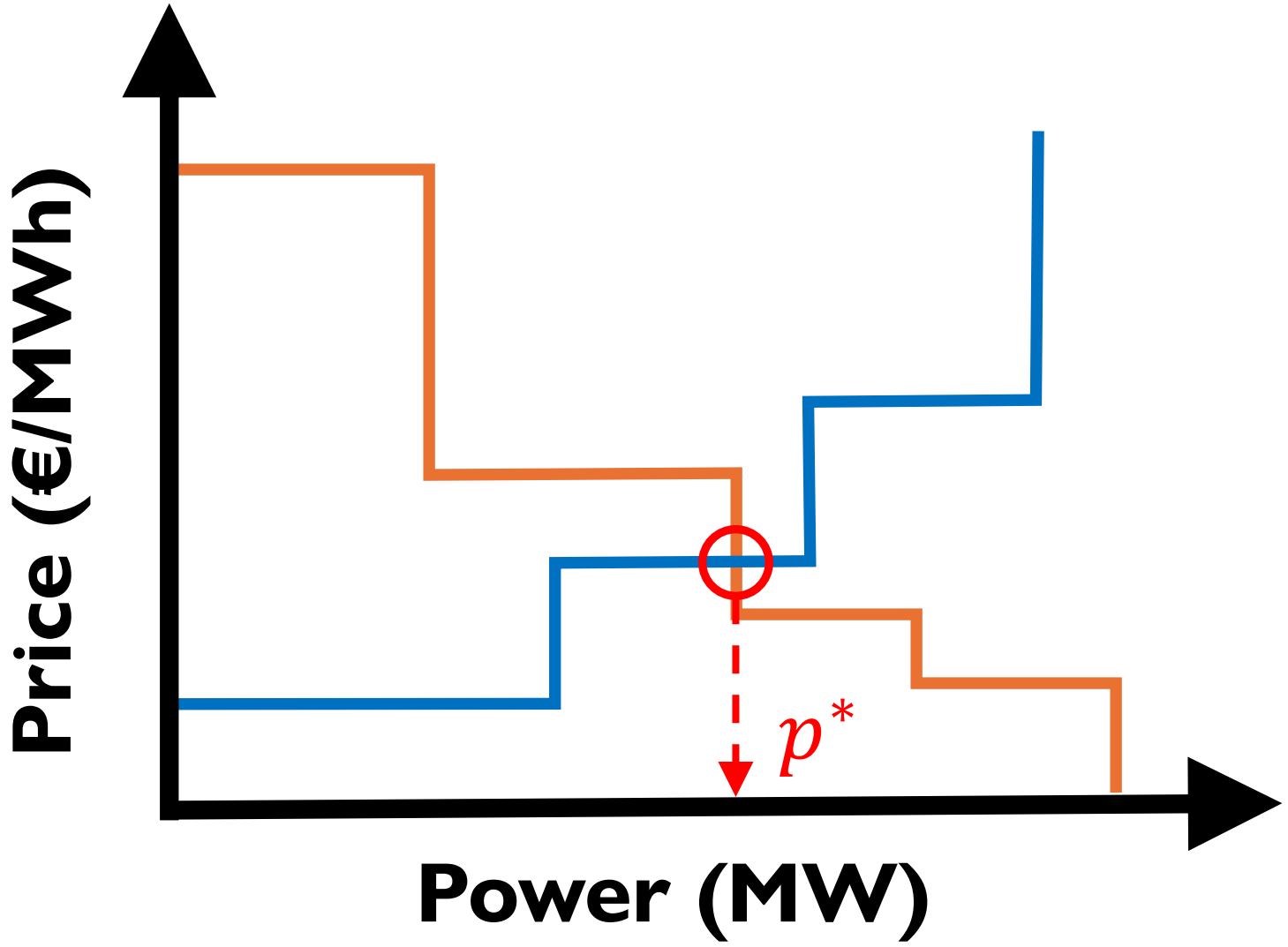
Day-ahead electricity market



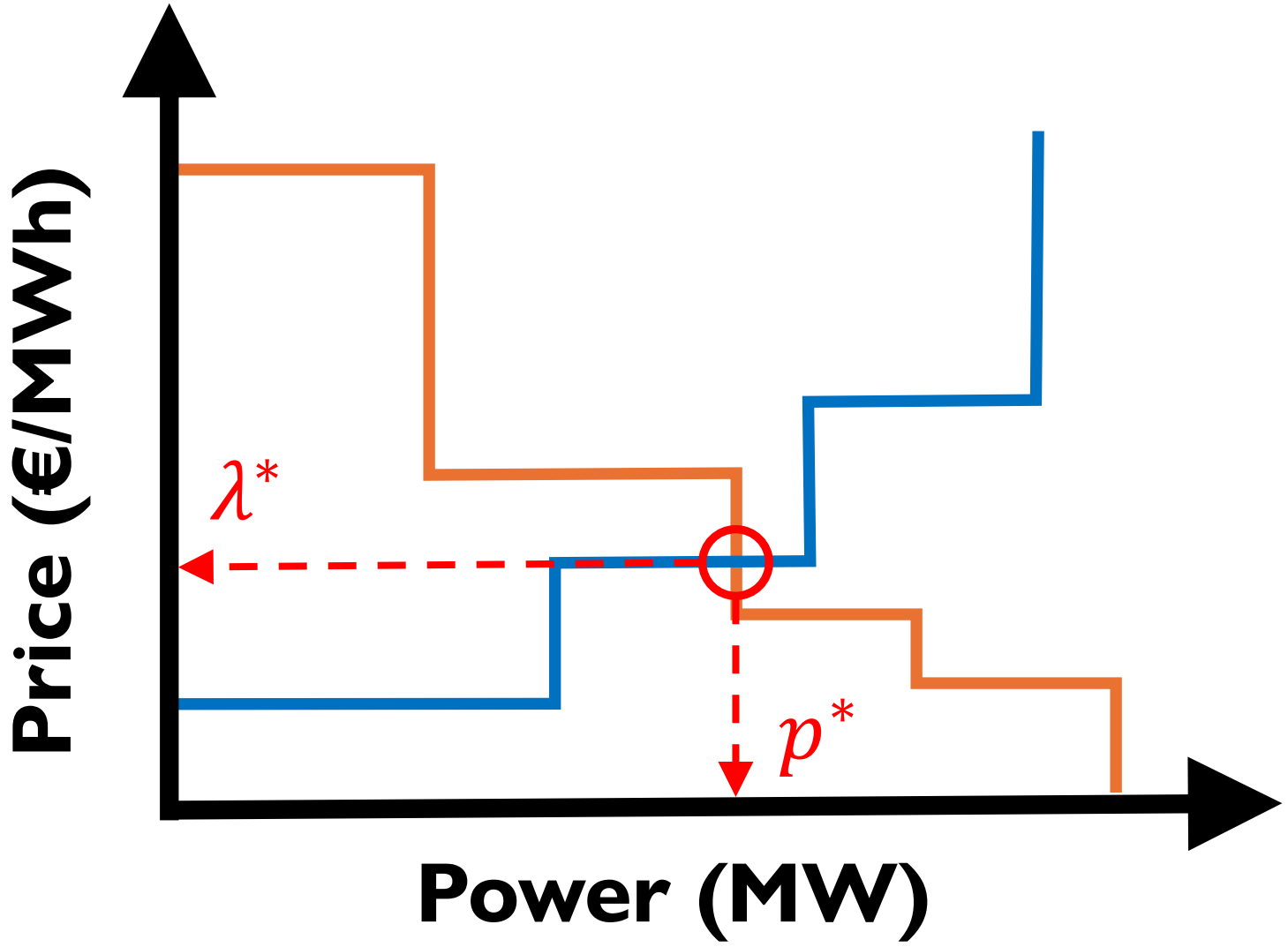
Day-ahead electricity market



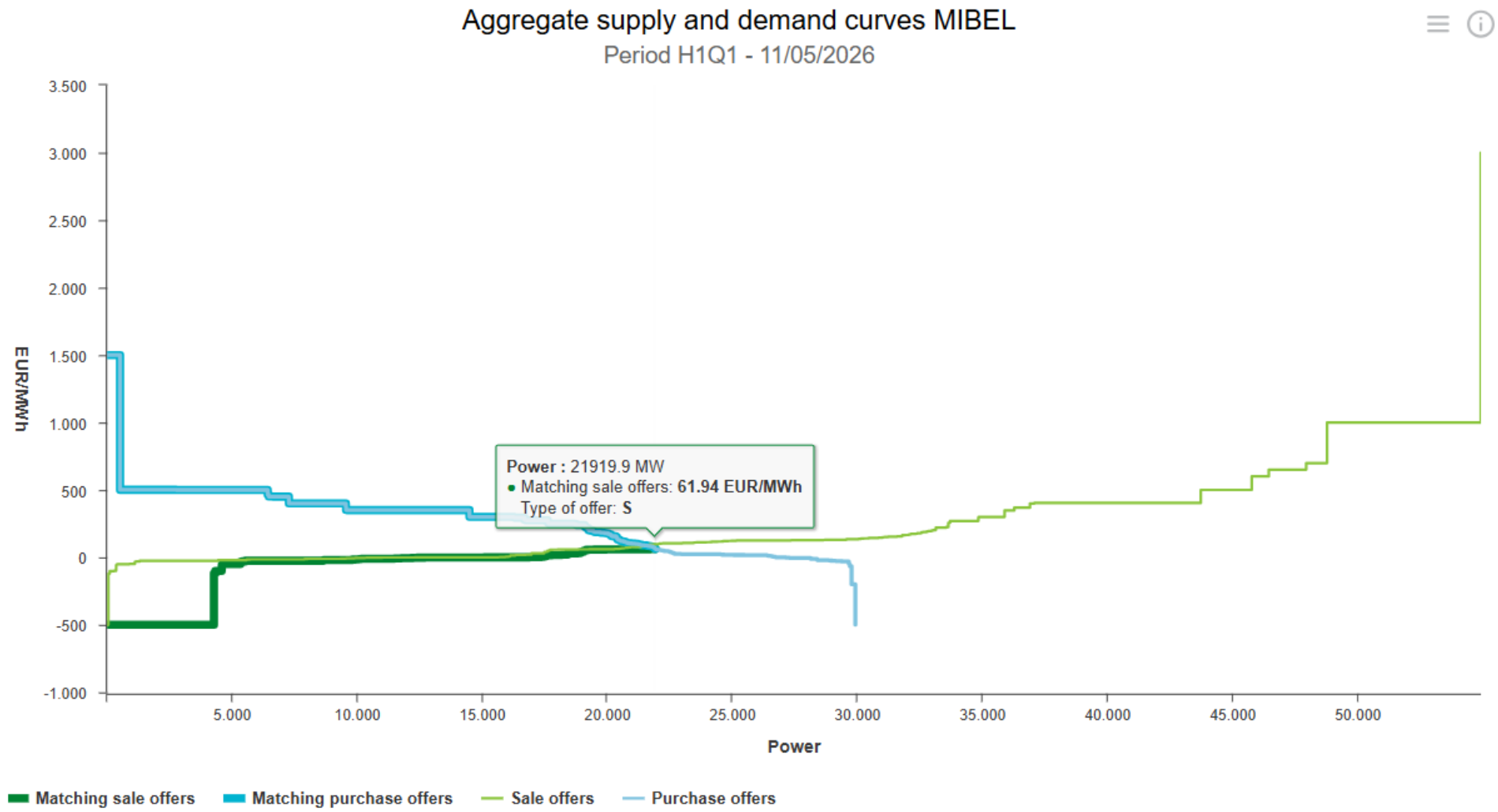
Day-ahead electricity market



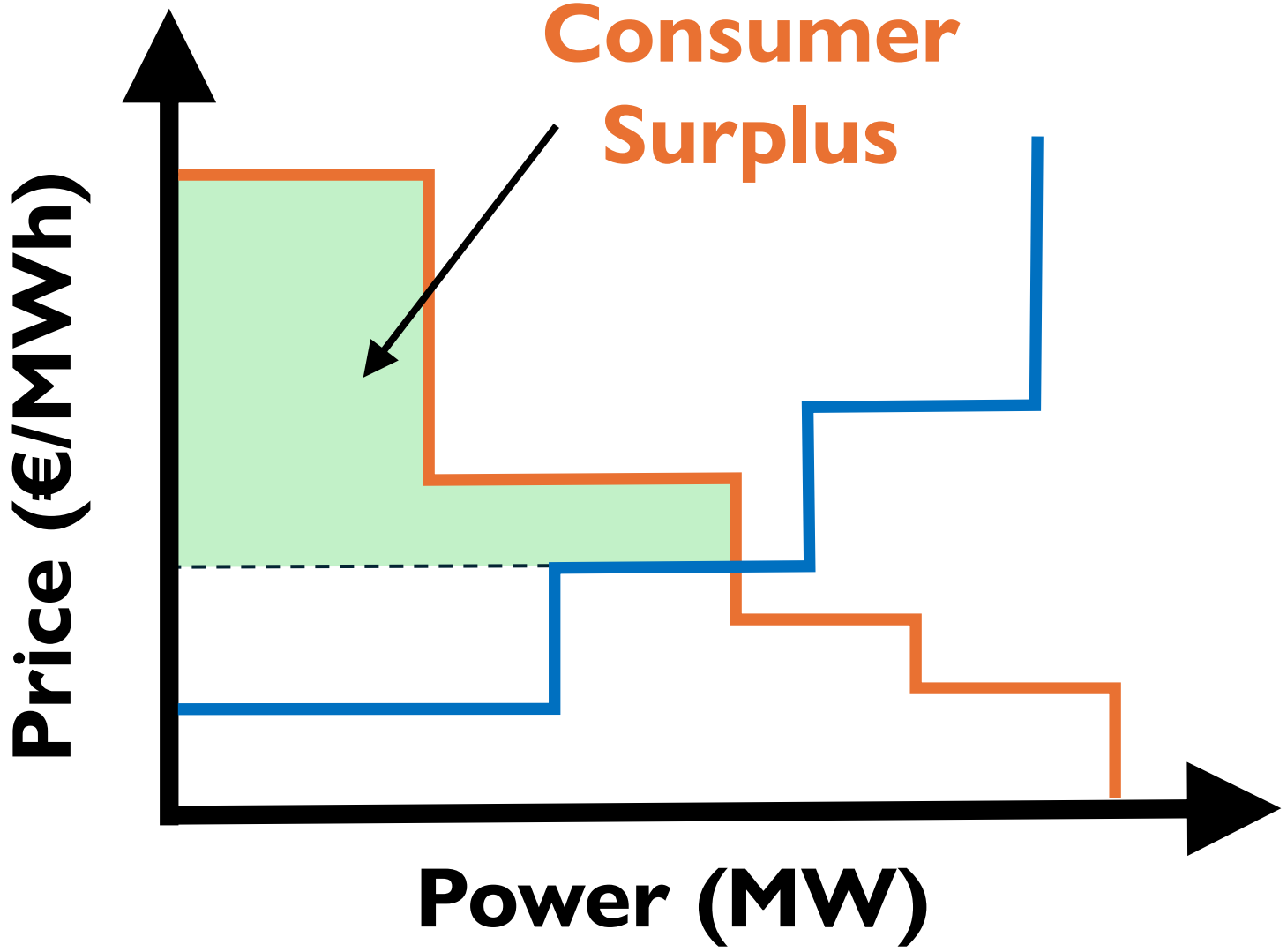
Day-ahead electricity market



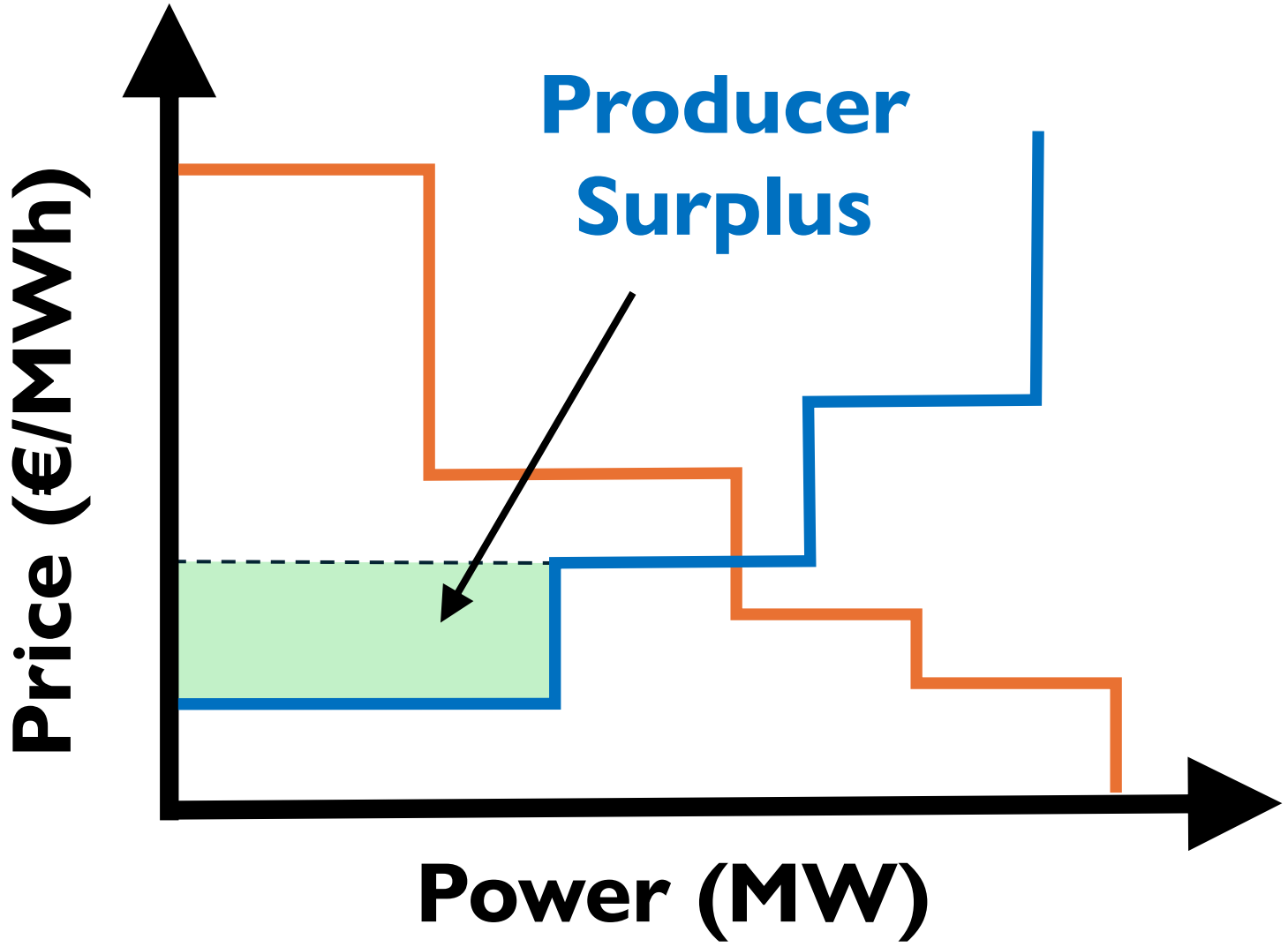
Day-ahead electricity market



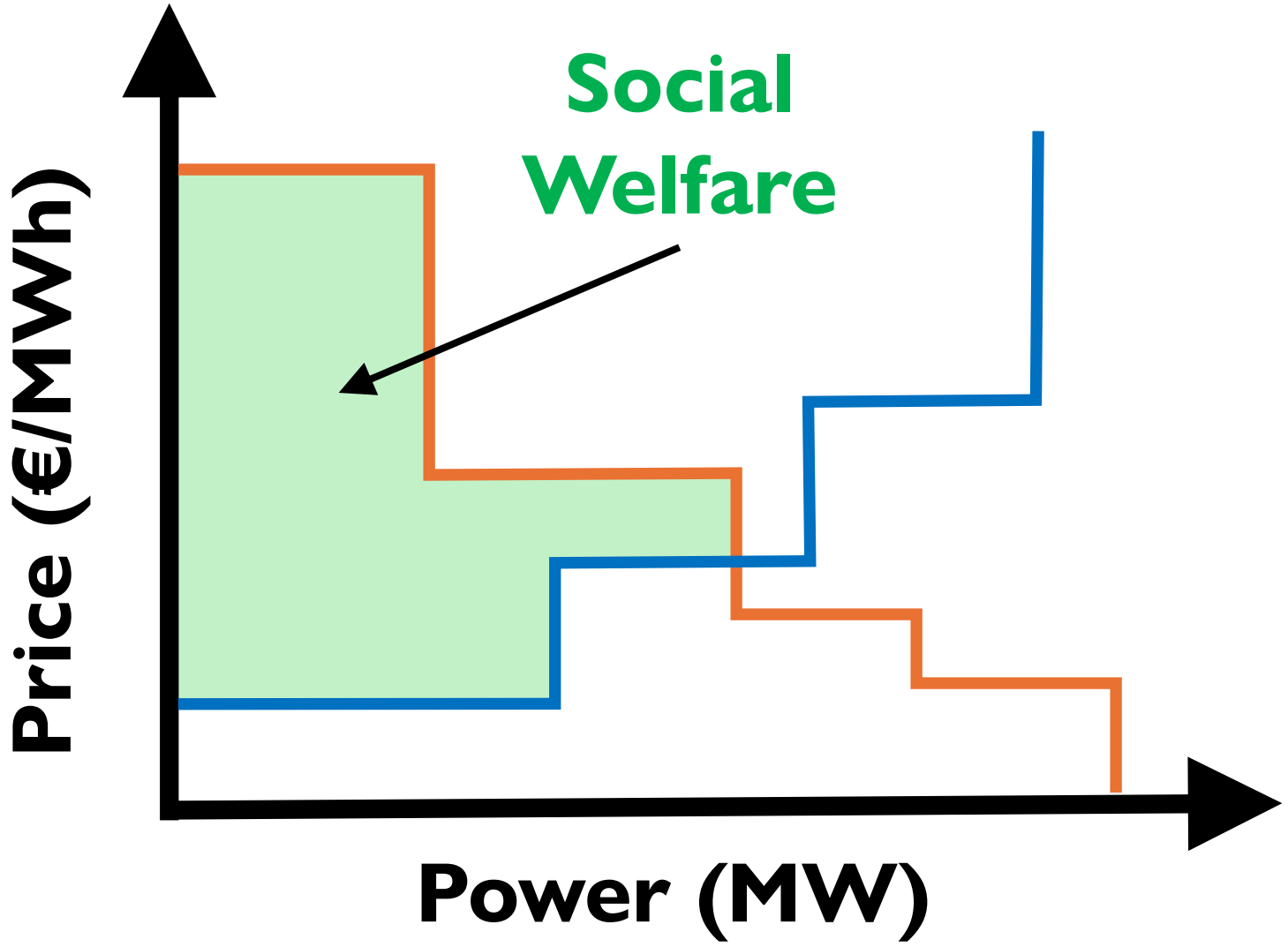
Day-ahead electricity market



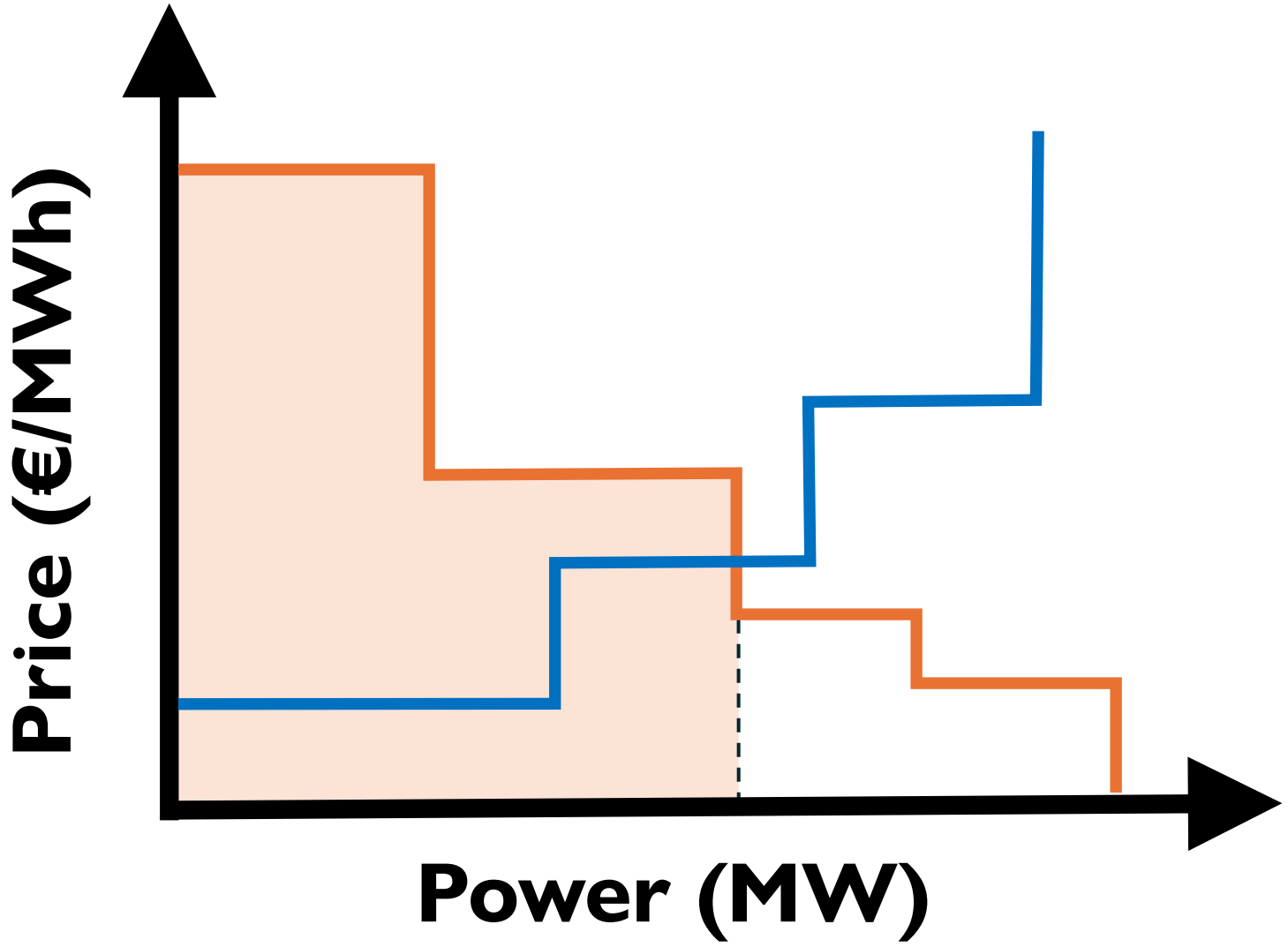
Day-ahead electricity market



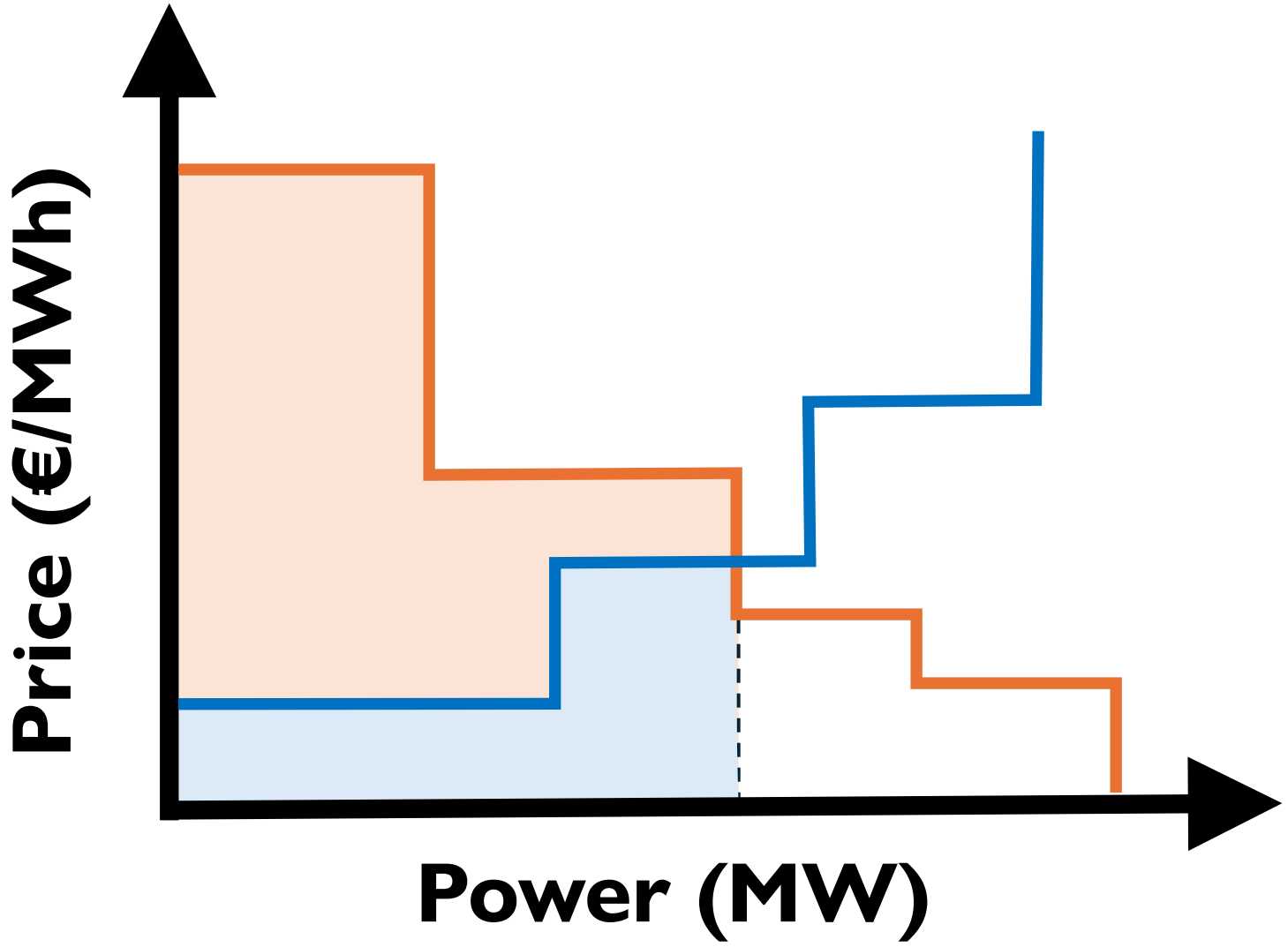
Day-ahead electricity market



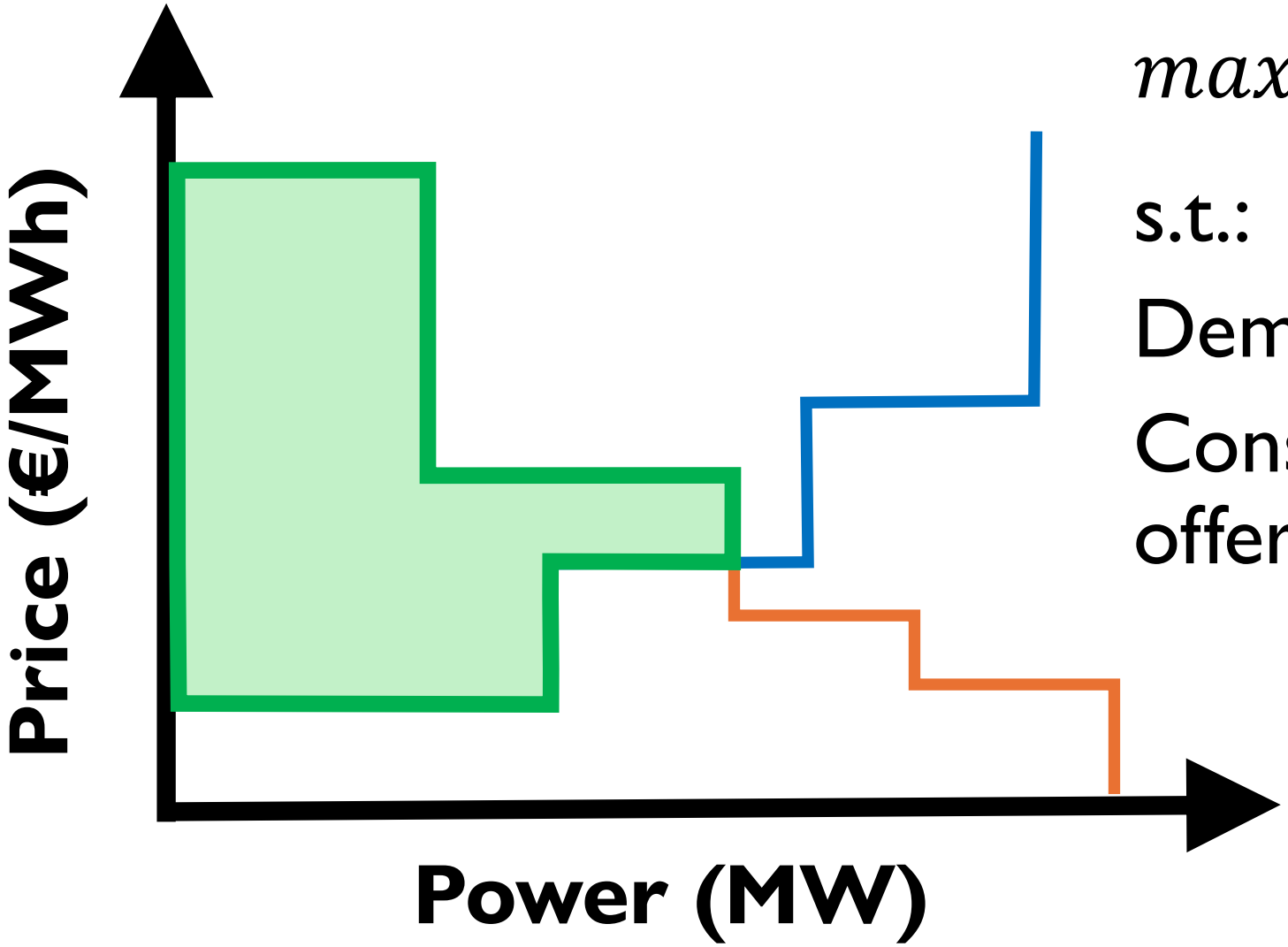
Day-ahead electricity market



Day-ahead electricity market



Day-ahead electricity market



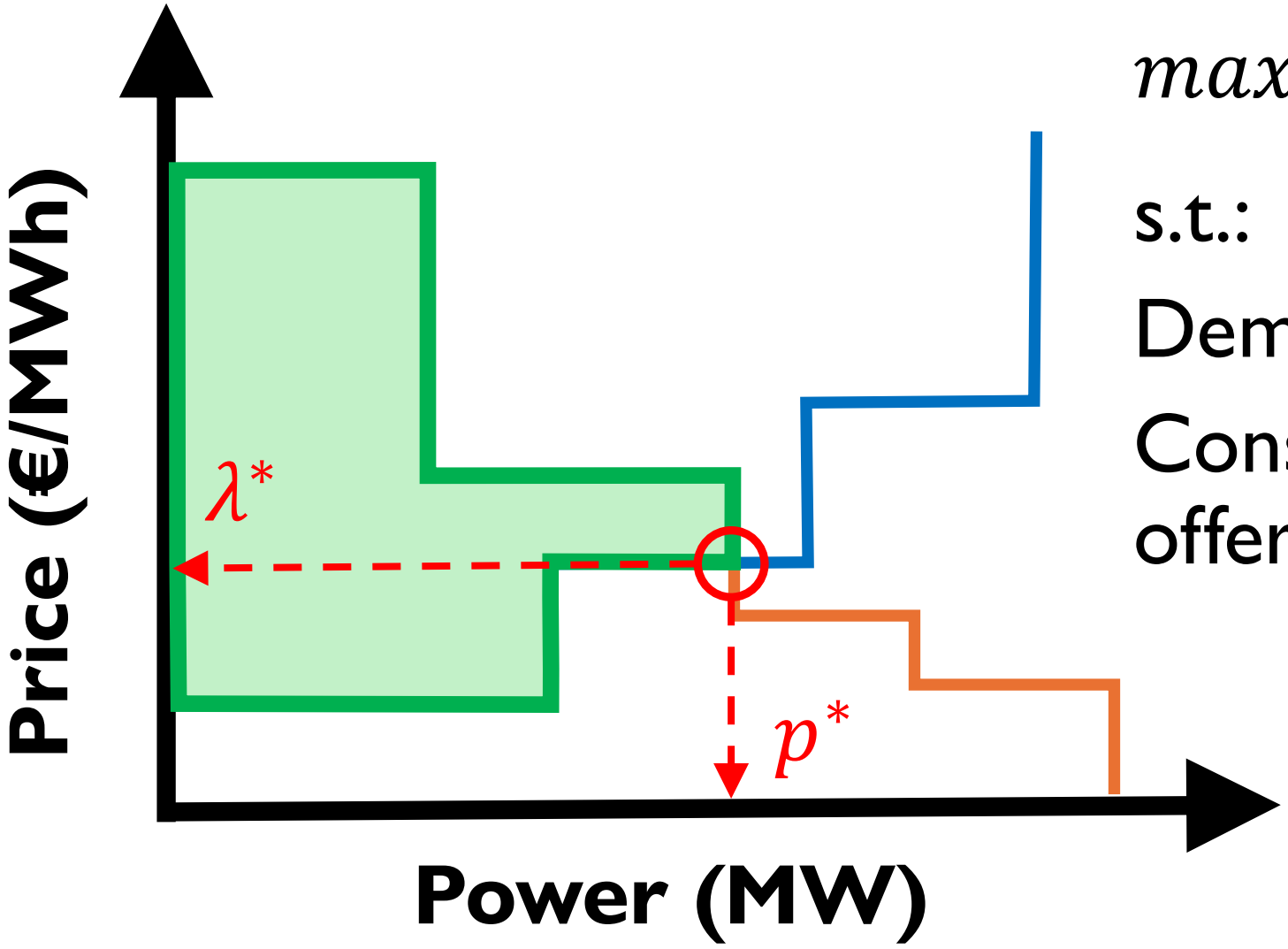
max ▢ = ▢ − ▢

s.t.:

Demand − Supply = 0 : λ

Constraints of the bids and offers

Day-ahead electricity market



max ■ = ■ − ■

s.t.:

Demand − Supply = 0 : $\boldsymbol{\lambda}$

Constraints of the bids and offers

Exercise 8 – Day-ahead electricity market

Consumer 1

- Bid 1: 240 MW, €25/MWh.
- Bid 2: 180 MW, €23/MWh.

Consumer 2

- Bid 1: 147 MW, €19/MWh.
- Bid 2: 110 MW, €18/MWh.

Exercise 8 – Day-ahead electricity market

Producer 1

- Offer 1: 217 MW, €18/MWh.
- Offer 2: 200 MW, €20/MWh.

Producer 2

- Offer 1: 220 MW, €21/MWh.
- Offer 2: 250 MW, €25/MWh.

Exercise 8 – Day-ahead electricity market

Determine the optimal solution of the day-ahead electricity market.

Solution:

Social Welfare = €2171.

$p^* = 420$ MW.

$\lambda^* =$ €21/MWh.

8. Reformulation techniques

Reformulation techniques

This section presents some reformulation techniques that I have used during my previous works:

- Slack variables.
- Product of two binary variables.
- Product of continuous and binary variables.
- Absolute value.
- Square root.

Slack variables

$$\min_{x,y} \quad z = f(x, y)$$

s.t.:

$$0 \leq x \leq 10$$
$$0 \leq y \leq 10$$
$$x + y = 25$$

Infeasible problem

Slack variables

$$\min_{x,y,\boldsymbol{w}} \quad z = Ax + By + \boldsymbol{Cw}$$

s.t.:

$$0 \leq x \leq 10$$
$$0 \leq y \leq 10$$
$$x + y + \boldsymbol{w} = 25$$

where C is a large constant such that $C > A$ and $C > B$.

Optimal solution:

$$x^* = 10$$
$$y^* = 10$$
$$w^* = 5$$

Slack variables

The use of slack variables is useful for example in transmission network expansion planning problems.

If the candidate transmission lines are not sufficient to satisfy the demand, then the slack variables are different from 0. Hence, the optimization problem must be solved again considering additional candidate transmission lines to ensure the feasibility of the problem, i.e., satisfy the demand.

Product of two binary variables – Case 1

$$\min_{a,b,x,y} \quad z = f(x,y)$$

s.t.:

$$\underline{X} \leq x \leq \overline{X}$$
$$\underline{Y} \leq y \leq \overline{Y}$$
$$a, b \in \{0,1\}$$
$$g(x,y) \leq D\boldsymbol{ab}$$

Nonlinear term: ab

Product of two binary variables – Case 1

$$\min_{a,b,x,y} \quad z = f(x,y)$$

s.t.:

$$\underline{X} \leq x \leq \overline{X}$$
$$\underline{Y} \leq y \leq \overline{Y}$$
$$a, b \in \{0,1\}$$
$$g(x,y) \leq D\boldsymbol{ab}$$

$$\min_{a,b,x,y} \quad z = f(x,y)$$

s.t.:

$$\underline{X} \leq x \leq \overline{X}$$
$$\underline{Y} \leq y \leq \overline{Y}$$
$$a, b \in \{0,1\}$$
$$\boldsymbol{g(x,y) \leq Da}$$
$$\boldsymbol{g(x,y) \leq Db}$$

Nonlinear term: ab

Product of two binary variables – Case 1

a	b	$g(x,y) \leq Da$	$g(x,y) \leq Db$	Result
0	0	$g(x,y) \leq 0$	$g(x,y) \leq 0$	$g(x,y) \leq 0$
0	1	$g(x,y) \leq 0$	$g(x,y) \leq D$	$g(x,y) \leq 0$
1	0	$g(x,y) \leq D$	$g(x,y) \leq 0$	$g(x,y) \leq 0$
1	1	$g(x,y) \leq D$	$g(x,y) \leq D$	$g(x,y) \leq D$

Product of two binary variables – Case 2

$$\min_{a,b,x,y} \quad z = f(x,y)$$

s.t.:

$$\underline{X} \leq x \leq \overline{X}$$

$$\underline{Y} \leq y \leq \overline{Y}$$

$$a, b \in \{0,1\}$$

$$h(x,y) = D\boldsymbol{ab}$$

Nonlinear term: ab

Product of two binary variables – Case 2

We cannot replace $h(x, y) = Dab$ with $h(x, y) = Da$ and $h(x, y) = Db$.

Try it if you do not believe me.

Product of two binary variables – Case 2

$$\min_{a,b,x,y} \quad z = f(x,y)$$

s.t.:

$$\underline{X} \le x \le \overline{X}$$
$$\underline{Y} \le y \le \overline{Y}$$
$$a, b \in \{0,1\}$$
$$h(x,y) = D\boldsymbol{ab}$$

Nonlinear term: ab

$$\min_{a,b,\boldsymbol{c},x,y,} \quad z = f(x,y)$$

s.t.:

$$\underline{X} \le x \le \overline{X}$$
$$\underline{Y} \le y \le \overline{Y}$$
$$a, b \in \{0,1\}$$
$$h(x,y) = D\boldsymbol{c}$$
$$\boldsymbol{c \le a}$$
$$\boldsymbol{c \le b}$$
$$\boldsymbol{c \ge a + b - 1, c \ge 0}$$

Product of two binary variables – Case 2

a	b	$c \leq a$	$c \leq b$	$c \geq a + b - 1$	c
0	0	$c \leq 0$	$c \leq 0$	$c \geq -1$*	0
0	1	$c \leq 0$	$c \leq 1$	$c \geq 0$	0
1	0	$c \leq 1$	$c \leq 0$	$c \geq 0$	0
1	1	$c \leq 1$	$c \leq 1$	$c \geq 1$	1

* Note that $c \geq 0$.

Product of continuous and binary variables

$$\min_{x,y} \quad z = f(x,y)$$

s.t.:

$$\underline{X} \leq x \leq \overline{X}$$

$$y \in \{0,1\}$$

$$g(x,y) \leq A\boldsymbol{xy}$$

Nonlinear term: $\boldsymbol{xy}$

Product of continuous and binary variables

Disjunctive linearization

&

Big-M method

Product of continuous and binary variables

$$\min_{x,y} \quad z = f(x,y)$$

s.t.:

$$\underline{X} \leq x \leq \overline{X}$$

$$y \in \{0,1\}$$

$$g(x,y) \leq A\boldsymbol{xy}$$

Nonlinear term: $\boldsymbol{xy}$

$$\min_{x,y,\boldsymbol{w}} \quad z = f(x,y)$$

s.t.:

$$\underline{X} \leq x \leq \overline{X}$$

$$y \in \{0,1\}$$

$$g(x,y) \leq A\boldsymbol{w}$$

$$\boldsymbol{-My \leq w \leq My}$$

$$\boldsymbol{-M(1-y) \leq x - w \leq M(1-y)}$$

where M is a large constant.

Product of continuous and binary variables

If $y = 1$:

$-M(1-1) \leq x - w \leq M(1-1) \rightarrow 0 \leq x - w \leq 0 \rightarrow$

$\rightarrow x - w = 0 \rightarrow \boldsymbol{w = x}$

$-M \cdot 1 \leq w \leq M \cdot 1 \rightarrow -M \leq w \leq M$

Note that $-M \leq w \leq M$ and $w = x$, i.e., $-M \leq x \leq M$.

The value of M must be selected such that $-M \leq x \leq M$ is not more restrictive than $\underline{X} \leq x \leq \overline{X}$.

Different M's can be used for upper and lower limits.

Product of continuous and binary variables

If $y = 0$:

$-M \cdot 0 \leq w \leq M \cdot 0 \rightarrow -0 \leq w \leq 0 \rightarrow \boldsymbol{w = 0}$

$-M(1 - 0) \leq x - 0 \leq M(1 - 0) \rightarrow -M \leq x \leq M$

As mentioned before:

- The value of M must be selected such that $-M \leq x \leq M$ is not more restrictive than $\underline{X} \leq x \leq \overline{X}$.
- Different M's can be used for upper and lower limits.

Product of continuous and binary variables

If $y = 1$, then $w = x$.

If $y = 0$, then $w = 0$.

This means that $\boldsymbol{w = xy}$.

Product of continuous and binary variables

Additional comments:

- Very large values of M may lead to computational issues.
- Constraints associated with M must be checked ex post $\rightarrow$ Trial-and-error method.
- The proper values of M can be known in some cases (see next slide).

Product of continuous and binary variables

$$\min_{x,y,w} \quad z = f(x,y)$$

s.t.:

$$\underline{X} \leq x \leq \overline{X}$$

$$y \in \{0,1\}$$

$$g(x,y) \leq Aw$$

$$\boldsymbol{\underline{X}} y \leq w \leq \boldsymbol{\overline{X}} y$$

$$\boldsymbol{\underline{X}}(1-y) \leq x - w \leq \boldsymbol{\overline{X}}(1-y)$$

Product of two continuous variables

Nonlinear term: xy in the objective function, an equality constraint or an inequality constraint.

We can discretize one of the continuous variables.

This approach is an **approximation**, not an exact reformulation!

Product of two continuous variables

For example, we can consider some discrete values of one of the continuous variables:

$$x = X_1 u_1 + X_2 u_2 + X_1 u_1 + X_3 u_3$$
$$u_1 + u_2 + u_3 = 1$$
$$u_1, u_2, u_3 \in \{0, 1\}$$

Then, $xy = X_1 u_1 y + X_2 u_2 y + X_1 u_1 y + X_3 u_3 y$. The nonlinear terms $u_i y$ of binary and continuous variables can be linearized as explained before.

Product of two continuous variables

We can also use a binary expansion linearization (**approximation**):

$$x = \underline{X} + \left(\overline{X} - \underline{X}\right) \sum_{k=1}^{K} 2^{-k} u_k$$

$$u_k \in \{0, 1\}, k = 1, \dots, K$$

Absolute value

$$\min_{x,y,w} \quad z = f(x,y) + D|\boldsymbol{w}|$$

s.t.:

$$\underline{X} \leq x \leq \overline{X}$$
$$\underline{Y} \leq y \leq \overline{Y}$$
$$\underline{W} \leq w \leq \overline{W}$$
$$g(x,y) \leq C|\boldsymbol{w}|$$

Nonlinear term: $|w|$

Absolute value

$$\min_{x,y,w} \quad z = f(x,y) + D|\boldsymbol{w}|$$

s.t.:

$$\underline{X} \le x \le \overline{X}$$
$$\underline{Y} \le y \le \overline{Y}$$
$$\underline{W} \le w \le \overline{W}$$
$$g(x,y) \le C|\boldsymbol{w}|$$

Nonlinear term: $|w|$

$$\min_{a,b,x,y,w} \quad z = f(x,y) + D\boldsymbol{a}$$

s.t.:

$$\underline{X} \le x \le \overline{X}$$
$$\underline{Y} \le y \le \overline{Y}$$
$$\underline{W} \le w \le \overline{W}$$
$$g(x,y) \le C\boldsymbol{a}$$
$$\boldsymbol{b} \in \{\boldsymbol{0}, \boldsymbol{1}\}$$
$$\boldsymbol{0} \le \boldsymbol{a} - \boldsymbol{w} \le \boldsymbol{M}(\boldsymbol{1} - \boldsymbol{b})$$
$$\boldsymbol{0} \le \boldsymbol{a} + \boldsymbol{w} \le \boldsymbol{M}\boldsymbol{b}$$

Absolute value

If $b = 1$:

$0 \leq a - w \leq M(1-1) \rightarrow 0 \leq a - w \leq 0 \rightarrow \boldsymbol{a = w}$

$0 \leq a + w \leq M \cdot 1 \rightarrow 0 \leq a + w \leq M \rightarrow$

$\rightarrow 0 \leq 2w \leq M \rightarrow 0 \leq w \leq M/2$

- Can w be < 0? $\boldsymbol{0 \leq w} \leq M/2 \rightarrow$ Not possible
- Can w be > 0? $0 \leq w \leq M/2 \rightarrow$ OK

Then, $b = 1$ if $w > 0$.

Absolute value

If $b = 0$:

$0 \leq a + w \leq M \cdot 0 \rightarrow 0 \leq a + w \leq 0 \rightarrow \boldsymbol{a = -w}$

$0 \leq a - w \leq M(1 - 0) \rightarrow 0 \leq a - w \leq M \rightarrow$

$\rightarrow 0 \leq -2w \leq M \rightarrow -M \leq 2w \leq 0 \rightarrow -M/2 \leq w \leq 0$

- Can w be > 0? $-M/2 \leq \boldsymbol{w \leq 0} \rightarrow$ Not possible
- Can w be < 0? $-M/2 \leq w \leq 0 \rightarrow$ OK

Then, $b = 0$ if $w < 0$.

Absolute value

If $w > 0$, then $a = w$.

If $w < 0$, then $a = -w$.

This means that $\boldsymbol{a = |w|}$.

Square root

$$\min_{x,y} \quad z = f(x,y) + \sqrt{x^2 + y^2}$$

s.t.:

$$h(x,y) = 0$$
$$g(x,y) \leq 0$$

Nonlinear term: $\sqrt{x^2 + y^2}$

This is an NLP problem.

Square root

$$\min_{x,y} \quad z = f(x,y) + \sqrt{\boldsymbol{x^2 + y^2}}$$

s.t.:

$$h(x,y) = 0$$
$$g(x,y) \leq 0$$

Nonlinear term: $\sqrt{x^2 + y^2}$

This is an NLP problem.

$$\min_{x,y,\boldsymbol{w}} \quad z = f(x,y) + \boldsymbol{w}$$

s.t.:

$$h(x,y) = 0$$
$$g(x,y) \leq 0$$
$$\boldsymbol{x^2 + y^2 \leq w^2}$$
$$\boldsymbol{w \geq 0}$$

Nonlinear terms: x^2, y^2, and w^2.

This is a QCQP problem.

Square root

The constraint $x^2 + y^2 \leq w^2$ acts as $x^2 + y^2 = w^2$ in practice because w is considered in the objective function (minimization problem). Hence, $w = \pm\sqrt{x^2 + y^2}$.

Moreover, constraint $w \geq 0$ should be considered to prevent the solution $w = -\sqrt{x^2 + y^2}$.

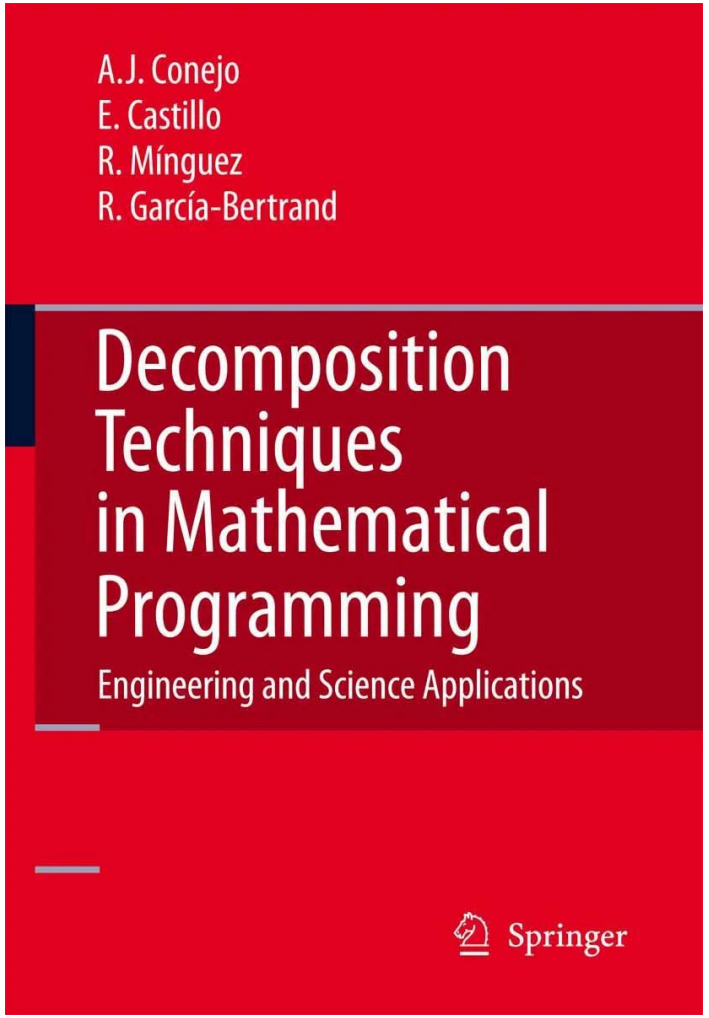
Therefore, the solution is $w = +\sqrt{x^2 + y^2}$.

Next lecture

- ~~Lecture 1: Introduction to Operations Research.~~
- **Lecture 2: Operation and Planning Problems.**
- Lecture 3: Operational Variability.
- Lecture 4: Uncertainty and Risk.
- Lecture 5: Further Applications.

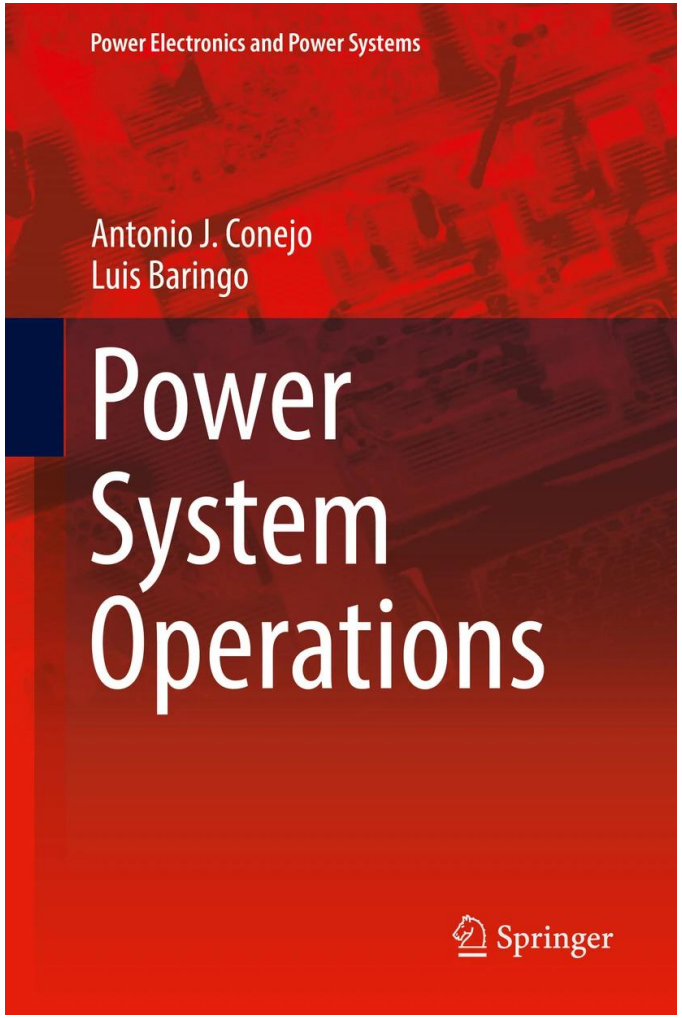
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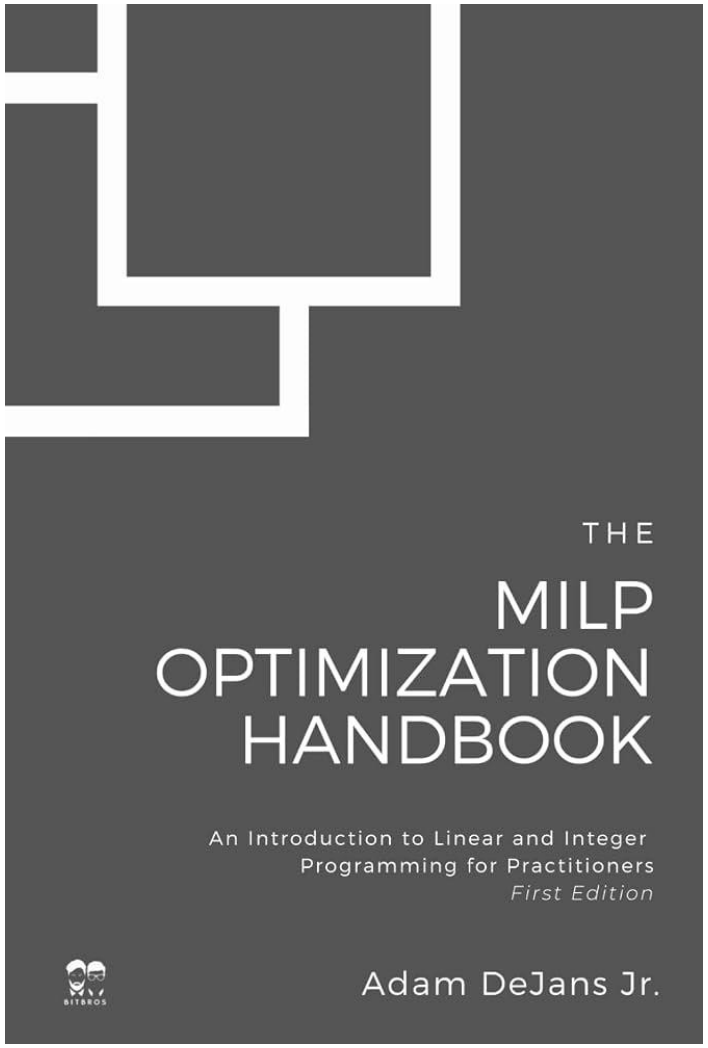
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